Volume limits for arborescent knots

Long twists and explicit conditions for ordered tangle trees

Finite knot invariants, hyperbolic filling, and the preferred cusp shape

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Abstract

We prove the volume conjecture for two specified classes of arborescent knot diagrams built by ordered stacking and quarter-turns of two-string tangles. The first is a recursive construction in which independently sufficiently large signed twist numbers give a hyperbolic knot and the volume limit. Its elementary pieces retain their original crossing circles, disk framings and endpoint order.

For ordered trees with one crossing-circle operator at every vertex, we give two explicit sufficient conditions. If d_v is the degree of a vertex and n_v is its signed crossing power, either

$$\sum_v \frac{d_v}{d_v^2 + n_v^2} \leq 2^{-25}, \qquad \text{or} \qquad \max_v \frac{d_v}{d_v^2 + n_v^2} \leq 2^{-120}$$

suffices whenever the original strand pairing gives a knot. The first condition gives the explicit bound $|n_v| \geq 8192\sqrt{E}$ for a tree with E edges. The second removes the sum over vertices and permits the bound $|n_v| \geq 2^{119}$ independently of the tree size. Both allow short or zero powers at sufficiently large-degree vertices. Every diagram, parity choice and integer power is fixed before the root-of-unity level N tends to infinity. The limits include both parities and composite levels, and only auxiliary circles are filled.

The proof starts with the finite matrices defining the invariant and retains their derivative data at repeated spectral roots. For the long-twist construction, positive multiplication coefficients and exact rational inequalities control the signed color terms. For the quantitative tree theorems, a literal projection cut gives two octahedra per edge. Positive deformations of these cells fit in the original cyclic order, and their actual primitive periods give smooth filling even when the total meridian logarithm has left a small neighborhood of zero. A separate metric condition, $\sqrt{d_v^2 + n_v^2} \geq 630$ at every vertex, constructs these geometric fillings without a summed condition. The local quantum argument evaluates each subtree as an exact polynomial and normalizes it by its own value at zero. The full Fourier calculation keeps the charge tails and boundary contributions, identifies a nonzero leading term, and matches its calibrated action with the volume of the original knot complement.

Both quantitative theorems give a uniform microscopic main-meridian profile for the zero-framed reduced ADO invariant, with its exact crossing and framing factor retained. Under the summed criterion we also identify the first correction with the preferred cusp shape of the filled knot. The construction hypotheses and the knot-pairing test are stated explicitly; the results do not classify all arborescent knots.

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Chapter 1

The knots, the theorems, and their proofs

1.1 What the theorem compares

The volume conjecture asks whether the growth of a sequence of finite knot invariants recovers hyperbolic volume. For each integer $N \geq 2$, the invariant is evaluated at a root of unity and produces a complex number. For a hyperbolic knot, the complement carries a complete finite-volume hyperbolic metric. The proposed comparison is

$$\frac{2\pi}{N} \log |\langle K \rangle_N| \longrightarrow \mathrm{Vol}(S^3 \setminus K).$$

The normalization of $\langle K \rangle_N$ matters: throughout this paper the unknot has value one, and Chapter 2 specifies the matrices and traces defining the invariant. Kashaev introduced the finite invariant in [1] and the volume proposal in [2]. Its relation with the root-of-unity colored Jones polynomial is described in [3].

The first theorem proves this limit for the recursive family of knot diagrams constructed below, once every designated twist region has sufficiently many twists. The family permits arbitrary finite branching and nesting, subject to its precise construction rule. Chapters 9 and 10 treat ordered trees with one crossing-circle operator at every junction. They give two explicit sufficient conditions involving degrees and signed crossing powers: one summed over vertices, and one imposed separately at each vertex. Both permit some short coefficients. We first explain the geometric operations and then state the construction scopes. Every construction records an actual diagram, including the order of its pieces and the positions where they are joined.

All these theorems take the same quantum limit. Choose a finite construction and signed integer twist numbers satisfying the applicable hypothesis, and hold them fixed. Then let N tend to infinity. Neither the diagram nor its twist numbers grow with N. In the first theorem the individual twists can have different signs and arbitrarily different large magnitudes. In the quantitative theorems their sizes and junction degrees enter the sum in Theorem 9.1 or the maximum in Theorem 10.1.

1.2 Building the diagrams from four-ended tangles

A *two-string tangle* is a pair of arcs in a ball whose four endpoints are fixed on its boundary. We draw its ball as a box, with two endpoints at the bottom and two at the top. Closed curves may also be present inside a box. In the augmented diagrams used in the proof, some of those curves are distinguished auxiliary circles. Composition can create further closed main components, so the eventual requirement that the filled diagram be a knot must be checked rather than assumed.

Label the boundary points in their cyclic order by

$$0 = \text{bottom left}, \quad 1 = \text{bottom right}, \quad 2 = \text{top right}, \quad 3 = \text{top left}.$$

We use three operations, all performed on the drawn tangles with these boundary positions retained.

Vertical stacking. Place a box F below a box G. Join the two top endpoints of F to the two bottom endpoints of G by short, uncrossed arcs: F_2 joins G_1 and F_3 joins G_0. Denote the resulting box by FG. Its exposed endpoints, in the order just defined, are (F_0, F_1, G_2, G_3). Changing parentheses only changes the heights of boxes; it does not interchange their order.

Quarter-turn. Rotate the entire box clockwise through 90 degrees in the diagram plane, carrying every strand and auxiliary circle with it. Write $\mathsf{T}F$ for the result. In terms of the old endpoints, its new ordered boundary is (F_1, F_2, F_3, F_0). The algebraic evaluation implements this operation with the ordinary cup and cap matrices. It is not an arbitrary permutation of matrix entries.

Cylindrical closure. Join the exposed bottom-left endpoint to the top-left endpoint outside the box, and do the same on the right, without adding crossings. Write $C(F)$ for this closed diagram. Thus the closing pairs are 0–3 and 1–2. In $C(FG)$ the two connections between F and G and the two exterior return arcs are all part of the definition. This closure differs from joining the two top endpoints together and the two bottom endpoints together, which we call a *cap closure*.

The elementary augmented box has two through strands and a small auxiliary circle surrounding them. The circle bounds a disk pierced once by each strand; this disk fixes the framing used for surgery. Denote the box by L, and denote its quarter-turn by $h = \mathsf{T}L$. Every occurrence of h in a construction is a separate copy with its own auxiliary circle. We call such an occurrence an *elementary piece*; later, when discussing its position in a construction tree, we also call it a *leaf*.

To obtain the final knot, replace each auxiliary circle by twists of the strands passing through its disk. Let R be one positive half twist of the two through strands; R^{-1} is the negative crossing. An integer power R^n is a stack of $|n|$ such crossings with the indicated sign. At elementary piece ℓ we retain $\varepsilon_\ell \in \{0, 1, -1\}$ half twists and perform the disk-framed integral filling with parameter p_ℓ. The filled piece is exactly

$$\mathsf{T}\big(R^{\varepsilon_\ell - 2p_\ell}\big). \tag{1.1}$$

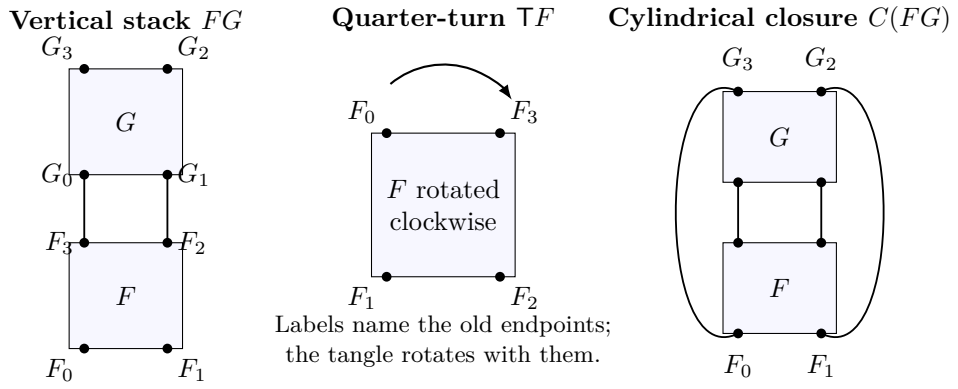


Figure 1.1: The actual connections defining the construction. Each box contains an arbitrary tangle; its interior is suppressed so that the boundary order is visible. The quarter-turn carries the tangle with the box. The exterior arcs in the last panel are the cylindrical closure.

A full twist contains two crossings, which accounts for the factor two. This formula fixes the sign convention for p_ℓ throughout the proof. After every auxiliary circle is filled, the remaining closed diagram has no auxiliary components. The main component is never filled; its complement is the manifold whose volume occurs in the theorem.

1.3 Which compositions are included

Start by stacking two elementary pieces. From an already constructed box, one may take a quarter-turn, stack it with another already constructed box, or attach one new elementary piece at its bottom or top. Repeat these operations any finite number of times. We call a box obtained in this way *completed*. This is a name for the construction rule, not an assertion that a hyperbolic metric has already been constructed. In particular, a single elementary piece is not completed.

Definition 1.1 (The recursive construction class)**.** The symbol E ranges over the completed boxes just described. Their concise recursive description is

$$E ::= hh \;\mid\; \mathsf{T}(E) \;\mid\; E_1E_2 \;\mid\; hE \;\mid\; Eh. \tag{1.2}$$

Here ::= means "can be built by one of the following operations," and the vertical bars separate the alternatives. Each written h denotes a distinct elementary piece. A closed diagram in the main theorem is $C(FG)$ with both F and G completed. Retain the original order of all endpoints, the crossing-disk framings, and one marked point on the main component for evaluating the reduced invariant.

For example, $C((hh)(hh))$ uses four elementary pieces, with two in each of the final boxes. A longer allowed construction is $C((h\mathsf{T}(hh))(hh))$: first build the two-piece box hh, turn it, attach one elementary piece, and close it against another two-piece box. Repeating $E_{r+1} = h\mathsf{T}(E_r)$ from $E_0 = hh$ gives arbitrarily long nested constructions. Branching is equally allowed, since any two completed boxes can be stacked. Starting instead with any stack h^s, $s \geq 2$, gives the same class.

The last condition in Definition 1.1 is important: both boxes joined by the final cylindrical closure must be completed. The proof uses it to place every stationary leaf parameter strictly above 1/4, as shown in Lemma 3.5. Chapters 2–8 retain this condition. The different class in Chapter 9 has its own finite-source estimates and geometric completion; it is not inserted into this recursive proof by changing the meaning of "completed."

The filled elementary pieces in (1.1) are rational tangles. Ordered stacking and quarter-turns of rational tangles produce Conway algebraic, or arborescent, links. Thus the construction gives a specified family within that class; it is not a classification of all arborescent knots. The choice of ε must make the final diagram one component and create no closed main component inside a box. Section 4.7 gives a finite endpoint-pairing test and proves that a suitable choice exists for every allowed construction. Changing p_ℓ adds full twists and therefore does not change this component test.

1.4 The long-twist theorem

Theorem 1.2 (Long-twist volume limit)**.** *Fix a finite expression $C(FG)$ as in Definition 1.1 and a half-twist pattern ε giving a knot. There is a finite number $P(F,G,\varepsilon)$ such that, for every independently chosen tuple of signed integers $\mathbf{p}=(p_\ell)$ with $|p_\ell|>P(F,G,\varepsilon)$, the resulting knot $K_{F,G,\varepsilon}(\mathbf{p})$ is hyperbolic and*

$$\lim_{N\to\infty}\frac{2\pi}{N}\log\left|\langle K_{F,G,\varepsilon}(\mathbf{p})\rangle_N\right|=\operatorname{Vol}\left(S^3\setminus K_{F,G,\varepsilon}(\mathbf{p})\right). \tag{1.3}$$

The level tends to infinity through all integers, including even and composite levels. The expression, signs and twists are fixed before that limit. The threshold depends on the fixed expression and sign pattern, but not on a comparison between the magnitudes or signs of the individual sufficiently long twists.

In concrete terms, one first chooses the entire finite arrangement of boxes and the parity of the crossings in each elementary piece. The theorem then supplies one threshold for that arrangement. Each filling integer may independently be positive or negative and exceed that threshold in absolute value. No ratio between different filling integers is bounded. Once these integers have been chosen, the conclusion concerns the single resulting knot and the sequence of its invariants as N grows.

The reduced invariant $\langle\cdot\rangle_N$ has unknot value one. Chapter 2 also compares it with the full modified one-cut normalization: that normalization adds the single factor i^{N+3}/N and a unit framing character, so the exponential limit is unchanged. This factor belongs only to the marked main component. The auxiliary circles are evaluated by ordinary traces.

The isolated closures $C(h^s)$, $s\geq 3$, form a separate inherited family; Section 6.4 works out $C(h^3)$ explicitly. The isolated $C(hh)$ is excluded: its even-level evaluation vanishes and its odd-level contribution lies on a flat boundary. General closures with a single elementary piece on one side, such as $C(hE)$, are also outside Theorem 1.2. For instance, $C(h\mathsf{T}(h^s))$ places the inner complete leaf parameters at 1/4, outside the interval used in the first proof. Such a circle-tree presentation can instead fall under Theorem 9.1 when its explicit degree and twist condition holds. The exclusions in this paragraph concern Theorem 1.2.

1.5 An explicit criterion for circle trees

The second construction uses the same actual stacking, quarter-turn and closing operations. At each vertex of a finite ordered tree put one crossing box with signed power n_v; its augmented version retains that box's crossing circle. Each tree edge is a quarter-turn of a child tangle. There are no unmarked product junctions, and terminal crossing circles are vertices of degree one. The cyclic order and the position of the crossing box among its children remain part of the diagram. Section 9.1 gives the full recursion.

If d_v is the degree of vertex v, Theorem 9.1 proves hyperbolicity and the same volume limit under the explicit condition

$$\sum_v \frac{d_v}{d_v^2 + n_v^2} \leq 2^{-25},$$

provided the original endpoint pairing gives a knot. The angle feasibility required by the proof is automatic for this literal tree construction. Every $|n_v| \geq 8192\sqrt{E}$ suffices when the tree has E edges, but individual powers may be short or zero at vertices of sufficiently large degree. Adding an identity braid at an unmarked junction also gives an application when the sum bound holds; its added crossing circle contributes $1/d_v$, rather than zero.

The quantitative theorem uses a direct cut of the actual augmented exterior into two octahedra per tree edge. Its positive geometric path keeps track of the original primitive slopes even when their accumulated logarithms are not small. The finite invariant has a uniform main-meridian profile, with its exact scalar retained; the first correction is the preferred cusp shape of the resulting knot complement. The size of the tree and every crossing power remain fixed as N grows.

Theorem 10.1 gives another sufficient condition for this same literal class:

$$\max_v \frac{d_v}{d_v^2 + n_v^2} \leq 2^{-120}.$$

Its proof evaluates each subtree as a polynomial in its remaining boundary variable and normalizes that polynomial by its exact value at zero. The next trace spends only the change of action at its current vertex. This gives a numerical bound independent of the number of vertices, although error constants and starting levels may depend on the fixed diagram. The separate geometric condition $\sqrt{d_v^2 + n_v^2} \geq 630$ constructs the positive metric without either quantum condition; it alone does not assert a volume limit.

1.6 A guide to the proof and its terminology

The long-twist proof first studies the augmented diagram, in which the auxiliary circles are still present, and then performs their fillings by an exact finite identity. A *color* is the integer label specifying the representation assigned to an auxiliary circle. At level N these labels range from 1 to N. Before filling, the evaluation therefore depends on a tuple of colors. We call this finite, color-dependent evaluation the *source*. The *complete-color source* means that every auxiliary color equals N; the term is distinct from completeness of a hyperbolic metric.

The algebra used to calculate a source has repeated roots. At a double root, both a polynomial's value and its derivative matter. We refer to the derivative data as the *first normal*

data, in the elementary algebraic sense of the coefficient of $z - c$ modulo $(z - c)^2$. Keeping these data is essential at both parities of N.

Expanding each elementary piece in a polynomial basis gives a finite sum. An admissible assignment of integer indices to that sum is a *state*. An internal *wire* is an index shared by two factors; it is not an additional physical strand. A *product gate* is the coefficient for one stacking operation with two input indices and one output index. After scaling indices by N, these coefficients have exponential growth described by a real function, the *action*. A state is *strict* when all its nonforced angle inequalities are strict. The maximizer of the complete-color action determines explicit ideal hyperbolic cells. The cells have quadrilateral faces along which they are glued. The order of the four vertices specifies the face map; Chapter 4 explains how these auxiliary gluing faces reconstruct the actual closed link diagram.

For colors other than N, the exact elementary evaluation splits into a principal part and a signed complementary part. We call them the *core* and the *alias*; the identities defining them appear in (2.17)–(2.18). Chapter 5 proves that every term containing an alias is exponentially smaller than the leading complete action, even after arbitrary allowed composition. This is where the finite rational certificates enter the proof.

Near the remaining strict maximum, the finite sum has a holomorphic asymptotic of the form $N^{3/2}a(\chi)e^{N\Phi(\chi)}$, with a nonzero. We call this local analytic contribution the *packet*. The word includes a specified finite sum and a controlled error; it does not mean a formal stationary-phase expression. The complex color parameters χ are also the parameters of a marked deformation of the augmented hyperbolic structure. Identifying their peripheral holonomies fixes the absolute volume, not just its differential.

Finally, the filling identity sums the source over all colors with oscillatory coefficients. We analyze the whole sum, including the boundary integrals created when contours move and every integer Fourier mode. These modes are called *charges*. The leading charges add without cancellation, and their critical action equals the volume after filling. This last step is necessary because that volume is smaller than the complete augmented volume; an estimate obtained by taking absolute values of the filling coefficients would not recover it.

1.7 Conventions and verification

All logarithms on a positive real interval are real. Complex logarithms and square roots continue those values on the explicitly chosen neighborhoods. We use $\operatorname{arccot} X \in (0, \pi)$ for real X and $\mathrm{i}^2 = -1$. Constants may depend on the fixed construction, its selected strict stationary point and a fixed analytic neighborhood, but not on N. A polynomial factor means a bound by N^C with such a fixed exponent.

The appendices give the finite identities and detailed analytic estimates for the long-twist argument. Chapter 9 gives the quantitative theorem and links its complete finite-sum, ordered-cut and geometric completion proofs. The certificate files contain rational candidate points and exact domain subdivisions; the verifier proves their supporting inequalities using outward rational bounds. Numerical optimization is not part of verification. Chapter 11 explains which claims are analytic proofs, which depend on these finite certificates, and which are merely checked by optional experiments.

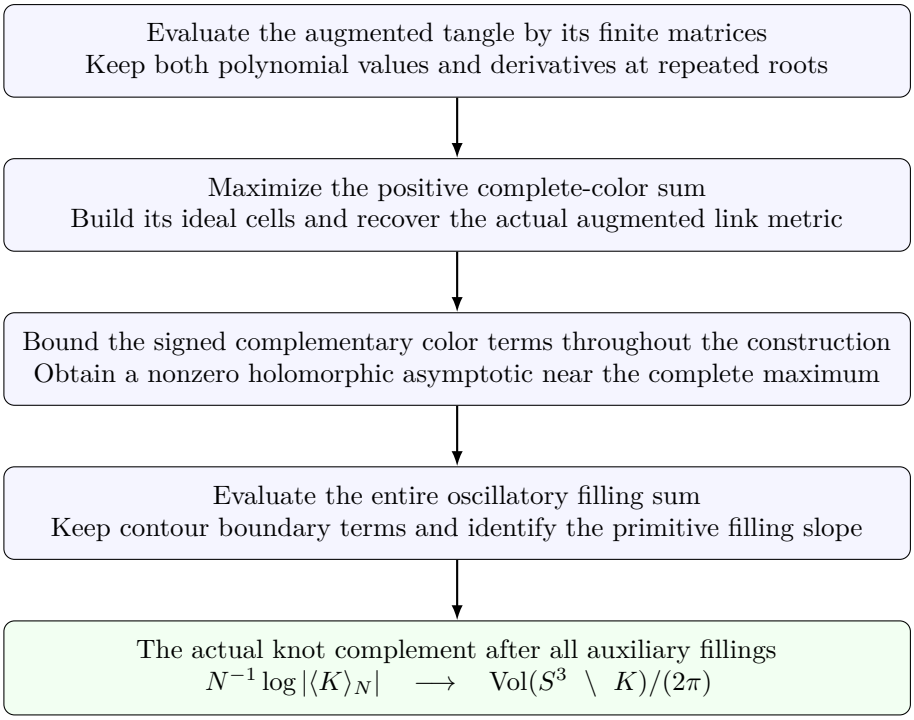


Figure 1.2: The four comparisons in the proof. The last comparison analyzes the full oscillatory filling sum and identifies its critical action with the volume of the resulting knot complement.

Chapter 2

Evaluating the invariant in a finite polynomial algebra

We now turn the tangle operations of Chapter 1 into finite matrix calculations. A strand carries a finite-dimensional vector space, a crossing acts on the two corresponding factors, and joining endpoints means contracting the appropriate indices. Closing an auxiliary circle takes a trace. Leaving one marked main strand open until the final contraction gives the reduced normalization.

The purpose of the polynomial algebra introduced below is to make these contractions compositional. Every two-strand operator that occurs here is a polynomial in one matrix, the pair Casimir Ω. Stacking becomes multiplication of polynomials, while the quarter-turn has a particularly simple action in the Newton basis. The last section then expresses an arbitrary filling power in the basis of colored auxiliary-circle operators. All of these statements are finite identities at each level; asymptotics begin only in Chapter 3.

2.1 Crossings and the one-cut normalization

Fix $N \geq 2$, and set

$$d = N - 1, \qquad q = e^{\pi \mathrm{i}/N}, \qquad Q = q^2, \qquad x = q^d, \qquad \kappa = q^{d^2/2}, \qquad \rho = q - q^{-1}. \tag{2.1}$$

There is no restriction to prime or odd N. The main strands carry the N-dimensional vector space V_N; this is what "complete color" means at this stage. The operators H, K, E, F encode its quantum-group action. They are concrete matrices, which we specify so that the crossing convention can be checked. One convenient choice of basis is $v_0, \dots, v_d$ and

$$Hv_j = (d - 2j)v_j, \quad K = q^H, \quad Ev_j = [j]_q^2 v_{j-1}, \quad Fv_j = v_{j+1}, \qquad [j]_q = \frac{q^j - q^{-j}}{q - q^{-1}}.$$

Missing endpoint vectors are zero. The finite universal crossing, including its Cartan factor, is

$$\mathscr{R} = q^{H \otimes H/2} \sum_{k=0}^{d} \frac{q^{k(k-1)/2} \rho^k}{[k]_q!} E^k \otimes F^k. \tag{2.2}$$

Every denominator is nonzero. The physical braided crossing is obtained with the fixed exchange and transpose convention; the inverse is the actual matrix inverse. The framing and

enhancement are part of the definition, not an unspecified factor appended to an asymptotic formula.

For calculations in the polynomial algebra it is useful to change basis on this same space. We call the new coordinates balanced; they do not define a different invariant. In this basis the operators are

$$\mathcal{E}|a\rangle = (Q^{-a} - 1)|a-1\rangle, \quad \mathcal{F}|a\rangle = (1 - Q^{a+1})|a+1\rangle, \quad K|a\rangle = xQ^{-a}|a\rangle. \tag{2.3}$$

The ordinary cup and cap, and the one marked cap, have entries

$$p_{ab} = Nx^{-b}\delta_{a+b,d}, \qquad \ell_{ab} = \frac{x^a}{N}\delta_{a+b,d}, \qquad m_{ab} = \frac{x^b}{N^2}. \tag{2.4}$$

The cup creates two endpoints and the cap joins them. Their two snake identities say that a bend followed by the opposite bend evaluates to a straight strand. The different marked cap implements the reduced invariant and satisfies $mp = 1$. The diagonal operator $g = K^{1-N} = (-1)^d K$ is the *pivot*, the correction inserted when an ordinary strand is closed by a trace. For a two-strand operator A commuting with the pair quantum-group action, the original cylindrical evaluation is

$$\phi(A) = \frac{1}{N}\,\mathrm{Tr}_{V_N \otimes V_N}\big[(I \otimes g)A\big]. \tag{2.5}$$

The factor $1/N$ in this expression is the single cut average. It is not repeated at the auxiliary circles.

Proposition 2.1 (Normalization)**.** *Let A_{raw} be the scalar obtained from the original crossings, turns, ordinary auxiliary traces and the single mark. In the enhanced reduced Kashaev convention,*

$$|\langle L\rangle_N| = |A_{\mathrm{raw}}(L)|.$$

In the full modified one-cut convention the modulus is $N^{-1}|A_{\mathrm{raw}}(L)|$. The changes of recorded component framing contribute only unit phases at the complete color.

Proof. The exact Fourier comparison is recorded in Appendix A.1. It is a simultaneous similarity of every adjacent braid generator and its inverse, together with an exact equality of enhancements. The factorial changes of basis cancel on every internal or traced wire. On the one remaining head the partial trace is scalar: it commutes with E, F, K, whose distinct weights and nonzero adjacent coefficients force scalarity. Hence the open-factor similarity cancels as well.

For clarity, the complete modified dimension is evaluated as a whole meridian germ. With the original meridian $M = -qq^{\alpha} = e^u$,

$$d_N(\alpha) = -q^{N(1-N)/2}\frac{q^{\alpha+1} - q^{-\alpha-1}}{q^{N\alpha} - q^{-N\alpha}} = \mathrm{i}^{N+3}\frac{\sinh u}{\sinh Nu}.$$

Its removable value at $u = 0$ is i^{N+3}/N. Exactly one component receives this factor. The twist at completeness is $q^{-d^2/2} = \kappa^{-1}$, of modulus one. These facts prove the asserted absolute normalization without estimating any large change-of-basis matrix. □

2.2 The full pair algebra

An operator on two strands must commute with the joint action of the quantum-group generators if it is to represent one of our invariant tangle boxes. The formulas below specify that joint action. The subsequent lemma identifies all such commuting operators; this is the meaning of the term *commutant*. The balanced pair generators and Casimir are

$$\mathcal{E}_{12} = \mathcal{E}_1 K_2 + \mathcal{E}_2, \quad \mathcal{F}_{12} = \mathcal{F}_1 + K_1^{-1}\mathcal{F}_2, \quad K_{12} = K_1 K_2, \quad \Omega = \mathcal{F}_{12}\mathcal{E}_{12} + qK_{12} + q^{-1}K_{12}^{-1}. \tag{2.6}$$

Let $\mathsf{C}_n(z) = 2T_n(z/2)$, where T_n is the usual Chebyshev polynomial. Thus $\mathsf{C}_0 = 2$, $\mathsf{C}_1 = z$ and $\mathsf{C}_{n+1} = z\mathsf{C}_n - \mathsf{C}_{n-1}$. Define

$$P_N(z) = \mathsf{C}_N(z) + 2 = \prod_{j=0}^{d}(z - c_j), \qquad c_j = q^{2j+1} + q^{-2j-1}, \qquad \mathcal{H}_N = \mathbb{C}[z]/(P_N). \tag{2.7}$$

The list of roots includes multiplicity. Since $c_{d-j} = c_j$, all distinct roots are double except the simple root -2 at odd N.

Lemma 2.2 (Full commutant)**.** *The endomorphisms of $V_N \otimes V_N$ commuting with the three pair generators are exactly $\mathbb{C}[\Omega] \simeq \mathcal{H}_N$.*

Proof. Let W_T be the span of $|a, T - a\rangle$. The K_{12} eigenvalue one occurs only on W_d. The matrices $\mathcal{E}_{12} : W_{T+1} \to W_T$ for $T < d$, and $\mathcal{F}_{12} : W_T \to W_{T+1}$ for $T \geq d$, have triangular maximal minors with nonzero diagonals. Consequently the entire pair is generated from W_d by these raising and lowering operators.

On W_d, the Casimir is a symmetric tridiagonal matrix with adjacent entries $-(1-Q^{a+1})^2 \neq 0$. Its first basis vector is cyclic: the kth power has last possible coordinate k, with coefficient equal to the nonzero product of the first k adjacent entries. An operator commuting with this block is therefore a unique polynomial of degree below N in that block. Subtracting the same polynomial in Ω from a pair intertwiner gives zero on W_d, hence zero everywhere by generation.

The elementary quantum-shift product gives $P_N(\Omega) = 0$. The cyclic block has minimal polynomial of degree N; its continuant, or the generic highest-weight decomposition followed by regular specialization, gives precisely (2.7). This proves the claim with all first normal directions intact. □

At a double root c, reduction modulo $(z - c)^2$ leaves

$$f(z) = f(c) + f'(c)(z - c) \pmod{(z - c)^2}.$$

Thus both the value and its first derivative are independent coordinates of the operator. Throughout the paper, retaining the "full algebra" or the "normal data" means retaining this derivative as well. A polynomial can vanish at every distinct root and still define a nonzero operator in $\mathcal{H}_N$. Evaluating only at roots would therefore lose part of the invariant.

2.3 The actual turn and the Newton basis

We choose a polynomial basis adapted to successive Casimir roots, including their multiplicities. Its normalization is designed so that turning a physical box reverses the basis index exactly.

The same basis makes the final closure a coefficient extraction. These are the two facts that will allow us to follow every internal summation index through a large construction. Put

$$s_r = \prod_{u=1}^{r} 2\sin\frac{\pi u}{N}, \quad s_0 = 1, \qquad B_r(z) = \prod_{j=0}^{r-1}(c_j - z), \qquad \widehat{B}_r(z) = \frac{B_r(z)}{s_r^2}, \quad 0 \le r \le d. \tag{2.8}$$

The quarter-turn is the literal contraction

$$\mathsf{T}(A) = (I^{\otimes 2} \otimes \ell)(I \otimes A \otimes I)(p \otimes I^{\otimes 2}).$$

Its entries in the recorded all-primal chart are

$$\mathsf{T}(A)_{(a,b);(c,e)} = x^{a-e} A_{(b,d-e);(d-a,c)}.$$

The cups and caps are those of (2.4). A different source-normalized rotation can carry an additional power of x; throughout this manuscript T is the unscaled genuine turn.

Proposition 2.3 (Newton turn and closure)**.** *In the full algebra,*

$$\mathsf{T}\widehat{B}_r = \widehat{B}_{d-r}, \qquad \phi\left(\sum_{r=0}^{d} \alpha_r \widehat{B}_r\right) = \alpha_d, \qquad \phi(\widehat{B}_i \widehat{B}_j) = \delta_{i+j,d}. \tag{2.9}$$

The finite verification is given in Appendix A.2. It uses the physical turn, its nonzero adjacent charge-block entries and the finite quantum-binomial recurrence. In particular, (2.9) is not a generic-q continuation of a formula checked only on distinct root values. The turn is linear and involutive in this basis, but is not multiplicative.

The same closure can be displayed directly in Hermite coordinates:

$$\phi(f) = (-1)^d \sum_{c \ mdouble} \big((4-c^2)f'(c) - cf(c)\big) + \mathbf{1}_{N \ modd} f(-2). \tag{2.10}$$

Equivalently,

$$\phi(f) = (-1)^d N^2 [z^d](f \bmod P_N). \tag{2.11}$$

For example, at a double root c, writing $Q_c = P_N/(z-c)^2$ gives $Q_c(c) = N^2/(4-c^2)$ and $Q_c'(c)/Q_c(c) = c/(4-c^2)$ from the Chebyshev differential equation. The residue of f/P_N is exactly the corresponding term of (2.10) after multiplication by $(-1)^d N^2$. Summing residues proves (2.11); the simple odd root has $P_N'(-2) = N^2$. The sole shortest tridiagonal path from the first to the last coordinate has product $(-1)^d N^2$, which also identifies this scalar with the original endpoint matrix element.

2.4 Positive multiplication with every normal retained

Suppose two stacked boxes contribute basis indices i and j. Their product is a linear combination of output indices r. The following coefficient formula gives both the allowed triples (i, j, r) and their weights. Calling this local operation a *gate* is only shorthand for this one multiplication; the geometric operation remains the vertical stack.

Theorem 2.4 (Multiplication law)**.** *If $i+j<N$, then*

$$\widehat{B}_i\widehat{B}_j=\sum_{r=\max(i,j)}^{i+j}\mu_{ijr}\widehat{B}_r,\qquad \mu_{ijr}=\frac{s_{i+j}s_r}{s_is_js_{r-i}s_{r-j}s_{i+j-r}}>0. \tag{2.12}$$

If $i+j\geq N$, the product is zero in the full algebra. In particular, a product gate has precisely the integer support

$$i+j\leq d,\qquad \max(i,j)\leq r\leq i+j. \tag{2.13}$$

Proof. At a generic parameter ζ, put $t=-(\zeta-\zeta^{-1})^2$. The unnormalized coefficients are

$$B_iB_j=\sum_r t^{i+j-r}\frac{[i]![j]![i+j]!}{[r]![r-i]![r-j]![i+j-r]!}\,B_r.$$

Multiplication by c_i-z and the identity $(c_i-z)B_r=B_{r+1}+(c_i-c_r)B_r$, with $c_i-c_r=t[r+i+1][r-i]$, reduce induction to

$$[i+1][i+j+1]=[r][r-j]+[r+i+1][i+j+1-r].$$

This follows by expanding the four powers of ζ. Endpoint coefficients follow from the same zero factors. At the principal root, all factorial indices in the case $i+j<N$ are below N. Since $[u]!=s_u/(2\sin(\pi/N))^u$, normalization by $s_i^2s_j^2/s_r^2$ gives (2.12).

For $i+j\geq N$, use the exact polynomial identity $B_N/B_j=B_{N-j}$, a consequence of $c_{d-u}=c_u$. Then B_iB_j is divisible by $B_N=(-1)^NP_N$ with multiplicities. This proves the zero assertion without dropping the first normal directions. □

2.5 The literal crossing circle and its colors

A color j on an auxiliary circle means that the circle carries a j-dimensional representation. The main strands still carry V_N. We must compute the resulting operator on those two main strands, including the trace around the auxiliary circle, before replacing that circle by twists. The answer is a Chebyshev polynomial in Ω, with the sign fixed by the pivot. For $1\leq j\leq N$, the auxiliary module is the ordinary j-dimensional highest-weight module. Its loop is the actual block-braiding word $[2,1,1,2]$, with the intermediate strand dimensions retained, followed by its ordinary pivot K_j^{1-N}. There is no auxiliary $1/j$ or $1/N$.

Proposition 2.5 (Circle identity)**.** *The resulting two-head operator is*

$$L_j=(-1)^{j-1}U_{j-1}(\Omega/2). \tag{2.14}$$

Here $U_0=1$, $U_1(z/2)=z$, and $U_{n+1}(z/2)=zU_n(z/2)-U_{n-1}(z/2)$.

Proof. At generic q, evaluate the closed monodromy on a target highest vector of weight n. In one direction the positive raising terms annihilate that vector; in the return direction a positive lowering term cannot return to its highest coefficient. The two Cartan factors and the auxiliary pivot give

$$\sum_{a=0}^{j-1}q^{(n+1)(j-1-2a)}=U_{j-1}((q^{n+1}+q^{-n-1})/2).$$

The generic pair highest vectors have distinct Casimirs and descendants of total dimension N^2, so this proves the full matrix identity. All finite crossing denominators have indices below N, and the identity specializes regularly. Finally $K_j^{1-N} = (-1)^{j-1}K_j$ at the root, proving (2.14), including its nilpotent coefficients. □

In particular $L_N = (-1)^d P_N'(\Omega)/N$. The trace identity

$$\phi(L_N f) = N \sum_{j=0}^{d} f(c_j) \tag{2.15}$$

counts the repeated roots. It follows first by Lagrange interpolation for a polynomial with distinct roots and then by polynomial continuation in its coefficients. At even levels $L_N^2 = 0$; at odd levels $L_N^2 = N^2 E_{-2}$, with the simple-node projector E_{-2}. Thus complete color alone cannot be inverted to reconstruct a filling.

Let $M = \lfloor d/2 \rfloor$ and $h_j = \mathsf{T} L_j$. The complete leaf has the exact positive expansion

$$h_N = \sum_{v=0}^{M} a_{N,v} \widehat{B}_v, \qquad a_{N,v} = \frac{N^2 s_{2v}}{s_v^4} > 0. \tag{2.16}$$

For $0 \le h \le M$, with $h_0 = 0$, the full color decomposition is

$$h_{N-h} = \sum_{v=h}^{M} \frac{N^2 s_{2v}}{s_v^2 s_{v-h} s_{v+h}} \widehat{B}_v + (-1)^N h_h. \tag{2.17}$$

For $1 \le h \le \lfloor N/2 \rfloor$,

$$h_h = \sum_{r=0}^{h-1} (-1)^{h-1+r} \frac{s_r^2 s_{h+r}}{s_{2r+1} s_{h-r-1}} \widehat{B}_{d-r}. \tag{2.18}$$

These identities separate two different ranges of Newton indices. The first term in (2.17), called the *core*, has positive coefficients before a half twist and occupies the lower indices. The second term, called the *alias*, is an exact signed contribution from the complementary color; formula (2.18) puts it in the higher Newton indices. An alias is therefore part of the finite invariant, not an error introduced by approximation. Appendix A.3 derives both formulas from the finite Chebyshev generating function. Chapter 5 will prove when the contribution of aliases is exponentially smaller after the complete tangle is assembled.

2.6 Half twists and exact filling

The preceding formulas describe a circle with no retained crossing. We next insert one positive or negative crossing in its original disk framing. This changes both the coefficients and their phases. We then expand a full integer twist power in the circle basis. Keeping these two steps separate ensures that the physical power is exactly $R^{\varepsilon-2p}$, with the retained half twist counted once. A retained half twist acts on a turned leaf by $\mathsf{T}_\varepsilon(f) = \mathsf{T}(R^\varepsilon \mathsf{T} f)$, for $\varepsilon = \pm 1$. Its exact transport is

$$\mathsf{T}_\varepsilon \widehat{B}_v = \kappa^\varepsilon (-1)^v \sum_{r=0}^{v} q^{\varepsilon v(r+1)} \frac{s_v}{s_r s_{v-r}} \widehat{B}_r. \tag{2.19}$$

The triangular formula retains the full Cartan phase and all normals. Its derivation uses the finite continuant expansion and root factorial reflection; the two signs are coefficient conjugates in the real Newton basis. A half-twisted alias may occupy every Newton coordinate, so the pure alias support cannot be reused after (2.19).

The N circle polynomials $L_1, \dots, L_N$ form a basis of $\mathcal{H}_N$. The full crossing has a Hermite polynomial $R = \kappa r_N(\Omega)$ with

$$r_N(c_k) = (-1)^k q^{k(k+1)}, \qquad \frac{r'_N(c_k)}{r_N(c_k)} = -\frac{d-2k}{2(q^{2k+1} - q^{-2k-1})}$$

at each double node, and the stated value at the odd simple node. Fourier–Hermite interpolation therefore gives, for every integer p,

$$R^{-2p} = \kappa^{-2p} \sum_{j=1}^{N} \beta_{j-1}^{-p} L_j, \qquad \sum_{j=1}^{N} |\beta_{j-1}^{-p}| \le 2N(1 + |p|). \tag{2.20}$$

The notation β_{j-1}^{-p} denotes the full interpolation coefficient, not a power of a scalar eigenvalue. Its value and derivative data are specified in Appendix A.4.

Insert (2.20) at every original auxiliary disk, tensoring with the spectators before taking the unchanged closure. If $\mathcal{A}_N(\mathbf{h})$ is the actual mixed source with colors $N - h_\ell$, then

$$\langle K(\mathbf{p}) \rangle_N = \chi_N \kappa^{-2\sum p_\ell} \sum_{j_1, \dots, j_H = 1}^{N} \left(\prod_{\ell} \beta_{j_\ell - 1}^{-p_\ell} \right) \mathcal{A}_N(N - j_1, \dots, N - j_H). \tag{2.21}$$

Here $|\chi_N| = 1$ in the reduced convention. This is an all-input finite identity. The remainder of the proof analyzes this complete sum, rather than replacing it by the complete-color term.

Chapter 3

From the finite sum to an angle maximization problem

At complete auxiliary color, every elementary coefficient and every multiplication coefficient is positive. We can therefore determine its exponential growth by maximizing the growth rate of an individual summand. This chapter makes that statement precise in three steps: determine the exact allowed integer indices, pass to their scaled compact polytope, and prove that the resulting action has a unique maximum away from every nonforced boundary. The same functions will then be recognized as hyperbolic volumes.

The distinction between an exact integer constraint and its limiting inequality matters here. We use $d = N - 1$ in every finite relation. The replacement by a continuous variable is made only after those relations have been solved, so no parity condition or half-lattice factor is lost.

3.1 Following the allowed indices through the construction

Expand an expression into its individual leaves, turns and binary products. A leaf has the positive support $[0, M] \cap \mathbb{Z}$. A turn replaces an index r by $d - r$. A product is governed by (2.13). Thus the source is a finite sum over a fixed tree of integer labels, with a positive coefficient at every admissible state in the complete pure case. The closure selects output d.

The finite support can be computed without evaluating any transcendental number. If a positive expression has interval support $[l, u] \cap \mathbb{Z}$, its turn has support $[d - u, d - l] \cap \mathbb{Z}$. If two children have supports $[l_1, u_1]$ and $[l_2, u_2]$, their product is zero when $l_1 + l_2 > d$. Otherwise its support is

$$[\max(l_1, l_2),\ \min(d, u_1 + u_2)] \cap \mathbb{Z}. \tag{3.1}$$

To verify that no integer in this interval is missing, fix such an output r. Choose $i \le r$ and $j \le r$ in the two intervals with $r \le i + j \le d$. Starting at the smallest permitted inputs and increasing them one at a time produces every intermediate sum up to $\min(d, u_1 + u_2)$. The lower bound $r \ge \max(l_1, l_2)$ ensures that both individual caps $i, j \le r$ leave enough room. The coefficient is then positive by Theorem 2.4.

At a cylindrical closing product, $i + j = d$ is forced and $\mu_{ij,d} = 1$. Delete this gate only after retaining that exact relation. It is the unique forced flat gate in the completed-root construction.

3.2 Strict output types

The word *strict* means that a state lies in the interior of every support inequality that has not been forced by closure. It is useful to classify a box by the possible values of its scaled output index. The letters in the following table name intervals, not new kinds of physical tangles. The table is a concise way of checking that the recursive construction leaves room for positive angles at every step. The limiting support inequalities for a retained product are

$$x > 0, \quad y > 0, \quad \max(x, y) < z < x + y < 1. \tag{3.2}$$

A strict leaf has $0 < x < 1/2$. A turn has output $1 - x$. For arbitrary syntax made from h, turn and product, the possible strict output intervals are

$$L = (0, 1/2), \qquad H = (1/2, 1), \qquad A = (0, 1), \qquad \varnothing.$$

Turn exchanges L and H and fixes A. Product has the following table; an empty child makes the product empty.

	L	H	A
L	A	H	A
H	H	$\varnothing$	H
A	A	H	A

For example, a high child forces the output above $1/2$, and two high inputs violate $x + y < 1$. For two low inputs and any prescribed $z \in (0, 1)$, choose both in $(z/2, \min(z, 1/2))$. The other entries follow by the same interval choices. At closure a strict reduced state exists exactly when the two intervals meet after reflection $y \mapsto 1 - y$.

Lemma 3.1 (Strict feasibility)**.** *Every completed expression has a strict state at every output $b \in (0, 1)$. Every completed/completed cylindrical closure has a strict reduced state.*

Proof. For hh, choose equal leaves $a \in (b/2, \min(b, 1/2))$. For a completed product choose its two outputs in the same interval and use induction in each child. For hE or Eh, use that number for the bare leaf and for the completed child's output. A turn uses the child's strict state at $1 - b$. At the closing product prescribe $1/2, 1/2$ and remove its forced gate. All remaining inequalities are strict. □

The distinction between strict support and every boundary coordinate is useful. At even N, hh has top support $2M = d - 1$, not d. This missing boundary coordinate does not affect the strict rounding argument below. Conversely, an empty strict interval does not imply a zero source at every level: $\mathsf{T}h\,\mathsf{T}h$ is zero at even levels but has a boundary-only odd contribution.

3.3 The primitive integer lattice

A lattice is called primitive here when the remaining free indices range over the ordinary integer lattice, with no hidden congruence restriction. We prove this by solving the closing relation with coefficient one. This gives both the correct number of summation variables and an admissible integer state near any fixed strict real state at every large level. Suppose there are H leaves. The uncontracted tree has $H - 1$ product gates. Remove the final forced gate

and contract every turn by $r \mapsto d - r$. There are then H leaf outputs and $H - 2$ retained product outputs, with one closing relation. Solving that relation for one output in one of the two disjoint root subtrees leaves

$$K = 2H - 3 \tag{3.3}$$

free integer coordinates. Every other index is an integer affine function of these coordinates with bounded, tree-dependent coefficients. No relation $2r = d$ is introduced.

Choose a strict reduced state and round its d-scaled free coordinates to integers. Reconstruct the eliminated indices exactly. The errors are bounded, whereas every strict inequality has a fixed positive slack times d. Therefore all constraints hold for every sufficiently large N, with either parity. For an open prescribed output, first fix r_N/d in its actual strict interval and use a continuous strict state at that output; then round the remaining coordinates. This gives a genuine all-level lower sequence. It does not rely on a subsequence whose denominator happens to divide d.

3.4 The action and its uniform finite approximation

The factorials in the finite coefficients are products of sines. Their logarithms are Riemann sums for a single integrable function. We use its integral J to express the exponential rates of an elementary piece and a stacking operation. The word *action* below means this logarithmic growth function, with the normalization fixed by the finite coefficients. Define

$$J(t) = \int_0^t \log(2 \sin \pi u)\, \mathrm{d}u, \qquad 0 \le t \le 1, \tag{3.4}$$

with its continuous endpoint values. Reflection gives $J(1 - t) = -J(t)$ and $J(1/2) = 0$. The leaf and gate actions are

$$g(x) = -2J(x) - 2J(1/2 - x) = J(2x) - 4J(x), \tag{3.5}$$

$$\begin{aligned} m(x, y, z) = {} & J(x + y) + J(z) - J(x) - J(y) \\ & - J(z - x) - J(z - y) - J(x + y - z). \end{aligned} \tag{3.6}$$

Let $\mathcal{A}_E$ be the sum of these terms over the leaves and retained gates, on the compact polytope defined by the non-strict support inequalities, the turn relations and the closing relation.

Lemma 3.2 (Uniform factorial estimate)**.** *Uniformly for* $0 \le r < N$*,*

$$\log s_r = NJ(r/N) + O(\log N).$$

For a fixed tree, every nonzero complete state of scaled labels $\mathbf{r}/d$ *has weight*

$$\log w_N(\mathbf{r}) = N\mathcal{A}_E(\mathbf{r}/d) + O_E(\log N). \tag{3.7}$$

Proof. Away from the endpoints, compare the sum of $\log(2 \sin \pi u)$ with its integral on consecutive mesh cells. Near zero and one, subtract $\log u$ or $\log(1 - u)$; the remaining function has bounded derivative, and the logarithmic part is controlled by factorials. The total discrepancy is $O(\log N)$, uniformly in the stopping index. Apply this estimate to (2.12) and (2.16). The shifts between r/N and r/d, and between M/N and $1/2$, change NJ by at most $O(\log N)$ because J has modulus of continuity $O(\delta |\log \delta|)$. There are a fixed number of factors. □

There are at most N^K states. The upper bound from (3.7), together with strict-state rounding at the maximizing state once strictness is proved, will give the complete-source limit.

3.5 Tetrahedral concavity and the boundary barrier

The geometric content of the action appears in its second derivatives and in its boundary behavior. Each local term is a sum of ideal-tetrahedron volumes. Their concavity will give uniqueness; their infinite one-sided derivative at a nontrivial flat angle will prevent a maximum from escaping to the boundary. We check both points because concavity alone would allow a boundary maximum. For $a+b+c=1$, $a,b,c\geq 0$, put

$$V(a,b,c) = -\pi\big(J(a)+J(b)+J(c)\big).$$

For positive angles this is the volume of the ideal tetrahedron with dihedral angles $\pi a, \pi b, \pi c$. It extends continuously by zero to the boundary. Its Hessian on the positive angle simplex is negative definite. This can be checked directly from $J''(t) = \pi\cot\pi t$: after eliminating c, the determinant is positive by the cotangent identity for three angles summing to π, while both diagonal entries of the volume Hessian are negative.

For a product write

$$p = z-x, \quad q = z-y, \quad r = x+y-z, \quad w = 1-x-y, \qquad p+q+2r+w = 1.$$

Substitution and reflection of J give

$$\pi g(x) = 2V(x, 1/2-x, 1/2), \tag{3.8}$$

$$\pi m(x,y,z) = V(y,x,w) + V(p,q,2r+w) + V(p+q,r,r+w). \tag{3.9}$$

Every triple in these expressions sums to one. Thus the entire action is concave on its original shared-label polytope.

Proposition 3.3 (Unique strict maximum)**.** *After deleting the forced closing gate, the complete action has a unique strict maximizer. Its Hessian on the primitive coordinates is negative definite. The fixed-output profile $f_E(b) = \max \mathcal{A}_E$ is real analytic on $0<b<1$ for every completed expression.*

Proof. Move from any boundary point toward the strict state of Lemma 3.1. A tetrahedron of type $(0, a, 1-a)$, $0<a<1$, has derivative $+\infty$ in this direction, since $-J(t) = -t\log t + O(t)$ near zero. A vertex-flat tetrahedron $(0,0,1)$ has a finite derivative: its two logarithmic singularities cancel. A positive tetrahedron also has a finite derivative. Hence no atom contributes a negative infinite term.

An endpoint wire descends through turns and products to a nonvertex-flat atom. At output zero both product inputs are zero; at output one they sum to one, giving $(y,x,0)$ unless an endpoint child permits further descent. A finite descent terminates at a leaf with atom $(0,1/2,1/2)$. Once all proper wires are interior, each remaining product boundary $p=0$, $q=0$, $r=0$ or $w=0$ exposes a nonvertex-flat atom in (3.9). A leaf at $1/2$ does the same. Thus every nonforced boundary point has a direction of positive infinite derivative and cannot maximize.

At a strict point the sum of the tetrahedral Hessians is strictly negative on every nonzero reduced tangent. Indeed, annihilating all leaf angle maps and all three product angle maps annihilates every leaf and product wire; turns then annihilate their wires. The only removed gate was already determined by its two children and the closing relation. Strict concavity gives uniqueness, and the implicit-function theorem on each fixed-output slice gives analyticity. □

Corollary 3.4 (Complete source limit)**.** *For the positive complete scalar $Z_{E,N}$,*

$$\lim_{N\to\infty}\frac{1}{N}\log Z_{E,N}=F_0,\qquad F_0=\max\mathcal{A}_E.$$

Proof. The number of states is polynomial, so Lemma 3.2 gives the upper bound. Round the strict maximizer in the primitive lattice. Its positive state weight gives the matching lower bound at every sufficiently large level. □

3.6 Boundary parameters and the leaf interval

For an open completed box, fix its output parameter b and maximize over all its other indices. The resulting function is the profile $f_E(b)$. Its derivative measures how the optimum changes when the output is varied. We package that derivative and b into three positive-branch coordinates (B,X,Y). Chapter 4 identifies their ratio with the shape of an ordered quadrilateral attachment face. For now they provide an efficient exact recursion for locating the stationary leaf parameters. The ordered output metric will be described in Chapter 4. Its useful algebraic coordinates are

$$B=\cot\pi b,\quad \varkappa=e^{-f_E'(b)},\quad X=\sqrt{\frac{1+B^2}{1+\varkappa}},\quad Y=\sqrt{\varkappa}\,X. \tag{3.10}$$

They satisfy $X^2+Y^2-B^2=1$. A leaf has $(B,X,Y)=(X,X,1)$. At a stationary product, the two children share X, and direct differentiation of (3.6) gives

$$B_o=\frac{B_1B_2+X^2-1}{B_1+B_2},\qquad Y_o=\frac{Y_1Y_2}{B_1+B_2},\qquad B_1+B_2>0. \tag{3.11}$$

A turn sends (B,X,Y) to $(-B,Y,X)$.

Lemma 3.5 (Bounds on the boundary parameters)**.** *The boundary parameters of every completed box satisfy $-Y<B<X$. In a completed/completed closure every completed descendant has $0<X,Y<1$, and every leaf satisfies*

$$\frac{1}{2}<X_\ell=\cot\pi a_\ell<1,\qquad \frac{1}{4}<a_\ell<\frac{\arctan 2}{\pi}. \tag{3.12}$$

Proof. The base hh has $B=X-1/(2X)$ and $Y=1/(2X)$. Turns preserve the claimed inequality. For a single attachment, (3.11) becomes

$$B_o=X-\frac{1}{X+B},\qquad Y_o=\frac{Y}{X+B}.$$

For $X\ge 1$, the child inequality forces $B>\sqrt{X^2-1}$ and the result follows. For $X<1$, let $h_0=\sqrt{1-X^2}$ and $\alpha=\arccos X$. Express a child as $(B,Y)=(h_0\cot\beta,h_0\csc\beta)$. A bare leaf has $\beta=\alpha$, a completed child has $\beta>\alpha$, and product adds the angles, with their sum below π. This proves the strict upper bound and the lower bound $B_o+Y_o=h_0\cot((\beta_1+\beta_2)/2)>0$. It also proves the completed/completed step when $X<1$. When $X\ge 1$, both completed children satisfy $\sqrt{X^2-1}<B_i<X$. Hence $0<X-B_i<X-\sqrt{X^2-1}\le 1$, so $(X-B_1)(X-B_2)<1$. Formula (3.11) then gives $B_o<X$, while $B_o>0$ proves the lower bound.

At closure the boundary triples of the two boxes are (B, X, Y) and $(-B, X, Y)$. Applying both inequalities gives $|B| < \min(X, Y)$; the hyperboloid relation implies $X, Y < 1$. In the angle chart, parent $Y < 1$ puts the sum of child angles below $\pi - \alpha$, while each child angle is at least α. Every completed child therefore has $Y < 1$, and turns exchange the bounds. Finally, at a single attachment $Y_o < 1$ gives $1 < 2X(X + B) < 4X^2$. At an hh base it gives $1/(2X) < 1$. Thus every leaf has $X > 1/2$, proving (3.12). □

For a fixed tree the inequalities give a compact neighborhood of every selected leaf inside $(1/4, 1/2)$. This is the precise input needed for the complex source analysis. It is not a uniform bound over an unbounded family of trees.

Chapter 4

Constructing the complete augmented hyperbolic metric

An ideal hyperbolic polyhedron has its vertices on the sphere at infinity. Truncating such a vertex by a horosphere gives a Euclidean polygon; after the face pairings, these polygons form the cross sections of the cusp ends. We shall build one polyhedron from the stationary angles and use two copies of it to recover the complement of the actual augmented diagram.

There are three separate verifications. The cells must fit isometrically along their ordered faces. Their labeled face identifications must recover the original link exterior. Finally, the Euclidean cusp polygons must glue without dilation, so that the metric is complete. A numerical agreement of volumes, or an agreement of face counts alone, would not establish these facts.

4.1 Why the local volume identity is not enough

The angle identity (3.9) proves concavity, but an abstract sum of tetrahedral volumes is not yet the volume of a link complement. We construct the cells, identify the ordered input maps, and then compare the closed colored boundary with the literal projection cut. This supplies both the topology and the degree of the geometric realization. The construction does not require a polyhedron existence theorem; Rivin's variational approach [4] provides useful context, but is not a premise of this explicit argument.

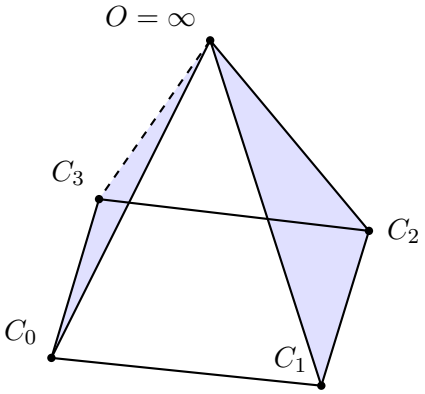
4.2 The leaf pyramid

We use the upper-half-space model of hyperbolic three-space, with boundary $\mathbb{C} \cup \{\infty\}$. An elementary piece will contribute an ideal pyramid. Its two shaded faces correspond to the two sides of a half crossing disk; the white faces come from regions of the projection sphere. These colors record different kinds of face pairings. The remaining quadrilateral is a gluing face used while the construction is still open. Its four ideal vertices are ordered: that order determines which vertex must be sent to which when two faces are identified. In upper half-space, place the apex at $O = \infty$ and the ordered vertices of the gluing face at

$$(C_0, C_1, C_2, C_3) = (1, e^{2\pi i a}, -1, -e^{2\pi i a}), \qquad 0 < a < 1/2. \tag{4.1}$$

Their convex hull is an ideal quadrangular pyramid. Its four apex edges are right angled. The base edge angles alternate πa and $\pi(1/2-a)$. A diagonal of the base gives the two tetrahedra of (3.8), so its volume is $\pi g(a)$. Color the side triangles (O, C_1, C_2) and (O, C_3, C_0) shaded; the other two side triangles are white. The base is the output gluing face.

Call edges $01, 23$ of an output gluing face *free*, and edges $12, 30$ *cap*. A leaf has free angles πa and cap angles $\pi(1/2-a)$. A completed cell will have free angles πb and cap angles $\pi(1-b)$.



quadrilateral gluing face

Schematic hyperbolic pyramid.

All five vertices are ideal. The shaded triangular faces are (O, C_1, C_2) and (O, C_3, C_0).

In the boundary plane: $C_0 = 1$, $C_1 = e^{2\pi i a}$, $C_2 = -1$, $C_3 = -e^{2\pi i a}$.

Figure 4.1: One leaf contributes two shaded crossing-disk triangles and one ordered quadrilateral gluing face. The one-leaf cap angle is $\pi(1/2-a)$; replacing it by the completed cap angle would give the wrong seam.

4.3 The multiplication prism

For a strict product define p, q, r, w as in Chapter 3. In the boundary plane set

$$v_0 = 1, \quad v_1 = e^{2\pi i p}, \quad v_2 = e^{2\pi i (p+r)}, \quad v_3 = e^{2\pi i (p+r+q)}. \tag{4.2}$$

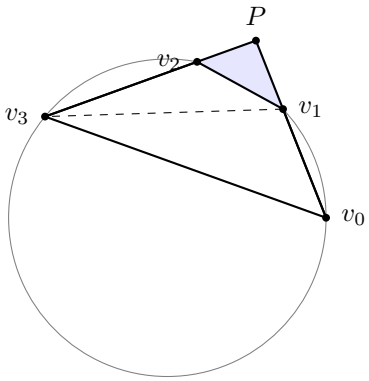
The consecutive half-arcs are $p, r, q, r+w$. Let P be the intersection of the lines (v_0, v_1) and (v_2, v_3). The Euclidean triangle (P, v_1, v_2) has positive angles $\pi y, \pi x, \pi w$, and the sine rule gives

$$P = v_1 + \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p}(v_1 - v_0) = v_2 + \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q}(v_2 - v_3). \tag{4.3}$$

In particular $|P| > 1$. Attach the tetrahedron (∞, P, v_1, v_2) to the quadrangular pyramid $(\infty, v_0, v_1, v_2, v_3)$, on opposite sides of their common face. The two coplanar face pairs merge. The result is a convex ideal triangular prism with white triangular faces (P, v_1, v_2) and (∞, v_0, v_3) and three quadrilateral gluing faces, with the indicated vertex orders

$$F_x = (\infty, v_0, v_1, P), \quad F_y = (\infty, v_3, v_2, P), \quad F_z = (v_0, v_1, v_2, v_3). \tag{4.4}$$

Its edge angles, divided by π, are



Boundary-plane construction.
$P = (v_0v_1) \cap (v_2v_3)$ lies outside the unit circle.
The sixth ideal vertex is ∞.
The three quadrilateral gluing faces have ordered vertices (∞, v_0, v_1, P), (∞, v_3, v_2, P) and (v_0, v_1, v_2, v_3).

$p = .12, \quad r = .10, \quad q = .17, \quad w = .51$

Figure 4.2: The exact boundary-plane construction of a strict prism at one rational angle tuple. The lines and circle are the defining support data; the drawing is not a perspective view of the three-dimensional hyperbolic cell.

Edges	Angle
$Pv_1,\ \infty v_0$	x
$Pv_2,\ \infty v_3$	y
$v_1v_2,\ v_0v_3$	$1 - z$
$P\infty$	w
v_0v_1	p
v_2v_3	q

The support half-spaces are the three vertical planes over the lines through P and over (v_3, v_0), and the hemispheres through the two remaining face circles. Formula (4.3) places the two attached pieces on the required sides. Equivalently, their seam is locally convex because its boundary angle sums are at most π. The volume is $\pi m(x, y, z)$: triangulate the pyramid across v_0v_2 and cancel the complementary intermediate Lobachevsky terms; the attached tetrahedron supplies $V(y, x, w)$.

4.4 Stationarity matches whole ordered faces

For cyclically ordered points on a circle or line, define

$$\varkappa(a, b, c, d) = \frac{(a - b)(c - d)}{(a - d)(b - c)} > 0.$$

Three ordered vertices can be sent to $(\infty, 0, 1)$, so this number determines the ordered ideal quadrilateral. A cyclic shift or the relevant reversal replaces it by its reciprocal. Direct

substitution gives

$$\varkappa(F_x) = \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p}, \qquad \varkappa(F_y) = \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q},$$
$$\varkappa(F_z) = \frac{\sin \pi p \sin \pi q}{\sin \pi z \sin \pi r}. \tag{4.5}$$

These are e^{-m_x}, e^{-m_y} and e^{-m_z}, respectively. The leaf metric is $\tan^2 \pi a = e^{-g'(a)}$.

The lower and upper stack maps are exactly

$$(C_0, C_1, C_2, C_3) \mapsto (v_0, v_1, P, \infty), \qquad (C_0, C_1, C_2, C_3) \mapsto (\infty, P, v_2, v_3). \tag{4.6}$$

Both invert the shape parameter of the gluing face. At each internal wire, stationarity says that the child output derivative and parent input derivative sum to zero. Hence $\varkappa_{\text{child}}\varkappa_{\text{parent}} = 1$, precisely the isometry condition for (4.6). This verifies the full face map, rather than only an edge-product equation.

4.5 Convex assembly and the closed cut

Attach child cells to the two prism inputs on opposite supported sides. A leaf cap edge meets a prism edge with total angle $\pi/2$; a completed cap edge gives π and merges two white faces. The common input edge has angle $\pi(w + x + y) = \pi$. The two free output seams have angles $\pi(p + x) = \pi z$ and $\pi(q + y) = \pi z$, while the cap output angles are $\pi(1 - z)$.

These inequalities prove local convexity at each finite seam. To justify global convexity directly, a length-minimizing sequence between two finite points of the union stays in a compact hyperbolic ball. A shortest path exists. Local convexity makes it an ambient geodesic at every finite point, so it is the unique ambient segment and lies in the union. The argument concerns finite points. To control the ideal boundary as well, observe that a strict quadrilateral gluing face is a maximal supported face. Cells attached on opposite sides can have no extra ideal intersection away from its supporting circle. An extra vertex on that circle would enlarge the supported face across one of its strictly nonflat edges. This is excluded by the strict output angles. Appendix D gives the corresponding incidence argument in detail.

There is also a combinatorial issue: repeated identifications must not collapse two corners of one leaf. At every open stage the four exposed corners are still unpaired original endpoints. A product consumes two disjoint pairs and never exposes those endpoints again; a turn only permutes the four remaining endpoints. Thus each consumed pair is permanent, and later gluing cannot extend it into a chain of identifications. The root pairs endpoints from the two disjoint final boxes. Each leaf therefore retains four distinct corners and its original apex. Together these geometric and combinatorial checks show that every completed box gives the claimed convex ideal cell, with right-angled edges outside the gluing face and two distinct shaded triangles per elementary piece.

At the final closure, the outputs are b and $1 - b$. Stationarity gives equal shape parameters for their ordered gluing faces. The gluing is

$$F : C_j \longmapsto G : C_{3-j} \qquad (j \bmod 4). \tag{4.7}$$

It implements both the stack interface and the two outside closing wires. Free and cap angles each sum to π for two completed children. After merging flat white seams, the result is a

convex right-angled ideal polyhedron P_E. It has $2H$ shaded triangles, $H+2$ white faces, $3H$ ideal vertices and $6H$ edges. These counts are checks, not the proof of topology.

For topology, cut the literal augmented diagram along its projection sphere and the recorded half crossing disks. In each projection ball the auxiliary semicircles are boundary-parallel along those half disks. Drilling them and then cutting returns a ball. A disk contributes precisely the two shaded triangles (O, C_1, C_2) and (O, C_3, C_0). The remaining regions are white. The original stack identifies lower corners $2, 3$ with upper corners $1, 0$ and preserves the exposed order (F_0, F_1, G_2, G_3). Turns rotate that order, and the final closure pairs 0 with 3 and 1 with 2. These rules determine every oriented white facial walk. They are exactly the rules induced by (4.6) and (4.7). The labeled sphere cellulation therefore agrees with the projection cut; its identification extends over each ball. Reversing the cuts gives the stated link exterior with degree one.

4.6 Complete cusps and half-twist partners

The polyhedron is only one of the two balls obtained by cutting along the projection sphere. We now describe how the two balls are reassembled. A complete cusp requires more than a total angle of 2π: the face maps must return pointwise around every edge, and their restrictions to the horospheres must have no dilation. The following pairing and length checks establish those conditions on the original labeled diagram. Take two oppositely oriented copies of P_E. Pair white faces between copies by their projection labels. In a pure leaf, pair its two shaded triangles within each copy, fixing O and sending $C_1 \mapsto C_0$, $C_2 \mapsto C_3$. The edge words alternate white and shaded gluings, contain four right angles, and return pointwise: the copied white maps and the inverse triangle maps cancel. Thus there is no translation along an edge hidden by an angle count.

At each ideal vertex, a horosphere cuts a Euclidean rectangle. Scale it so that its two opposite shaded sides have length one. Every shaded triangle then has the same canonical corner horocycles. All shaded and white pairings preserve the decoration. Uniform shrinking makes the horospheres disjoint. The links glue without dilation to the actual compact cusp tori, while their complement is compact. The resulting metric is complete and has volume

$$\operatorname{Vol}(Y_E^0) = 2\operatorname{Vol}(P_E) = 2\pi F_0. \tag{4.8}$$

A half twist rotates one side of the twice-punctured disk by π, exchanging both punctures and the upper/lower half disks. Hence the shaded maps become

$$S_0^+ \to S_1^- : \ (O, C_1, C_2) \mapsto (O, C_3, C_0), \qquad S_1^+ \to S_0^- : \ (O, C_3, C_0) \mapsto (O, C_1, C_2).$$

They go across the two copies, not within one copy. White maps are unchanged. The explicit edge matrices in Chapter 7 show pointwise cancellation of all three odd edge words. The length-one triangle decorations are also preserved, so every fixed half-twist pattern has the same complete volume $2\pi F_0$. The two half-twist signs have the same unmarked pairing but different collar paths; those paths are kept for filling.

4.7 A constructive knot choice

The hyperbolic construction initially keeps all main components. To obtain a knot after the auxiliary fillings, we must choose the retained half twists so that the final main diagram is

connected. This can be checked without any geometry. A box connects its four boundary points in one of three pairings; the table below follows the two actual identity arcs of a stack. It also records the only way stacking creates a closed internal main component. For the four tangle endpoints $(0, 1, 2, 3)$ in cyclic order, write

$$I = (0, 3)(1, 2), \qquad U = (0, 1)(2, 3), \qquad X = (0, 2)(1, 3).$$

A pure leaf gives U and an odd leaf gives X. Turn exchanges I, U and fixes X. Tracing the two stack wires gives

lower/upper	I	U	X
I	I	U	X
U	U	U plus an internal circle	U
X	X	U	I

Only $U \cdot U$ creates an internal component. The base hh offers $\{I, U\}$ without such a component. Every product of two feasible two-element subsets of $\{I, U, X\}$, after deleting the forbidden choice, still offers at least two elements; the same holds when one child is the leaf set $\{U, X\}$. Turn permutes the choices. Induction gives at least two loop-free choices for every completed expression. At the root choose U or X, whose cylindrical closure is one component, and backtrack the finite choices. Either sign realizes an odd leaf. Full twists have even endpoint permutation, so all subsequent auxiliary fillings preserve the one main component.

Chapter 5

Controlling the signed complementary color terms

The positive complete-color calculation does not by itself estimate the invariant after filling, because the exact filling formula uses every auxiliary color. For those colors an elementary piece contains the core and alias terms defined in Chapter 2. After a retained crossing, their coefficients can oscillate. We need a bound on the entire contribution of every selection containing at least one alias.

The bound is obtained by comparing logarithmic coefficient rates with tangent planes at the complete stationary state. Internal tangent terms cancel when boxes are stacked. A positive loss left at one elementary piece or one small block then survives all later compositions. The reason for treating blocks is that an individual alias need not lie below its leaf tangent throughout the permitted interval.

5.1 Coefficient envelopes on the true support

Let C_j^{ε} and A_j^{ε} denote, respectively, the core polynomial and the alias polynomial of an elementary piece of color j with retained half-twist sign ε. For a polynomial P, the notation $[\widehat{B}_r]P$ means its coefficient in the Newton basis, not an evaluation at a root. Write each colored elementary piece as core plus alias, using (2.17) and then applying (2.19) when required. A low physical color has no core. A high physical color has the displayed positive core before its half twist. The resulting signed arrays are always propagated through the full Newton multiplication law.

The endpoint-uniform factorial estimates and the selected complete half-twist recurrence give the following coefficient bounds, uniformly in every physical color and Newton index:

$$\begin{aligned} &|[\widehat{B}_r]C_j^{\varepsilon}| \le \exp\{N[\widetilde{g}(r/N) + e_N]\}, && 0 \le r \le M, \\ &|[\widehat{B}_r]A_j^{\varepsilon}| \le N^C e^{Nk(r/N)}, && 0 \le r \le d, \\ &\quad \widetilde{g}(u) = g(\max(u, 1/4)), \qquad k(u) = -2J(u/2). \end{aligned} \tag{5.1}$$

Here $e_N \to 0$ uniformly over colors and indices. No polynomial prefactor is asserted for the core through spectral collisions. Zero coefficients satisfy the same bounds. The first profile is constant below $1/4$ and agrees with g above it; its derivative joins continuously at zero.

The second follows directly from the triangular transport: the logarithmic rate of a term with upper index $v/N = w$ is

$$J(w) - J(u) - J(w-u), \qquad w \geq \max(u, 1/2).$$

Its maximum is at $w = (1+u)/2$. Reflection and duplication of J reduce that value to $-2J(u/2)$. The untransformed alias coefficients in (2.18) are bounded by $N/2$, so they have no additional exponential cost. The detailed finite derivation and the core recurrence estimate are in Appendix B.1.

The gate domain is the entire tetrahedron

$$D = \{(u, v, w) : u, v \geq 0,\ \max(u, v) \leq w \leq u + v \leq 1\}. \tag{5.2}$$

In particular, an aliased input can exceed $1/2$. All three functions $\widetilde{g}$, k and m are concave on their stated domains. For k, this is $k''(u) = -(\pi/2)\cot(\pi u/2) < 0$; for m, it follows from (3.9). This global concavity is what makes a finite supporting-plane certificate possible.

5.2 Why a block estimate is needed

At a pure leaf a, the tangent deficit against an alias is

$$I(a) = \min_{0 \leq u \leq 1} \{g(a) + g'(a)(u - a) - k(u)\}.$$

It can be negative in part of the interval allowed by (3.12). Thus an induction that assigns a positive loss to every individual leaf would fail. The multiplication gate provides a compensating contribution. We first treat an entire hh seed, then a new leaf together with the gate attaching it to a completed child.

Lemma 5.1 (Two-leaf block)**.** *At the pure stationary hh block, write*

$$X = \cot \pi a, \quad \cot \pi b = X - \frac{1}{2X}, \quad F_2(b) = 2g(a) + m(a, a, b), \quad \lambda = 2\log(2X^2).$$

For $1/2 \leq X \leq 1$, each of the two alias dual envelopes satisfies

$$\max_{D,\ v \leq 1/2} \{k(u) + \widetilde{g}(v) + m(u, v, w) - \lambda w\} \leq F_2(b) - \lambda b - \frac{1}{100}, \tag{5.3}$$

$$\max_{D} \{k(u) + k(v) + m(u, v, w) - \lambda w\} \leq F_2(b) - \lambda b - \frac{1}{100}. \tag{5.4}$$

Proof. Let $P(\lambda)$ be the pure dual maximum with both inputs at most $1/2$. At (a, a, b), $m_u = m_v = -2\log X$, $m_w = 2\log(2X^2)$ and $g'(a) = 2\log X$. Concavity therefore gives $P(\lambda) = F_2(b) - \lambda b$.

Choose a rational feasible pure point p and a rational strict point v for an alias action H. The pure value bounds P below; concavity gives the global upper tangent $H(w) \leq H(v) + \nabla H(v) \cdot (w - v)$. The vertices of D are

$$(0, 0, 0),\ (1, 0, 1),\ (0, 1, 1),\ (1/2, 1/2, 1/2).$$

Cutting by the core condition $v \le 1/2$ gives instead

$$(0,0,0),\ (1,0,1),\ (1/2,1/2,1/2),\ (0,1/2,1/2),\ (1/2,1/2,1).$$

Partition $\lambda/\log 2 \in [-2,2]$ into sixteen intervals of length $1/4$. The certificate supplies one pure point and two alias tangent points on each interval. For every vertex and both interval endpoints it verifies

$$A(p) - \lambda p_w - H(v) - \nabla H(v)\cdot(w-v) + \lambda w_w > \frac{1}{100}.$$

The expression is affine in the slope and in w, so these checks cover the whole interval and polytope. There are 288 rationally enclosed inequalities. Their least certified lower bound is

$$\frac{21000210290230963809378576600597934432 9}{50000000000000000000000000000000000000000000} > \frac{1}{25}.$$

Appendix C proves the enclosure rules and specifies the finite data and replay. No optimizer, sampled covering, or assertion about the alias maximizer is used in verification. □

5.3 A leaf attached to an arbitrary completed child

Suppose the new elementary piece is stacked with a box that has already been constructed. We retain all the internal structure of that box through its profile f_E and its output tangent. The comparison below therefore has one leaf variable, one output variable for the existing box, and the output of their stack. It is not an estimate for two bare leaves. Let (B,X,Y) be the selected boundary triple of the child and (B_p,X,Y_p) the boundary triple of the parent. Put

$$\begin{aligned} a = \operatorname{arccot} X/\pi, \quad b = \operatorname{arccot} B/\pi, \quad c = \operatorname{arccot} B_p/\pi, \qquad B_p = X - \frac{1}{X+B}, \\ \mu = f_E'(b) = 2\log(X/Y), \qquad \lambda = f_{hE}'(c) = \mu + 2\log(X+B). \end{aligned} \tag{5.5}$$

The bounds of Lemma 3.5 give $1/2 < X < 1$ and $1/(2X) - X < B < X$. Define

$$P(\mu,\lambda) = \max_{D,\ u\le 1/2} \{g(u) + m(u,v,w) + \mu v - \lambda w\}, \tag{5.6}$$

$$Q(\mu,\lambda) = \max_{D} \{k(u) + m(u,v,w) + \mu v - \lambda w\}. \tag{5.7}$$

Only the new leaf is restricted below half in P. The child variable v retains its whole interval; it is represented by its output tangent, not by a second bare leaf.

Lemma 5.2 (Attachment gap)**.** *At every proper attachment in the completed-root grammar,*

$$P(\mu,\lambda) - Q(\mu,\lambda) \ge \frac{1}{2500}. \tag{5.8}$$

The pure maximum in (5.6) *is attained at* (a,b,c) *from* (5.5)*.*

Proof. The three derivatives there are $m_u = -2\log X$, $m_v = -\mu$, $m_w = \lambda$. Thus the pure objective is stationary, and concavity gives its global maximum. The same is true with $\widetilde{g}$ replacing g, because $a > 1/4$ and the two functions agree near a.

For $1/2 < X \le 2/3$, both slopes belong to $[-2\log 2, 0]$. Indeed, either child or parent has $Y < 1$, so $2\log(X/Y) > 2\log X > -2\log 2$, while $Y^2 \ge 1 - X^2 > X^2$ gives the upper bound zero. On this entire square an exact certificate proves $P - Q > 1/1000$. It uses 55 dyadic rectangles, a rational pure point and a global alias tangent on each rectangle. For the four slope corners and four vertices of D, it checks

$$g(p_1) + m(p) - \{H(v) + \nabla H(v) \cdot (w - v)\} \\ + \mu(p_2 - w_2) - \lambda(p_3 - w_3) > \frac{1}{1000}, \qquad H = k + m. \quad (5.9)$$

The 880 inequalities and exact recursive square coverage imply the global assertion. The least lower bound is

$$\frac{5543719176001987489251797268402076337 3}{10^{40}} > \frac{1}{1000}.$$

This uses the same outward rational engine as the block certificate.

For $X \ge 2/3$, the elementary one-leaf estimate is

$$k(u) \le g(a) + g'(a)(u - a) - \frac{1}{2500}, \qquad 0 \le u \le 1. \quad (5.10)$$

Here is a direct proof of its endpoint margin. For $a \ge \arctan\sqrt{2}/\pi$, the minimum deficit occurs at $u = 1$, and it equals

$$g(a) + (1 - a)g'(a) = \frac{2}{\pi}\,\mathrm{Ti}_2(X) + \log X, \quad \mathrm{Ti}_2(X) = \int_0^X \frac{\arctan t}{t}\,\mathrm{d}t.$$

It decreases with a. At $X = 2/3$, four alternating terms give $\mathrm{Ti}_2(2/3) > 1708762/2679075$; the logarithm series gives $\log(3/2) \le 177391/437500$. With $\pi < 22/7$, the difference is greater than $57269/136687500 > 1/2500$. For $1/4 \le a \le \arctan\sqrt{2}/\pi$, the interior minimizing alias index is $u = (2/\pi)\arcsin(\tan^2(\pi a)/2) > a$, so the deficit decreases with a and its endpoint is larger than the same bound. The exact rational checks are included in the reproduction suite.

Finally add the tangent of the concave gate and the child linear term. The three variable coefficients cancel by stationarity, proving (5.8). □

5.4 The whole-tree induction

For a fixed choice of which elementary pieces contribute aliases, evaluate that contribution through the original construction tree. Such a choice is called an *alias subset*, or sector. There are only finitely many subsets because the construction is fixed. We first bound each subset separately, retaining an affine tangent in the open output index; only at the final closure are those output terms canceled. For every completed subtree at its selected output b_E, write $\lambda_E = f'_E(b_E)$. We prove, uniformly over all colors and all output indices,

$$|[\widehat{B}_r]E_S| \le \exp(N\,[f_E(b_E) + \lambda_E(r/N - b_E) - \delta_S + e_N])\,, \quad e_N \to 0, \quad (5.11)$$

where S is a specified alias subset, $\delta_\varnothing = 0$, and $\delta_S > 0$ when $S \ne \varnothing$. Lemma 5.1 supplies the base. A turn reverses the index, negates the slope, and adds only the fixed affine correction

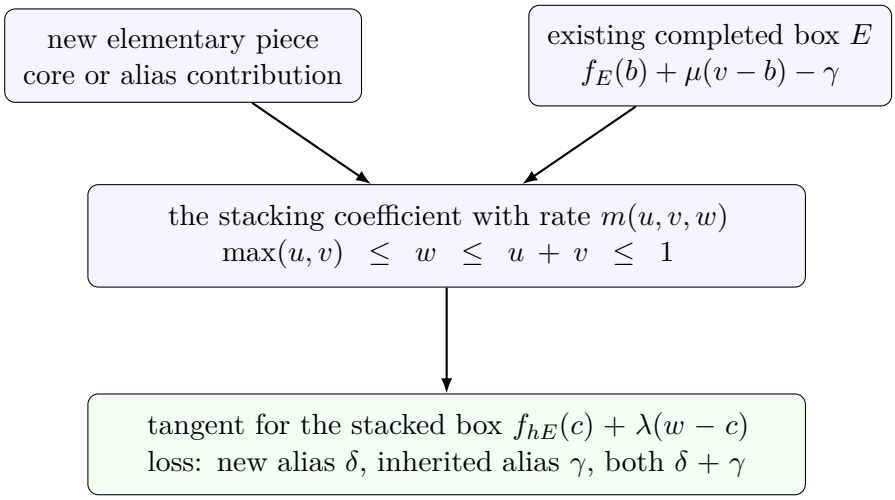


Figure 5.1: The child retains its output tangent and its original index. The new estimate is a leaf–gate comparison; it does not substitute a leaf for the child.

from $d/N = 1 - 1/N$. A completed/completed product adds both child tangents and the gate tangent. The internal derivatives cancel, while any inherited loss remains.

For hE, the exact split is

$$C_{hE} = C_h \star C_E, \qquad A_{hE} = A_h \star C_E + C_h \star A_E + A_h \star A_E,$$

where $\star$ is (2.12) with $i + j \le d$. A new alias loses at least $1/2500$ by Lemma 5.2; a child alias carries its old loss; when both occur the losses add. A core leaf uses the tangent of $\widetilde{g}$ with zero new loss. Thus the parent may take the minimum of its child loss and $1/2500$. The order Eh has the same algebraic bound without moving a coupon through a turn.

At the root, the two output tangents cancel on $i + j = d$. The finite affine shifts contribute $O(1)$ to the exponent. There are only polynomially many wire states and finitely many alias subsets for a fixed tree. We obtain constants $\gamma > 0$ and C such that the total aliased source satisfies

$$|\mathcal{A}_N - \mathcal{A}_N^{\mathrm{core}}| \le N^C e^{N(F_0 - \gamma)} \tag{5.12}$$

uniformly over all physical colors.

The all-core majorant also has a unique maximum at the complete strict state. On its high-index neighborhood it agrees with the strictly concave complete action. If another global maximum existed, concavity would make the segment joining it to the strict maximum constant, contradicting the negative Hessian near that maximum. Compactness therefore gives a fixed loss outside a fixed state window. Inside the window, the exact local core bound has a strict loss when any complementary color h_ℓ is at least a fixed positive multiple of N. Combining the two alternatives yields

$$\max_\ell h_\ell \ge \delta N \quad \Longrightarrow \quad |\mathcal{A}_N(\mathbf{h})| \le N^C e^{N(F_0 - \gamma_\delta)}. \tag{5.13}$$

The uniform global envelope and the strict color estimate are proved in Appendix B.2. Equations (5.12) and (5.13) are the global estimates required for filling. They concern the entire physical color range, not only a formal small-color series.

Chapter 6

A holomorphic asymptotic near the complete maximum

The preceding chapter removes the alias terms and the states far from the complete maximum with an exponential error. What remains is a finite sum over a fixed neighborhood of that maximum. We now continue its color parameters to complex values and compute its leading term. This finite holomorphic function is the *packet* used later in the filling integral.

The order of the construction matters. We first define the finite function by continuing the exact elementary coefficients, with their original phases. We then compare it with analytic index interpolants to exponential accuracy and apply a lattice saddle argument. An error of order N^{-1} on the real grid would not suffice: after an oscillatory filling transform, such an error could be larger than the desired leading answer.

6.1 A finite holomorphic object

Let $\mathbf{k} \in \mathbb{Z}^K$ be the primitive wire coordinates from (3.3). Every other wire has the exact form

$$r_w = N\ell_w(\mathbf{x}) + \delta_w, \qquad \mathbf{x} = \mathbf{k}/N, \tag{6.1}$$

with an integral affine function ℓ_w and a fixed integer δ_w. In particular, each use of $d = N - 1$ contributes to the offsets. Choose a closed rectangular box about the strict complete maximizer, lying wholly inside the retained support. By Lemma 3.5, every leaf lies in a fixed compact subinterval of $(1/4, 1/2)$.

For pure leaves, the exact color coefficient on this box is

$$\frac{N^2 s_{2r}}{s_r^2 s_{r-h} s_{r+h}}, \qquad t = h/N.$$

Use the canonical factorial continuation for the paired arguments $r - h, r + h$. For a piece with a retained crossing, use its finite continuant sum, meaning the solution of the displayed second-order recurrence fixed by the finite upper endpoint. Retain the continuous quadratic phase q^{h^2}, often called a chirp, and the recurrence branch selected by that exact sum. The precise construction, including the endpoint source anchor and exponential interpolation error, is proved in Appendix B.1. In particular, no interpolation is chosen merely because it agrees to order N^{-1} on the real grid.

Define $C_N(\mathbf{t})$ by the *finite* rectangular wire sum of the exact continued-color leaf factors and the original integer gate coefficients. At each fixed integer wire index, only the color is continued: the signed leaf remains its finite source sum with the canonical chirp. Index interpolants are introduced afterward as a proof device and replace this exact window only with exponentially small error. This is the definition in (B.42). It is holomorphic on a fixed complex polydisk and separately even in every t_ℓ. It agrees with the original source in the small physical color corridor up to the exponentially small errors established in Chapter 5 and the local interpolation proof.

6.2 The selected local action

The pure action depends on a scaled Newton index x and a scaled color deficit t. A retained half twist changes the relation between this pure parameter and the complex parameter χ that will become a peripheral coordinate. The following implicit change of variables describes that relation. Its branch is fixed by the finite recurrence, so both its derivative and its additive constant are determined. For a pure leaf the action is

$$g(x,t) = J(2x) - 2J(x) - J(x-t) - J(x+t). \tag{6.2}$$

For a signed half twist, the selected action f_ε is an analytic shear of this action. Put

$$\chi = t - \frac{\varepsilon g_t(x,t)}{2\pi\mathrm{i}}, \qquad f_\varepsilon(x,\chi) = g(x,t) - \frac{\varepsilon g_t(x,t)^2}{4\pi\mathrm{i}}, \qquad \varepsilon \in \{0,\pm 1\}. \tag{6.3}$$

For $\varepsilon = 0$ this is the identity. For $\varepsilon \neq 0$, the implicit map is nonsingular near the selected complete leaf. The finite recurrence selects this branch and fixes its additive constant; the relation is not used to choose an arbitrary solution after taking a semiclassical limit.

Let $F_\varepsilon(\mathbf{x},\boldsymbol{\chi})$ be the sum of the f_{ε_ℓ} and the unchanged gate actions. The implicit-function theorem gives a unique nearby stationary point $\mathbf{x}_*(\boldsymbol{\chi})$, and define

$$\Phi_\varepsilon(\boldsymbol{\chi}) = F_\varepsilon(\mathbf{x}_*(\boldsymbol{\chi}),\boldsymbol{\chi}).$$

The same relation holds after eliminating all internal wires:

$$\chi_\ell = t_\ell - \frac{\varepsilon_\ell \partial_{t_\ell}\Phi_0}{2\pi\mathrm{i}}, \qquad \Phi_\varepsilon(\boldsymbol{\chi}) = \Phi_0(\mathbf{t}) - \sum_\ell \frac{\varepsilon_\ell(\partial_{t_\ell}\Phi_0)^2}{4\pi\mathrm{i}}.$$

At fixed χ, differentiation of (6.3) gives $(f_\varepsilon)_x = g_x$ and $(f_\varepsilon)_\chi = g_t$. Thus the wire stationary equations are the pure equations evaluated at the implicitly changed colors. This proves the global shear by the envelope theorem, with the complete value F_0 as its anchor.

6.3 Offsets, powers of the level, and the Gaussian

The exponential action alone is not enough to prove noncancellation. We also need a nonzero leading coefficient and its exact power of N. Integer shifts such as $d = N - 1$ affect that coefficient even though they disappear from the limiting action. We keep these shifts before

applying the Gaussian integral in the free lattice coordinates. On a compact complex strict window, the factorial expansion is

$$G_N(N\alpha + c) = \sqrt{2N \sin \pi\alpha}\, e^{NJ(\alpha)} (2 \sin \pi\alpha)^c (1 + O(N^{-1})). \tag{6.4}$$

The square roots and logarithms continue from positive real arguments. The integer shift c is essential. For a signed leaf the corresponding offset factor is $e^{\delta \partial_x f_\varepsilon}$, evaluated at the changed critical point, not at the complete point or with g in place of f_ε.

Each leaf has prefactor $N^{1/2}$; each of the $H-2$ retained gates has prefactor $N^{-3/2}$. Hence the summand has N^{3-H}. A $K = 2H-3$ dimensional lattice Gaussian contributes $N^{K/2}$, leaving $N^{3/2}$. More explicitly, if

$$I_N(\mathbf{x}, \boldsymbol{\chi}) = N^{3-H} b_\varepsilon(\mathbf{x}, \boldsymbol{\chi}) e^{N F_\varepsilon(\mathbf{x}, \boldsymbol{\chi})} (1 + O(N^{-1})),$$

then the leading packet amplitude is

$$a_\varepsilon(\boldsymbol{\chi}) = \frac{(2\pi)^{K/2} b_\varepsilon(\mathbf{x}_*(\boldsymbol{\chi}), \boldsymbol{\chi})}{\sqrt{\det[-\partial_{\mathbf{x}}^2 F_\varepsilon(\mathbf{x}_*, \boldsymbol{\chi})]}}. \tag{6.5}$$

Every factor in b_ε is specified by (6.4), the finite signed recurrence, and the offsets (6.1).

Theorem 6.1 (Exact corridor packet)**.** *For the fixed completed-root expression there are a fixed complex polydisk, a nonzero holomorphic a_ε, and a separately even holomorphic Φ_ε such that*

$$C_N(\boldsymbol{\chi}) = N^{3/2} a_\varepsilon(\boldsymbol{\chi}) e^{N\Phi_\varepsilon(\boldsymbol{\chi})} (1 + O(N^{-1})) \tag{6.6}$$

uniformly there. The physical source has the comparison

$$|\mathcal{A}_N(\mathbf{h}) - \eta_N C_N(\mathbf{h}/N)| \le N^C e^{N(F_0 - \gamma)}, \qquad 0 \le h_\ell \le \delta N, \tag{6.7}$$

where $\eta_N = \exp(\mathrm{i}\pi d \sum_\ell \varepsilon_\ell/4)$ is the retained unit phase. At zero,

$$\Phi_\varepsilon(0) = F_0, \qquad \partial_{\chi_\ell} \partial_{\chi_j} \Phi_\varepsilon(0) = -\frac{2\pi X_\ell}{1 - \varepsilon_\ell \mathrm{i} X_\ell} \delta_{\ell j}. \tag{6.8}$$

The statement holds at both parities and all sufficiently large levels.

Proof. The local finite interpolation and its exponentially small error are proved in Appendix B.1. Apply the finite complex lattice Laplace lemma of Appendix B.1.8 to the full rectangular window. That lemma uses exact cotangent residues, bounds every vertical side and nonconstant cotangent mode, and translates the ordinary integral to the unique complex critical point. It gives (6.6) and (6.5).

At completeness, the signed multiplier relative to the positive amplitude is

$$\prod_{\varepsilon_\ell \neq 0} \sqrt{\sin \pi a_\ell} \exp(-\mathrm{i}\varepsilon_\ell \pi a_\ell/2),$$

which is nonzero. The positive amplitude is positive by the strict Gaussian. For pure colors, separate evenness makes the mixed wire–color derivative vanish at zero, and direct differentiation of (6.2) gives $-2\pi X_\ell$. The shear transforms it to (6.8). Its real part is strictly negative. The source estimates of Chapter 5, together with the finite-window local error, prove (6.7). All lattice offsets are fixed integers and no parity-dependent lattice density occurs. □

After shrinking the polydisk, this implies a source-specific Gaussian upper bound

$$|C_N(\mathbf{x}+\mathrm{i}\mathbf{y})| \le N^C \exp N\{F_0 - a_0\|\mathbf{x}\|^2 + B_0\|\mathbf{y}\|^2\}, \qquad a_0 > 0. \tag{6.9}$$

This bound is needed for the finite Fourier contours. A pointwise asymptotic on the physical grid would not imply it.

6.4 An exact three-leaf calibration

For $C(h^3)$ choose the three leaf indices i, j, k as free coordinates. The surviving gate output is $d-k$, so its limiting output is $1-x_3$ with integer offset -1. The complete phase is

$$F(\mathbf{x}) = \sum_{\ell=1}^{3} g(x_\ell) + m(x_1, x_2, 1-x_3).$$

Its stationary point is $x_\ell = a = \arctan 2/\pi$. The negative Hessian is

$$A = \frac{25\pi}{4}(I_3 + \mathbf{1}\mathbf{1}^T), \qquad \sqrt{\det A} = \frac{125\pi^{3/2}}{4}.$$

The three unshifted leaf amplitudes are $\sqrt{10}/8$, the unshifted gate amplitude is $5\sqrt{10}/16$, and the gate offset is $e^{-m_z} = 4$ at completeness. Thus $b = 125/512$ and $a_0 = \sqrt{2}/64$. The complete rate is $F_0 = 2v_8/\pi$, so the augmented volume is $4v_8$. At a changed color the offset is e^{-m_z} evaluated at the changed critical point; it is not the constant four.

The alias proof for this isolated case uses the exact symmetric kernel

$$\phi(\widehat{B}_i\widehat{B}_j\widehat{B}_k) = \frac{s_{i+j}s_{i+k}s_{j+k}}{N s_i s_j s_k s_{i+j+k-d}}$$

on $i+j, i+k, j+k \le d$ and $i+j+k \ge d$, and zero elsewhere. It follows from (2.12) and $s_r s_{d-r} = N$. For one or two aliases, algebraic reassociation groups the aliased leaves into a two-leaf block and uses Lemma 5.1 at $X = 1/2$; the remaining core tangent cancels the output slope, with only $\log 4/N$ from the closing shift. For three aliases, $k \le v_3/\pi$ and $m \le 3v_3/\pi$, while $6v_3 < 2v_8$. The exact rational margin and its elementary bounds are included in the reproduction suite. This is a source identity, not a tangle move. The one-leaf cap seam in the geometric closure is $\pi/2$, as in Chapter 4. These observations supply the same packet and filling argument for the isolated three-leaf case.

Chapter 7

Peripheral curves and the volume after filling

We now identify the complex action from Chapter 6 with a deformation of the actual augmented link complement. A cusp cross section is a torus. A small loop going around the corresponding link component is its meridian, and the original crossing disk specifies the longitude used in the surgery convention. These curves must be primitive: together they form an integral basis, rather than nontrivial multiples of a basis.

A *marked deformation* keeps those curves, their orientations, and the continuously selected holonomy logarithms. The complex-valued function G below is a volume primitive: its derivative records the peripheral holonomies, and an explicit boundary correction recovers the volume of the representation. The complete metric fixes its additive constant. This information is what connects the critical point of the filling integral to the knot complement with the intended surgery slopes.

7.1 A metric modulus is not yet a peripheral coordinate

The real boundary shape parameter $e^{-f'_E(b)}$ describes an ordered ideal quadrilateral. To identify the filling action with volume, one must also construct the nearby representation on the same link exterior, specify the two auxiliary peripheral curves, and fix the additive constant of its regulator. Matching only a Hessian or an unmarked cusp shape would not suffice.

Appendix D gives the local rational face maps and their compositions. The construction starts at the complete metric of Chapter 4. It continues the leaf apex flags and the prism coordinates in the selected source variables. The input and output maps obey the nine local prism compatibility identities, so the white-face map is independent of the sequence of glued cells. The pure and signed shaded pairings give pointwise edge relations. The main cusps stay parabolic. The resulting framed representation is holomorphic on the same sufficiently small parameter neighborhood used by the packet.

Proposition 7.1 (Marked source primitive)**.** *For the completed-root source, with the original disk framings and half-twist signs, the selected deformation has centered primitive auxiliary logarithms*

$$U_\ell = -2\pi\mathrm{i}\chi_\ell, \qquad V_\ell = -2\partial_{\chi_\ell}\Phi_\varepsilon. \tag{7.1}$$

Its absolute complex volume primitive can be chosen as

$$G = 2\pi\mathrm{i}\Phi_\varepsilon, \qquad dG = \frac{1}{2}\sum_\ell V_\ell\, dU_\ell, \qquad \Im G(0) = \mathrm{Vol}(Y_E^\varepsilon) = 2\pi F_0. \tag{7.2}$$

The half-twist contribution is the specified quadratic correction $-\varepsilon_\ell V_\ell^2/8$ in G. The physical filling power is $\varepsilon_\ell - 2p_\ell$.

Proof. The local rational identities and original peripheral paths are proved in Appendix D. Differentiating the shear (6.3) gives the same boundary covectors as the pure source after the primitive meridian shear. Across each paired quadrilateral gluing face the two internal covectors cancel, leaving only the auxiliary boundary terms. Thus the differential is (7.2). The complete polyhedron fixes its imaginary constant, so no arbitrary additive volume normalization remains. The sign of the meridian shear is determined by the original collar path, not merely by the unmarked half-twist pairing. □

For perspective, in the leaf coordinate $E = e^{2\pi\mathrm{i}a}$ the complete disk longitude has translation $-2(E+1)$. The two signed meridian collar paths have translations -2 and $2E$, differing by that longitude. Each pair spans the area of the two cusp rectangles and is primitive. Their relation to the quantum sign is checked with the recorded crossing convention. This explains why the two signs can have the same complete volume but different filling equations.

7.2 Geometric completion for long primitive slopes

Solving a matrix equation that kills a peripheral element does not alone construct a smooth filled metric. We need to show that the geometric end closes around a solid-torus axis with angular period exactly 2π. The marked curves and small logarithms make that check possible. We first solve the equations near the complete structure and then use the explicit end construction in Appendix D. The local deformation is a representation of the actual augmented-link group near its oriented complete holonomy, with a bounded finite collection of framed vertices and face maps. The strict complete shapes remain nondegenerate in a small neighborhood. The primitive equations

$$U_\ell + p_\ell V_\ell = -2\pi\mathrm{i} \tag{7.3}$$

have a unique nearby solution for all independently sufficiently large signed p_ℓ. One can see the local existence by writing $1/p_\ell$ as independent small parameters and applying the implicit-function theorem to the cusp equations. The sign and the integer on the right side are fixed by the primitive collar, not chosen after solving a formal equation.

The geometric completion argument in Appendix D constructs a smooth solid-torus completion at each filled cusp on this branch. The remaining main cusp stays complete. It applies on one fixed neighborhood, hence requires only a lower bound on each $|p_\ell|$, with no comparison between them. The resulting complete metric is on the topology obtained by the specified auxiliary disk fillings. After the knot pairing choice and all such fillings, that topology is $S^3 \setminus K_{F,G,\varepsilon}(\mathbf{p})$.

Set

$$S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}) = \Phi_\varepsilon(\boldsymbol{\chi}) + \frac{\pi\mathrm{i}}{2}\sum_\ell \frac{(\chi_\ell - 1)^2}{p_\ell}. \tag{7.4}$$

Its critical equations are exactly (7.3), by (7.1). The regulator with the solid-torus correction gives

$$2\pi\Re S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}_*) = \mathrm{Vol}(S^3 \setminus K_{F,G,\varepsilon}(\mathbf{p})). \tag{7.5}$$

This is an equality on the selected, correctly marked branch. Chapter 8 proves that the full finite invariant has a nonzero leading contribution with precisely this action.

Chapter 8

Evaluating the complete oscillatory filling sum

The geometric saddle has now been identified, but the invariant is still the full finite sum over auxiliary colors in (2.21). This chapter proves that the saddle actually contributes to that sum with a nonzero coefficient, and that all other contributions are smaller. In particular, we do not infer the knot volume from the complete-color growth rate.

We convert the exact Hermite interpolation coefficients into a Fourier expansion. Its integer indices are called charges. The zero charge vanishes by an exact boundary identity; the charges with absolute value one give the leading term. To justify that conclusion, we control both the infinite charge tails and the boundary integrals introduced by moving contours. The latter are the *endpoint currents* mentioned below.

8.1 Replacing the color sum by its controlled local part

For the fixed diagram, Chapters 5 and 6 give a global physical-color gap, an exponentially accurate separately even corridor packet, and a uniform holomorphic asymptotic. In this chapter write $b = H$ for the number of filled circles, $\varphi = \Phi_\varepsilon$, and

$$\varphi''(0) = 2\pi\mathrm{i}\,\mathrm{diag}(\tau_1, \dots, \tau_b), \qquad \tau_\ell = \frac{\mathrm{i}X_\ell}{1 - \varepsilon_\ell \mathrm{i}X_\ell}, \qquad \Im\tau_\ell > 0.$$

The different cusps need not have equal τ_ℓ. The packet exponent is $\mu = 3/2$.

The coefficient mass bound in (2.20) shows that the global source error and the corridor comparison error cost only a polynomial in N for fixed twists. They remain exponentially below e^{NF_0}. This removal takes place before folding colors: an alias need not be even and is not artificially made so. It remains to analyze the exact even corridor.

Let $H_N = \lfloor \delta N \rfloor$, and define its finite Fourier polynomial

$$\mathcal{F}_N(\mathbf{z}) = \sum_{|h_\ell| \le H_N} C_N(\mathbf{h}/N) e^{2\pi\mathrm{i}\mathbf{h}\cdot\mathbf{z}}. \tag{8.1}$$

This is an entire, one-periodic function in each variable and is separately even. It is finite at every N; no infinite-lattice extension has been made.

8.2 An exact Fourier bound away from zero frequency

We record the finite contour estimate that controls the physical endpoints. In one coordinate suppose

$$|C_N(x+\mathrm{i}y)| \le N^C e^{N(F-ax^2+By^2)}$$

on a fixed tube. For a fixed $\phi_0 > 0$ and fixed nonnegative integer r,

$$\left| \sum_{h=-H_N}^{H_N} h^r C_N(h/N) e^{\mathrm{i}\phi h} \right| \le N^{C_r} e^{N(F-g)}, \qquad \operatorname{dist}(\phi, 2\pi\mathbb{Z}) \ge \phi_0, \tag{8.2}$$

for some $g > 0$.

To prove it, reduce ϕ to $[\phi_0, 2\pi - \phi_0]$ and use the rectangle in the unscaled variable z with real sides $\pm(H_N + 1/2)$ and imaginary sides $\pm\sigma N$. The exact residue formula is

$$\frac{1}{2\pi\mathrm{i}} \oint z^r C_N(z/N) e^{\mathrm{i}\phi z} (\pi \cot \pi z - \mathrm{i}\pi)\, \mathrm{d}z.$$

The kernel equals $2\pi\mathrm{i}/(e^{2\pi\mathrm{i}z} - 1)$. On the upper and lower horizontal sides the products with $e^{\mathrm{i}\phi z}$ are bounded by $4\pi e^{-\phi\sigma N}$ and $4\pi e^{-(2\pi-\phi)\sigma N}$. On a vertical half-integer side the bound is $2\pi e^{-\phi_0|\Im z|}$. Choose σ first so that $B\sigma^2$ is smaller than both half the horizontal loss and half the real boundary loss. All four sides then have a strict exponential gap. This proves (8.2); fixed derivatives are treated by the factor h^r. The argument is uniform in the other real colors and in a smaller complex neighborhood of any remaining parameters.

8.3 The total derivative and the zero-charge cancellation

The derivative terms in Hermite interpolation are especially important at this step. Together with the values at the roots, they combine into one total derivative on a finite midpoint grid. This exact identity determines the endpoint behavior and removes the zero Fourier mode. It is proved before any stationary-phase approximation is made. The exact Fourier–Hermite interpolation of Appendix A.4 has a useful combined form. In one color, for the normalized power r_N^{2P}, put

$$U_N(z) = e^{2\pi\mathrm{i}PNz^2} \sinh(2\pi\mathrm{i}z) \mathcal{F}_N(z), \qquad z_k = (k + 1/2)/N - 1/2.$$

Then the folded color pairing is exactly

$$\sum_{h=0}^{H_N} \beta_{N-h-1}^P C_N(h/N) = \frac{(-1)^d \kappa^{-P}}{2\pi\mathrm{i}N^2} \sum_{k=0}^{N-1} U_N'(z_k). \tag{8.3}$$

The derivative contains both the h term and the linear normal term from the Hermite data. Removing either would change this identity.

The endpoint values satisfy

$$U_N(1/2) = U_N(-1/2) = 0, \qquad U_N'(1/2) = U_N'(-1/2).$$

The periodic extension of U'_N is continuous and piecewise smooth. Two integrations by parts give an absolutely summable Fourier series at each fixed N. Midpoint Fourier selection followed by one integration by parts gives

$$(-1)^d \kappa^{-P} \sum_{m\neq 0} (-1)^{md} m \int_{-1/2}^{1/2} e^{2\pi \mathrm{i} N(Pz^2-mz)} \sinh(2\pi \mathrm{i} z) \mathcal{F}_N(z)\,\mathrm{d}z. \tag{8.4}$$

There is no zero charge: it vanishes from the two exact endpoint zeros. For an even source the terms at m and $-m$ are equal, because both m and the hyperbolic sine change sign. This equality precedes any saddle approximation.

Taking $P_\ell = -p_\ell$ in every color and restoring the Cartan factors gives the full corridor expression

$$\chi_N \eta_N (-1)^{bd} \kappa^{-\sum p_\ell} \sum_{\mathbf{m}\in(\mathbb{Z}\setminus\{0\})^b} (-1)^{d\sum m_\ell} \prod_\ell m_\ell \times \int_{[-1/2,1/2]^b} e^{-2\pi \mathrm{i} N \sum_\ell (p_\ell z_\ell^2 + m_\ell z_\ell)} \prod_\ell \sinh(2\pi \mathrm{i} z_\ell) \mathcal{F}_N(\mathbf{z})\,\mathrm{d}\mathbf{z}. \tag{8.5}$$

This is a sum of commuting color-register functionals. No physical box is commuted through another box.

8.4 Uniform charge tails and their endpoint currents

A Fourier series can converge for each N without having a tail estimate uniform enough for an exponential limit. We therefore bound the tails directly. Moving the interval downward or upward adds two vertical boundary segments. Their difference vanishes to second order at height zero; this cancellation produces a summable power of the charge. We retain the pair together rather than discard either boundary segment. Absolute convergence at each N is not a uniform exponential tail estimate. We prove the latter before using a Gaussian approximation. By (8.2), $\mathcal{F}_N$ and its first few derivatives are exponentially small on fixed complex strips based at $z = \pm 1/2$. Fix a small strip height σ using only this source estimate.

For $m > 0$, shift the one-dimensional Fourier integral down by σ. On the horizontal side the source's physical support gives a bound with exponent

$$F_0 - 2\pi\sigma(m - |P| - \delta).$$

The two vertical sides must be paired. Their common Fourier endpoint phase is $(-1)^{mN}$, and their difference is

$$V_N(v) = U_N(1/2 - \mathrm{i}v) - U_N(-1/2 - \mathrm{i}v), \qquad 0 \le v \le \sigma.$$

Periodicity of the finite Fourier polynomial and the exact endpoint relations give $V_N(0) = V'_N(0) = 0$. Keeping the chirp's height-dependent cost, rather than bounding it at the top of the strip, gives

$$|V_N(v)| \le \mathrm{poly}(N,P) v^2 e^{N(F_0-g_1)+2\pi|P|Nv}.$$

Therefore

$$\left| m \int_0^\sigma e^{-2\pi m N v} V_N(v)\,\mathrm{d}v \right| \le \frac{\mathrm{poly}(N,P) m}{N^3 (m-|P|)^3} e^{N(F_0-g_1)}, \qquad m > |P|. \tag{8.6}$$

This is summable for $m > 2|P| + 2\delta + 2$. The upper shift treats negative m. Neither vertical current has been set to zero.

In several variables, remove tails one at a time while all other chirps remain on real grids or finite real integrals. If D_i is the exact sampled derivative functional and D_i^{lo} its finite-charge restriction, use the exact identity

$$\prod_i D_i - \prod_i D_i^{\mathrm{lo}} = \sum_i \left(\prod_{j<i} D_j^{\mathrm{lo}} \right) (D_i - D_i^{\mathrm{lo}}) \left(\prod_{j>i} D_j \right).$$

The low-charge functionals have finite supremum norm; the sampled ones and their fixed derivatives cost powers of N and the fixed slopes. All spectator chirps have modulus one. Thus each term has the exponential gap of (8.6), without a factor $e^{N|p_j|}$ from a different cusp. This is the point that permits independently unequal long slopes.

8.5 The central Fourier packet

The implicit-function theorem defines $\boldsymbol{\xi}(\mathbf{z})$ by

$$\nabla\varphi(\boldsymbol{\xi}) + 2\pi\mathrm{i}\mathbf{z} = 0.$$

Set

$$\Psi(\mathbf{z}) = \varphi(\boldsymbol{\xi}(\mathbf{z})) + 2\pi\mathrm{i}\boldsymbol{\xi}(\mathbf{z}) \cdot \mathbf{z}, \qquad B(\mathbf{z}) = \frac{(2\pi)^{b/2} a_\varepsilon(\boldsymbol{\xi}(\mathbf{z}))}{\sqrt{\det[-\varphi''(\boldsymbol{\xi}(\mathbf{z}))]}}. \tag{8.7}$$

Both functions are separately even, $B(0) \neq 0$, and

$$\Psi(\mathbf{z}) = F_0 - \pi\mathrm{i} \sum_\ell \frac{z_\ell^2}{\tau_\ell} + O(\|\mathbf{z}\|^4). \tag{8.8}$$

Exact half-grid cotangent rectangles applied to (8.1) turn it into N^b times its finite real integral, with exponentially small boundary and nonconstant-mode errors. Translate that integral to the parallel real contour through $\boldsymbol{\xi}(\mathbf{z})$. The strict complex Gaussian estimate controls the connectors. Gaussian integration gives, uniformly on a fixed complex polydisk,

$$\mathcal{F}_N(\mathbf{z}) = N^{\mu+b/2} B(\mathbf{z}) e^{N\Psi(\mathbf{z})} (1 + O(N^{-1})). \tag{8.9}$$

This argument applies to the exact C_N, not to a pointwise real-grid approximation whose error might have uncontrolled Fourier behavior.

8.6 Local charges for independently signed slopes

After the tail estimate, only finitely many charges remain for the fixed filling tuple. Their saddles must be compared uniformly when the filling integers have different signs and magnitudes. We use the reciprocal slopes $1/p_\ell$ and the ratios m_ℓ/p_ℓ as independent parameters. This gives one neighborhood in which the saddle equations and the charge comparison are valid, instead of assuming the slopes are comparable. For a nonzero integer vector $\mathbf{m}$, define

$$T_{\mathbf{m}}(\mathbf{z}) = \Psi(\mathbf{z}) - 2\pi\mathrm{i} \sum_\ell (p_\ell z_\ell^2 + m_\ell z_\ell).$$

Write $e_\ell = 1/p_\ell$ and $t_\ell = m_\ell/p_\ell$. The saddle equations are

$$z_\ell = -\frac{t_\ell}{2} + \frac{e_\ell \Psi_\ell(\mathbf{z})}{4\pi \mathrm{i}}. \tag{8.10}$$

At $\mathbf{e} = 0$ their derivative in $\mathbf{z}$ is the identity, so one fixed polydisk in $(\mathbf{t}, \mathbf{e})$ has a unique holomorphic solution. Completing the square yields

$$T_{\mathbf{m}}(\mathbf{z}(\mathbf{t}, \mathbf{e})) = \frac{\pi \mathrm{i}}{2} \sum_\ell \frac{t_\ell^2}{e_\ell} + \widehat{B}(t_1^2, \dots, t_b^2, \mathbf{e}), \tag{8.11}$$

where the remainder is holomorphic even at $e_\ell = 0$. For real parameters the first term is purely imaginary. Put $b_\ell = \pi \Im \tau_\ell / |\tau_\ell|^2 > 0$. Equation (8.8) gives

$$\partial_{t_\ell^2} \Re \widehat{B}(0, 0) = -b_\ell/4.$$

On one smaller fixed real box these derivatives are at most $-b_\ell/8$. Integrating along the squared coordinates proves

$$\Re T_{\mathbf{m}}(\mathbf{z}_{\mathbf{m}}) \le \Re T_{\mathbf{1}}(\mathbf{z}_*) - \sum_\ell \frac{b_\ell}{8} \frac{m_\ell^2 - 1}{p_\ell^2} \tag{8.12}$$

for all local nonzero charges. The strict loss holds unless every $|m_\ell| = 1$. This is an exact analytic inequality on a fixed parameter box, not a conclusion from a truncated expansion with comparable slopes.

Separate evenness writes $\Psi_\ell = z_\ell Q_\ell(z_1^2, \dots, z_b^2)$. Thus

$$z_\ell = -\frac{t_\ell}{2(1 - e_\ell Q_\ell/(4\pi \mathrm{i}))}.$$

For a nonzero charge, $z_\ell \neq 0$; on the small neighborhood $\sinh(2\pi \mathrm{i} z_\ell) \neq 0$. The real part of the translated-contour Hessian stays negative definite. Hence the local Gaussian coefficient is nonzero for negative as well as positive slopes.

There are finitely many charges left after the tails were removed. A charge outside the local ratio box has some $|m_i/p_i| \ge t_0$. Split its real interval at $|z_i| = t_0/4$. Outside, the source Fourier bound has a fixed loss. Inside, $|m_i + 2p_i z_i| \ge |p_i| t_0/2$; a small imaginary displacement of the appropriate sign decreases the filling phase. The displacement can be chosen within the fixed source tube, and its joining segments are estimated with the same nonstationary inequality and the endpoint Fourier loss. Spectator coordinates are still real. This supplies a fixed exponential loss for every such nonlocal charge. For local saddles, the connector height is $O(|p_i|^{-1} |m_i/p_i|)$ by the displayed factorization, which keeps their boundary losses below the selected saddle. These estimates are detailed in Appendix B.3.

8.7 Noncancellation and the final action

The 2^b charges with $|m_\ell| = 1$ have identical complex critical action. Under $(m_\ell, z_\ell) \mapsto (-m_\ell, -z_\ell)$, the product $m_\ell \sinh(2\pi \mathrm{i} z_\ell)$, the parity character and the integral are unchanged. They add exactly. For these charges the factor $(-1)^{bd+d\sum m_\ell}$ in (8.5) is one.

Combining (8.9) with the b-dimensional Gaussian gives

$$\langle K(\mathbf{p})\rangle_N = \chi_N \eta_N \kappa^{-\sum p_\ell} N^{3/2} D(\mathbf{p}) e^{NT_{\mathbf{1}}(\mathbf{z}_*)}(1 + O_{\mathbf{p}}(N^{-1})), \tag{8.13}$$

where

$$D(\mathbf{p}) = \frac{2^b(2\pi)^{b/2} B(\mathbf{z}_*) \prod_\ell \sinh(2\pi \mathrm{i} z_{*,\ell})}{\sqrt{\det[-T_{\mathbf{1}}''(\mathbf{z}_*)]}} \neq 0. \tag{8.14}$$

The Gaussian branches are fixed by the translated original real contours. The selected real action tends to F_0 as all $1/p_\ell \to 0$. Enlarge the fixed slope threshold so that it lies above every source, endpoint, nonlocal-charge and interpolation error level. Equation (8.13) then holds for the whole source, not only for the corridor.

Finally $\Psi_\ell = 2\pi \mathrm{i} \xi_\ell$ and the charge-one saddle equation give $\xi_\ell = 1 + 2p_\ell z_{*,\ell}$. Substitution into the Legendre expression shows

$$T_{\mathbf{1}}(\mathbf{z}_*) = \varphi(\boldsymbol{\xi}) + \frac{\pi \mathrm{i}}{2} \sum_\ell \frac{(\xi_\ell - 1)^2}{p_\ell} = S_{\varepsilon,\mathbf{p}}(\boldsymbol{\xi}).$$

Equations (7.5), (8.13) and Proposition 2.1 prove Theorem 1.2. The construction fills precisely the auxiliary cusps. The surviving cusp is the knot cusp, so the right side is the knot complement volume.

Chapter 9

An explicit volume theorem for ordered trees of crossing circles

The number of twists at a region is not the only quantity that can make filling controllable. In the construction below, a crossing circle of large degree meets many geometric pieces. Its complete cusp makes even a small integer twist correspond to a long normalized filling slope. This gives a numerical sufficient condition for the volume conjecture in which some twist coefficients may be short or zero.

9.1 The knot construction

A two-string tangle is drawn in a ball with four ordered boundary points: lower left, lower right, upper right, upper left. A product places one ball above another and joins their two adjacent endpoints by uncrossed arcs. The order of the factors is retained. The operation T rotates an entire ball through the specified quarter-turn in the diagram plane. Cylindrical closure joins lower left to upper left and lower right to upper right outside the final ball.

Fix a finite tree $\mathcal{T}$ with at least one edge, together with the cyclic order of the edges and the tangle frame at each vertex. There is one original crossing circle at every vertex, including the terminal vertices. There is no unmarked product junction. Each edge represents an actual quarter-turn of a child tangle, rather than a crossing of two drawn edges.

Here is a precise recursive way to specify the diagrams. Choose a terminal vertex as root. At a nonroot vertex v, put the integer braid R^{n_v} between the two ordered lists of its child tangles:

$$W_v = \left(\prod_{w \in I_v^-} T(W_w) \right) R^{n_v} \left(\prod_{w \in I_v^+} T(W_w) \right). \tag{9.1}$$

The lists I_v^- and I_v^+ specify the position of the crossing box in the original cyclic order. Empty lists are allowed. If w is the root's unique neighbor, close $R^{n_0} T(W_w)$ cylindrically. This is an actual link diagram, not a tensor network considered only up to scalar identities. Other choices of cutting section are related by the physical closed-collar moves, with their induced rotations and reversals of factor order retained.

The augmented diagram replaces each R^{n_v} by its crossing circle and a retained half twist $\epsilon_v \in \{0, \pm 1\}$, where $n_v \equiv \epsilon_v \pmod 2$. Filling that circle on its original disk-framed slope

$\mu_v + p_v\lambda_v$, with

$$n_v = \epsilon_v - 2p_v, \tag{9.2}$$

restores exactly the indicated integer braid. Here μ_v goes once around the auxiliary circle and λ_v follows its punctured disk. The main strands are not filled.

We require that the closed main strands form one knot K. This includes excluding a closed main component created inside an intermediate box. The test uses only the strand pairings and the parities of the n_v; large magnitudes alone do not guarantee it. Write d_v for the degree of v in the circle tree and $E = |E(\mathcal{T})|$. This graph degree is unrelated to the quantum level N.

9.2 The quantitative theorem

Let $F_N(M)$ be the zero-framed reduced Akutsu–Deguchi–Ohtsuki (ADO) invariant of the actual knot, normalized to one on the unknot. Its marked meridian coordinate is M, and at $q_N = e^{\pi i/N}$ its complete value is the reduced Kashaev invariant $K_N(K) = F_N(1)$. Moving the meridian on the scale considered here means setting $M = q_N^s$, with s in a fixed complex disk. Define

$$\mathfrak{S}(\mathcal{T}, \mathbf{n}) = \sum_v \frac{d_v}{d_v^2 + n_v^2}, \qquad H_N(s) = \frac{\sin(\pi s)}{N \sin(\pi s/N)}, \quad H_N(0) = 1. \tag{9.3}$$

Let $w(D)$ be the oriented writhe of the actual filled diagram, using the same positive checked-crossing convention as R. The exact framing and crossing factor is

$$A_N(s) = q_N^{a(D)s^2/2}, \qquad a(D) = -w(D) - \sum_v n_v. \tag{9.4}$$

The power uses the fixed logarithm $\log q_N = \pi i/N$. All crossing and strand-turn factors are included in (9.4); it is not a chosen normalizing phase.

Theorem 9.1 (A degree and twist criterion)**.** *Fix the ordered tree, its diagram frames, and its signed integer powers before letting N tend to infinity. Suppose the main strands form a knot and*

$$\mathfrak{S}(\mathcal{T}, \mathbf{n}) \le 2^{-25}. \tag{9.5}$$

Then the knot complement is hyperbolic, and

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N(K)| = \mathrm{Vol}(S^3 \setminus K). \tag{9.6}$$

For every fixed $R < 1$, the stronger statements

$$\frac{F_N(q_N^s)}{F_N(1)A_N(s)H_N(s)} = 1 + O_{\mathcal{T},\mathbf{n},R}(N^{-1}), \qquad |s| \le R, \tag{9.7}$$

and

$$\frac{2\pi}{N} \log |F_N(q_N^s)| \longrightarrow \mathrm{Vol}(S^3 \setminus K) \tag{9.8}$$

hold uniformly on that disk. In particular these values are nonzero there for all sufficiently large levels. Both level parities and composite levels are included. The tree, its degrees, its twist numbers and the radius R do not grow with N.

The constant in (9.5) is conservative. It makes the condition explicit; no optimality is asserted. There is no additional strict-angle hypothesis for this literal class. Indeed an unturned elementary circle has attainable output interval $(1/2, 1)$, and a turned subtree has interval $(0, 1/2)$. The exact product interval shows that one upper-interval factor together with any number of lower-interval factors again has the full upper interval, in any fixed association. The turn changes it to the lower interval. At the terminal root the two complementary intervals give a strict closing pair. This inductive construction supplies the strict augmented geometry used by the finite-source proof. In the two-circle case the excluded two-odd pattern is not a knot; the knot test retains the necessary even, or pure, leaf.

Two consequences make the range of (9.5) concrete.

- For a tree with E edges, every tuple satisfying $|n_v| \geq 8192\sqrt{E}$ satisfies (9.5), because $\sum_v d_v = 2E$.
- Start with any ordered tree on m vertices, with arbitrary fixed powers at those vertices. Attach B new terminal circles at each vertex and give each added circle a power of magnitude at least B. The resulting sum is strictly less than $2m/B$. Hence $B \geq 2^{26}m$ suffices, with the original powers still arbitrary, including zero. The knot/parity test remains necessary.

An unmarked product junction can also be included by inserting the literal identity braid R^0 at its prescribed position in the ordered stack. Add a crossing circle there in the augmented diagram and fill it with $\epsilon = p = n = 0$. The filled diagram is unchanged. The augmented exterior is different, and the added degree-d circle contributes $1/d$ to (9.3). This gives a direct application to constructions with sufficiently large-degree unmarked junctions when the same total bound holds. It does not treat such junctions as free or assert that a low-degree unmarked junction satisfies (9.5).

All these are finite constructions. Their potentially large size is fixed before the quantum limit. Related results use several invariant and root conventions; the coverage discussion records their precise scopes.

9.3 Why the original exterior has the required geometric pieces

Cut the augmented diagram along the projection sphere and the original crossing disks. The two resulting balls carry white projection faces and shaded disk triangles. At a leaf with half-segments (v_0, v_1, v_2, v_3), the shaded triangles are (O_v, v_1, v_2) and (O_v, v_3, v_0), where O_v is the auxiliary-circle end. The literal strand matching fixes every cyclic vertex order and hence every white face boundary.

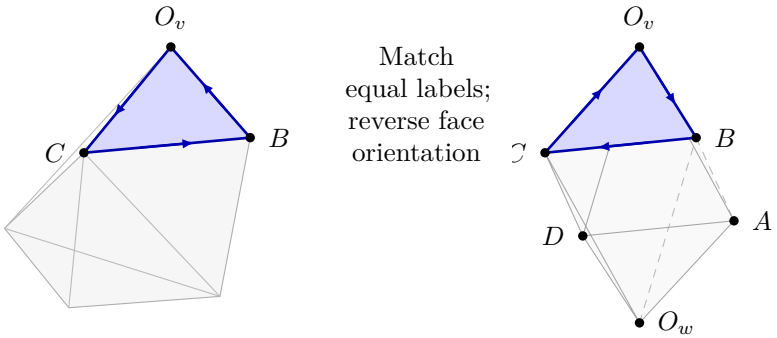
Adding one terminal child supplies an explicit local cell attachment. Let h_v denote the quarter-turned crossing-circle tangle. Write its parent's unturned box as $P_v = T^{-1}h_v$, so its exposed tuple is (v_3, v_0, v_1, v_2). If the old downstream matches are $v_1 - x_1$ and $v_2 - x_2$, insertion of h_w immediately after P_v replaces them by

$$v_1 - w_1, \quad v_2 - w_0, \quad w_2 - x_1, \quad w_3 - x_2. \tag{9.9}$$

Set $C = (v_1, x_1)$, $B = (v_2, x_2)$, $D = (v_1, w_1)$, and $A = (v_2, w_0)$. An octahedron with opposite vertices O_v, O_w and equator A, B, C, D has shaded faces

$$(O_v, B, C), \quad (O_v, D, A), \quad (O_w, D, C), \quad (O_w, B, A). \tag{9.10}$$

Glue its first face to the old face (O_v, C, B), with the opposite boundary orientation. The other three faces are precisely the new shaded faces prescribed by (9.9). The adjacent white sectors also concatenate in the literal strand order. Thus this is an attachment in the original cut, not a recognition based on equal volumes.



Existing projection ball
shaded face (O_v, C, B)

New octahedron
for edge vw
shaded face (O_v, B, C)

The glued triangle becomes internal. The three other shaded faces of the octahedron retain the circle v and introduce the circle w.

Figure 9.1: Attaching one octahedron in a projection half of the actual augmented tangle diagram. Equal labels are identified, and the blue faces have opposite boundary orientations. The triangle becomes internal; the other three shaded disk faces retain the parent circle and introduce the child circle, as listed in (9.10). Only the attachment faces are blue. The labels O_v, O_w belong to the original auxiliary crossing circles. The same attachment is made in the other projection half.

The left-insertion version attaches along the other parent triangle. It allows the parent box to remain at its specified position among its children. Start from one tree edge and insert the remaining children in the appropriate left and right order, carrying each rotated child ball with its frame. Each projection half is a tree of E octahedra attached along $E - 1$ triangles. Its remaining shaded faces are exactly the original two per circle.

The internal interfaces give the additional disk pages. At a circle of degree d_v, the d_v incident blocks occur in the original cyclic order. One distinguished seam retains the original half twist; the other seams are the untwisted insertion maps. In particular the construction does not put one copy of the retained crossing on every incident edge. It also retains the actual half-turn produced when a turn is moved across the closing sphere. Restoring the labeled faces recovers the original augmented link and its primitive disk paths. The ordered decomposition proof gives the full half-segment and face-word calculation.

9.4 Positive cells and primitive filling periods

For an edge with endpoint angular variables t, u, the finite quantum sum has the following analytic phase. Its derivatives determine the geometric logarithms below:

$$\begin{aligned} \psi(t,u) &= \operatorname*{stat}_x [b(x,t) + b(x,u)], \\ b(x,t) &= \int_0^x \log(2\cos 2\pi v + 2\cos t)\, dv - 2J(x), \\ J(x) &= \int_0^x \log(2\sin \pi v)\, dv. \end{aligned} \tag{9.11}$$

Branches continue $x = 1/4, t = u = 0$. In particular

$$\tan^2(\pi x) = \frac{1}{1 + \tan^2(t/2) + \tan^2(u/2)}, \qquad \psi_t = -\frac{1}{\pi}\operatorname{arsinh}(\tan(t/2)\cos(u/2)). \tag{9.12}$$

Let $\Psi = \sum_{vw} \psi(\tau_v, \tau_w)$. The first Fourier term of the exact crossing traces has phase

$$\Psi(\boldsymbol{\tau}) + \frac{i}{4\pi}\sum_v n_v \tau_v^2 - i\sum_v \tau_v. \tag{9.13}$$

To locate its geometric branch, vary its last coefficient continuously from zero to one. For $0 \le c \le 1$ solve

$$\Psi_{\tau_v} + \frac{i n_v}{2\pi}\tau_v = ic, \qquad \boldsymbol{\tau}(0) = 0. \tag{9.14}$$

The quantitative estimate (9.5) gives a unique branch with

$$|\tau_v(c)| \le \frac{2\pi c}{(1 - 1/50)\sqrt{d_v^2 + n_v^2}} < 0.00111, \qquad \Im\tau_v(c) < 0 \quad (c > 0). \tag{9.15}$$

This is a bound on each local angular parameter. It is not a bound requiring the total primitive meridian logarithm to stay near zero.

Each edge block admits eight positively oriented ideal tetrahedra whenever both angular parameters have modulus at most 0.01. Their complete shapes are i, and their deformed shapes stay within distance 0.4 of i. On a disk page put $Q = e^{i\tau_v}$. The three complex ratios on its incoming edges are $(-Q^{-1}, -Q, -Q^{-1})$; the outgoing ratios are their inverses. These ratios keep both angle and shear, and depend only on this circle's parameter. Consequently adjacent blocks fit even if their other endpoints have different parameters. At the distinguished odd seam the original cross-copy disk maps exchange the prescribed punctures. Their full matrix edge words are identities. Positivity and continuation fix every interior edge angle at 2π.

The actual meridian crosses the consecutive disk-page sectors once. A pure edge contributes $-2\pi\psi_{\tau_v}$; the single retained half twist contributes $-i\epsilon_v\tau_v$. The original primitive logarithms are therefore

$$U_v = -2\pi\Psi_{\tau_v} - i\epsilon_v\tau_v, \qquad V_v = 2i\tau_v, \qquad U_v + p_v V_v = -2\pi i c. \tag{9.16}$$

These are continued logarithms of actual paths. Their last identity cannot be obtained by reducing U_v to a principal logarithm.

For completeness, the link of an auxiliary ideal vertex is a torus with a nonsingular similarity structure. Since $\Re V_v > 0$, its holonomy has a second fixed point q_v after the marked point has been sent to infinity. The developing image avoids q_v: a zero of the invariant vector field $(z - q_v)\partial_z$ has positive index, whereas a torus has total index zero. Hence $\log(z - q_v)$ develops a complete Euclidean torus. Near $c = 0$, q_v tends to infinity and the periods on the fixed original paths tend to zero. Their difference from (9.16) is a continuous integer multiple of $2\pi i$, and is therefore zero along the whole path.

The slope in (9.16) has cone angle $2\pi c$. Its independent longitude has positive real period, so it bounds a punctured metric tube with a positive core length. At $c = 1$, attaching the core produces a smooth solid torus on precisely the original primitive slope. The main cusp stays a complete parabolic cusp. This constructs a complete finite-volume hyperbolic metric on the original knot complement, including the short coefficients permitted by (9.5). The local angular cells give the explicit vertex and face formulas, and the geometric completion proof retains the original odd seams and logarithmic periods throughout this path.

9.5 The action and the invariant have the same normalization

The same actual cells fix the additive constant of the complex action. With $V_v = 2i\tau_v$, set

$$G = 2\pi i \left(\Psi - \sum_v \tau_v \Psi_{\tau_v} \right) + \frac{1}{2} \sum_v \epsilon_v \tau_v^2. \tag{9.17}$$

Then $dG = \frac{1}{2} \sum_v V_v \, dU_v$. At zero, its imaginary part is the volume of the original complete augmented link. The marked boundary formula is

$$\mathrm{Vol} = \Im G - \frac{1}{2} \sum_v \Re U_v \Im V_v. \tag{9.18}$$

Let $\mathcal{C}(c)$ be (9.13) with the last term multiplied by c, evaluated at (9.14). Substitution of the original filling relation in (9.17)–(9.18) gives

$$\mathrm{Vol}_c = 2\pi \Re \mathcal{C}(c). \tag{9.19}$$

At $c = 1$, the right side is thus the volume of the knot complement just constructed.

The finite invariant selects this action without an assumption about saddle dominance. The polynomial expansion of the finite invariant assigns an integer label $0 \le k < N$ to every internal edge. Its normalized label is $x = k/N$; the high range is $k > \lfloor (N - 1)/2 \rfloor$. The complete contribution of an edge is maximized at $x = 1/4$. Outside $[1/5, 3/10]$, that edge loses a fixed exponential amount; a high index loses a fixed amount as well. The total discarded source has an absolute gap $\pi/200$, independently of graph size, although its polynomial prefactor can depend on the fixed tree and powers. The exact finite angular sum has a Gaussian bound weighted by the vertex degrees. Its real decay constant can be taken as $1/400$ and its imaginary growth constant as 70.

Preconditioning (9.14) by $(d_v - in_v)/(2\pi)$ gives a contraction with constant at most $1/25$. The same estimates compare every retained nonzero Fourier charge with charge one, with a strict loss unless every charge has magnitude one. The exact crossing-trace identity has already paired the reflected positive and negative modes, and the resulting coefficient of the

first charge is exactly one. Exact contour identities bound the omitted charge tails and their joining segments by an absolute exponential error. The possible drop of the first action is at most $\pi(2550/2401)\mathfrak{S}$, which (9.5) places below half the available error gap. Thus the selected term is nonzero for both parities of N.

The physical closing trace contributes exactly H_N, and all remaining crossing, changes of tensor order and zero-framing factors combine into the exact A_N in (9.4). The bounded insertion on each retained edge is $1 + O(N^{-1})$; the conditioned saddle evaluation therefore gives (9.7). The full constant and the scalar are retained when passing to (9.6)–(9.8). Neither the volume of a comparison diagram nor the complete augmented volume replaces the filled knot volume. The quantitative finite-sum proof contains the uniform contraction, Fourier-tail and joining-contour estimates; the elementary Newton bounds give an independent derivation of the graph-independent decay constants. The exact scalar proof retains the physical strand orientations and framing in (9.4).

9.6 A geometric form of the numerical condition

At the complete auxiliary cusp the primitive shape is $2ic_v/(1 - i\epsilon_v c_v)$, where $c_v = 1/d_v$. The normalized length of a slope means its Euclidean length divided by the square root of the cusp area. For the actual primitive filling slope this gives

$$\mathcal{L}_v^2 = \frac{d_v^2 + n_v^2}{2d_v}, \qquad \mathfrak{S} = \frac{1}{2}\sum_v \mathcal{L}_v^{-2}. \tag{9.20}$$

Thus (9.5) is equivalently $\sum_v \mathcal{L}_v^{-2} \le 2^{-24}$. This identity explains why high degree can compensate for a small crossing power. The proof above establishes the completion for these particular ordered exteriors; it does not import an unrelated normalized-length criterion or classify all short fillings.

9.7 The preferred cusp and the first correction

The remaining cusp belongs to the knot itself. Normalize its meridian translation to one. Its preferred longitude, which has linking number zero with the knot, then translates by a complex number τ_K, the preferred cusp shape. This geometric normalization is needed to interpret the first correction in the quantum profile.

Let x_e^* be the branch in (9.12) evaluated at $\boldsymbol{\tau}(1)$. The finite edge-insertion calculation retains one further term:

$$\frac{F_N(q_N^s)}{F_N(1)A_N(s)H_N(s)} = 1 - \frac{\pi s^2}{N}\sum_e \cot(\pi x_e^*) + O_{\mathcal{T},\mathbf{n},R}(N^{-2}). \tag{9.21}$$

The finite insertion calculation, derived in Sections 5–6 and 8 of the joint wire calculation, has an integrand independent of s, multiplied by $1 - \pi s^2 \sum_e \cot(\pi x_e)/N + O(N^{-2})$, uniformly with derivatives. At the nondegenerate joint saddle, the normalized expectation of this coefficient differs from its critical value by $O(N^{-1})$. Its explicit factor $1/N$ makes that an $O(N^{-2})$ error. The common saddle corrections cancel in the ratio, and the other charges and contour terms are exponentially smaller by the quantitative estimate. This proves the remainder in (9.21) under the present summed criterion.

The geometric translation calculation gives, on the same filled branch,

$$\tau_K = a(D) + 2i \sum_e \cot(\pi x_e^*). \tag{9.22}$$

This identification follows from the actual preferred longitude; it is not a definition of the cusp shape from (9.21).

Let λ_{aug} be the preferred main longitude in the augmented exterior and μ_K its meridian. For the fixed filling integers retain the boundary word $\lambda_{\mathrm{fill}} = \lambda_{\mathrm{aug}} \mu_K^{\sum_v p_v \ell_v^2}$, where ℓ_v is the actual linking number with crossing circle v. Before the endpoint this is simply a fixed curve on the unfilled main boundary. At $c = 1$, surgery homology identifies it with the preferred longitude of the smooth filled knot complement.

Here is why its translation formula applies beyond the initial small-deformation neighborhood. The edge blocks form a positive ideal triangulation of the original augmented exterior. Near completeness, prescribing the main meridian logarithm and the primitive auxiliary longitude logarithms determines its marked shape germ uniquely. The marked multicusp gluing theorem, Sections 2–3 and 5, proves precisely these mixed peripheral coordinates and the corresponding decorated representation statement. Its logarithmic gluing Jacobian has the form $A + B\,\mathrm{diag}(1/(z_j - 1))$. The edge and peripheral rows are Lagrangian, whereas positivity of the shapes gives $\Im(1/(z_j - 1)) < 0$; pairing a supposed kernel vector with its complex conjugate proves that the Jacobian is invertible.

Developing this triangulation with the earlier marked representation and its actual cusp flags therefore identifies the two representation germs. The exact circle-variable elimination identifies their retained physical-edge labels by (9.12), before any edge is eliminated. The preferred-longitude face-word calculation then supplies the translation $a(D) + 2i \sum_e \cot(\pi x_e)$ for this fixed boundary word near completeness. Both sides continue holomorphically on the small angular chart: the main meridian remains a nonidentity parabolic, so the commuting longitude has a uniquely defined translation in its unit-meridian frame. This ratio has no logarithmic integer ambiguity. Their equality thus continues along the whole positive charge path. At the smooth endpoint the fixed word is the preferred longitude, giving (9.22). The same original surgery homology gives exactly the integer $a(D) = -w(D) - \sum_v n_v$, with no unrecorded framing shear.

Put $z = 2\pi i s$ and

$$\mathcal{H}_N(z) = \frac{\sinh(z/2)}{N \sinh(z/(2N))}, \qquad \mathcal{H}_N(0) = 1. \tag{9.23}$$

Combining the exact factor (9.4), the expansion (9.21), and the geometric identity (9.22) gives

$$\frac{F_N(e^{z/(2N)})}{F_N(1)\mathcal{H}_N(z)} = 1 + \frac{\tau_K z^2}{8\pi i N} + O_{\mathcal{T},\mathbf{n},R}(N^{-2}), \qquad |z| \le 2\pi R < 2\pi. \tag{9.24}$$

All fixed derivatives have the corresponding limits on smaller disks. The shape in this formula belongs to the filled knot complement, not the complete augmented link. The exact scalar in (9.4) remains part of the statement; its quadratic expansion is the integer contribution to the preferred cusp in (9.24).

9.8 The pointed quantum A-period

Fix a genuine all-level quantum A-operator P annihilating the knot's reduced ADO family. The finite initial vector c_0 of ordinary-color polynomials selects its solution. The pointed operator construction, Sections 2–3, gives an ordered recurrence row $r_N(q)$ with

$$\mathscr{P}_N(P, c_0) = \mathrm{CT}_{t=0}\big[r_N(q_N(1+t))c_0(q_N(1+t))\big] = K_N(K). \tag{9.25}$$

The constant Laurent coefficient is taken after multiplying the whole row by the initial vector. This retains cancellations at singular recurrence steps and the required derivatives of the fixed initial data.

Corollary 9.2. *Under the hypotheses of Theorem 9.1,*

$$\lim_{N\to\infty} \frac{2\pi}{N} \log|\mathscr{P}_N(P, c_0)| = \mathrm{Vol}(S^3 \setminus K). \tag{9.26}$$

Proof. The exact reconstruction (9.25) holds at every sufficiently large level. Substitute it in (9.6). □

This is a limit for the operator with its specified initial data. Identifying the selected geometric factor of its classical symbol remains a separate problem and is not used in this corollary.

Chapter 10

A local filling condition from subtree polynomials

The summed condition in Chapter 9 compares an error for the complete augmented diagram with the total change of action caused by filling. That comparison is useful when the sum of the changes is small. A tree also permits a different calculation. Evaluate a terminal subtree first, keep its value at zero exactly, and estimate the polynomial left on its one boundary. The next trace then needs to absorb only its own change of action. This chapter proves that this procedure gives a sufficient condition at each vertex, with no sum over the vertices.

We keep exactly the ordered tangle construction of Section 9.1. There is one crossing circle at every vertex of a finite tree, including its terminal vertices. The cyclic orders, positions of the crossing boxes, quarter-turns, disk framings and retained half twists are part of the diagram. Each tree edge is an actual quarter-turn. Write $d_v \geq 1$ for a vertex degree and $n_v = \epsilon_v - 2p_v$ for its signed crossing power. The component test must give one main knot, including no closed main component inside an intermediate tangle.

10.1 The local volume theorem

The reduced invariant, the root of unity, and the exact scalar factors have the same normalization as before:

$$\begin{aligned} q_N &= e^{\pi i/N}, & K_N(K) &= F_N(1), \\ H_N(s) &= \frac{\sin \pi s}{N \sin(\pi s/N)}, & A_N(s) &= q_N^{a(D)s^2/2}, \\ a(D) &= -w(D) - \sum_v n_v. \end{aligned} \tag{10.1}$$

The removable value of H_N at zero is one. The exponent $a(D)$ uses the oriented writhe of the actual filled diagram and its original checked-crossing convention.

Theorem 10.1 (A local degree and twist condition)**.** *Fix the ordered tree, its strand frames, its knot parity pattern, and all its signed integer powers. Suppose*

$$\kappa = \max_v \frac{d_v}{d_v^2 + n_v^2} \leq 2^{-120}. \tag{10.2}$$

Then the actual filled knot complement is hyperbolic. For every fixed $0 \le R < 1$,

$$\frac{F_N(q_N^s)}{F_N(1)A_N(s)H_N(s)} = 1 + O_{\mathcal{T},\mathbf{n},R}(N^{-1}), \qquad |s| \le R, \tag{10.3}$$

and

$$\frac{2\pi}{N}\log|F_N(q_N^s)| \longrightarrow \mathrm{Vol}(S^3 \setminus K) \tag{10.4}$$

uniformly on that disk. These values are nonzero for all sufficiently large levels. The limit includes both parities and composite levels. The tree, the powers and the disk are fixed before N *tends to infinity.*

The numerical test is independent of the number of vertices. Polynomial prefactors, error constants and starting levels can depend on the entire fixed tree and its powers. Thus the theorem does not assert a limit for a sequence of diagrams growing with N. The knot condition also retains the two-circle qualification from Chapter 9: the two-odd pattern is not a knot, so an admissible two-circle knot has a pure leaf.

The constants in the summed and local tests are different. Neither test is being used as an optimal characterization. Both remain sufficient results. A simple consequence of the local test is

$$\frac{d}{d^2+n^2} \le \frac{1}{2|n|} \quad (n \neq 0): \qquad |n_v| \ge 2^{119} \text{ for every } v \quad \Longrightarrow \quad (10.2).$$

This bound no longer grows with the number of edges. A zero power is permitted if its degree is at least 2^{120}, provided the other vertices satisfy their own inequalities and the diagram is a knot. These large constants make the scope explicit; they are not estimates of the smallest admissible twists.

10.2 What a subtree leaves on its boundary

Choose a terminal root. A nonroot vertex has one edge toward its parent and a finite ordered list of child subtrees. After evaluating all the crossings and internal edges in those child subtrees, each child leaves a polynomial in the one variable at its boundary. Writing that variable as $-2\cos t$ is convenient for the exact root-of-unity trace. We denote the polynomial value by $M_{w,N}(t)$. It is even and periodic as a function of the angle t.

The fixed Newton polynomials e_k and their full pairing are defined in Section E.1. The full edge kernel is

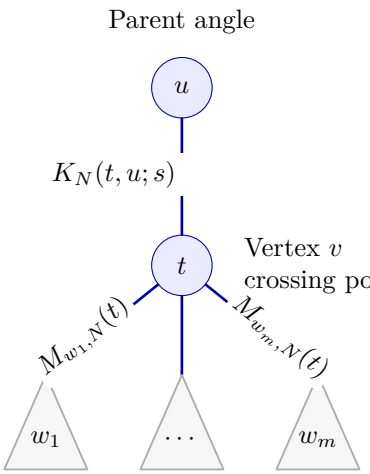
$$K_N(t,u;s) = \sum_{k=0}^{N-1} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N(s)}{W_k(s)}. \tag{10.5}$$

Every Newton index is included. In particular the boundary polynomial is not obtained by discarding high indices before the subtree has been evaluated.

Let $\mathcal{D}^{(t)}_{n_v,\sigma_v,N}$ be the actual trace after multiplication by the full crossing power at v. It acts in $-2\cos t$, takes the remainder in the prescribed Casimir quotient, and retains values and first derivatives at repeated roots. The orientation type determines $\sigma_v \in \{0, s, -s\}$. The edge orientation scalar γ_{e_v} belongs to that edge once. The exact outgoing polynomial is

$$M_{v,N}(u) = \gamma_{e_v}(s)\,\mathcal{D}^{(t)}_{n_v,\sigma_v,N}\left[K_N(t,u;s)\prod_{w \text{ child of } v} M_{w,N}(t)\right]. \tag{10.6}$$

Only K_N depends on u. Hence $M_{v,N}(u)$ has trigonometric degree at most $N-1$, independently of the size or depth of the subtree. The trace may receive a polynomial of much larger degree in $-2\cos t$; its quotient operation handles that degree without changing the outgoing degree bound.



The edge keeps every index $0 \le k < N$:

$$K_N(t,u;s) = \sum_{k=0}^{N-1} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N}{W_k}.$$

The trace at v acts only in t:

$$M_{v,N}(u) = \gamma_{e_v}\,\mathcal{D}_v^{(t)}\left[K_N(t,u;s)\prod_{j=1}^{m} M_{w_j,N}(t)\right].$$

Trigonometric degree in u: at most $N-1$.

Factor out the exact value at zero:

$$\widehat{M}_{v,N}(u) = \frac{M_{v,N}(u)}{M_{v,N}(0)}.$$

Its local phase and its real decay control the next trace. The denominator includes all crossing and orientation factors from this subtree.

Figure 10.1: Evaluating a subtree of the original finite contraction. The blue edge retains the full Newton kernel. Each lower triangle denotes a child subtree with its existing polynomial boundary value. The trace at v acts only on t, leaving an even trigonometric polynomial in the parent angle u. Its value at zero is factored out exactly before estimating the next trace.

The last root trace acts on the product of its child polynomials. The original closing and framing factors, reference factors and one reduced cut factor are then restored. This is a change in the order of a finite tensor contraction. It does not commute tangles or change any physical quarter-turn.

10.3 Why the normalization is local

The induction proves that $M_{v,N}(0)$ is nonzero and that, near the origin,

$$\frac{M_{v,N}(u)}{M_{v,N}(0)} = \frac{\alpha_v(u)}{\alpha_v(0)}e^{Nh_v(u)}(1+O(N^{-1})). \tag{10.7}$$

Here α_v is holomorphic and nonzero, $h_v(0) = h_v'(0) = 0$, and h_v is even. The important uniform property is its curvature:

$$\left|h_v''(u) + \frac{1}{2\pi}\right| \le \frac{1}{40\pi}, \qquad |u| \le 2^{-12}. \tag{10.8}$$

It depends on the single edge joining the subtree to its parent. It does not grow with the number of vertices already evaluated.

To see why this is possible, let $\psi(t,u)$ be the exact edge action of (E.16)–(E.19). The conditional action at a vertex is

$$A_v(t,u) = \psi(t,u) + \sum_w h_w(t).$$

Its t curvature is close to $-d_v/(2\pi)$. After including the crossing chirp and the first Fourier charge, the stationary equation has the form

$$\left(\frac{d_v - in_v}{2\pi}\right) t = -i + E_{v,t}(t,u), \qquad |E_{v,tt}| \le \frac{d_v}{40\pi}.$$

The stationary value, with its value at $u=0$ subtracted, is the new phase h_v. Differentiating a stationary value gives a Schur complement:

$$h_v'' = \psi_{uu} - \frac{\psi_{ut}^2}{A_{v,tt} + in_v/(2\pi)}.$$

The denominator grows with d_v and $|n_v|$, whereas the two derivatives in the numerator come from one edge. The explicit edge bounds therefore preserve (10.8).

The exact denominator in (10.7) is essential. On a smaller zero-free disk,

$$\frac{1}{N}\log\frac{M_{v,N}(u)}{M_{v,N}(0)} = h_v(u) + \frac{1}{N}\log\frac{\alpha_v(u)}{\alpha_v(0)} + O(N^{-2}).$$

The next trace retains this entire expression. Its error changes an analytic amplitude; it does not add the difference between a subtree's complete and filled actions to each of its ancestors.

10.4 From a local saddle to the full finite trace

A local expansion alone cannot exclude a larger contribution far from the origin. The induction also proves a real angular envelope for the exact polynomial,

$$\left|\frac{M_{v,N}(u)}{M_{v,N}(0)}\right| \le CN^A e^{-N2^{-18}(1-\cos u)}, \qquad u \in [-\pi,\pi].$$

The degree bound following (10.6) converts this to a linear bound on complex strips by the subharmonic maximum principle. Near the origin the phase curvature gives a quadratic bound instead. Combining the two proves one fixed complex-strip estimate for every subtree.

The exact crossing trace is then treated on its translated root grid. Its Fourier coefficients retain the Chebyshev factors $U_{\ell-1}(\cos 2\pi\sigma_v)$, the parity signs and all endpoint currents. The first coefficient is exactly one. The conditional critical values are strictly ordered with the charge, and all large charges and contour connectors have a fixed conditional loss. Condition (10.2) makes the current vertex's first-charge action drop smaller than that loss. It does not compare this loss with a sum over descendants.

Finally the real envelope is proved afresh from the actual trace: the full polynomial norm of a fixed crossing power, multiplication, quotient reduction and the closing functional cost only a polynomial in N. The edge kernel supplies angular decay at the outgoing variable. Dividing by the newly proved nonzero value at zero spends only the same local action drop. This closes the induction. Appendix G gives all constants, finite contours, collision limits and normalizations.

10.5 A separate vertexwise metric criterion

The geometric completion requires less than the local quantum condition. The following result is useful independently.

Theorem 10.2 (Positive completion from vertexwise data)**.** *For the same actual ordered construction, suppose*

$$L_v = \sqrt{d_v^2 + n_v^2} \geq 630 \qquad \text{for every } v. \tag{10.9}$$

Then the charge equations

$$\Psi_v(\boldsymbol{\tau}) + \frac{in_v}{2\pi}\tau_v = ic, \qquad \Psi = \sum_{vw} \psi(\tau_v, \tau_w), \qquad 0 \leq c \leq 1 \tag{10.10}$$

have a unique solution in $|\tau_v| \leq 10^{-2}$. It is holomorphic in c near the closed unit disk and satisfies

$$|\tau_v(c)| \leq \frac{2\pi c}{L_v - d_v/39000}, \qquad \Im\tau_v(c) < 0 \quad (c > 0). \tag{10.11}$$

The actual cells complete to cone solid tori of angle $2\pi c$, and to smooth solid tori at $c = 1$. The main cusp remains complete. When the component test gives a knot, the endpoint is its complete finite-volume hyperbolic complement.

The sign in (10.11) is proved, rather than inferred from the smallness of the parameters. Put

$$F(t, u) = 2\,\mathrm{arsinh}(\tan(t/2)\cos(u/2)), \qquad E = F - t.$$

On the radius 10^{-2} bidisk, the explicit derivatives give $|E_t|, |E_u| < 1/39000$. Since $E(0, u) = 0$, write $E = tH$ with H holomorphic and $|H| < 1/39000$. The charge equation is

$$\left(d_v - in_v + \sum_{w \sim v} H(\tau_v, \tau_w)\right)\tau_v = -2\pi ic.$$

Its denominator has strictly positive real part. This gives the sign and a contraction whose row bound is $2/39000$, independent of the tree size. The proof, including a finite variant with different radii at different vertices, is in Section G.7.

The original primitive periods are

$$U_v = -2\pi\Psi_v - i\epsilon_v\tau_v, \qquad V_v = 2i\tau_v, \qquad U_v + p_vV_v = -2\pi ic. \tag{10.12}$$

Appendix F proves that these are actual logarithmic periods along a positive path. At the initial cusp their integral ambiguities vanish; continuity preserves that choice. At $c = 1$ the primitive filling slope therefore makes one full turn. The conclusion concerns the original marked exterior, even if a total meridian logarithm has moved far from zero.

10.6 The original volume and the two sufficient tests

Back-substitution of the subtree stationary points gives the simultaneous equations (10.10) at $c = 1$. The constants in the subtree phases cancel exactly, leaving

$$\mathcal{C}(c) = \operatorname*{stat}_{\boldsymbol{\tau}} \left[\Psi + \frac{i}{4\pi}\sum_v n_v\tau_v^2 - ic\sum_v \tau_v\right].$$

The primitive-period calibration in Section F.3.6 gives $\mathrm{Vol}_c = 2\pi\Re\mathcal{C}(c)$. This identifies the action selected by the finite trace with the volume of the actual filled complement. It is the geometric step in Theorem 10.1.

Each edge sum contributes $N^{-3/2}$, and each circle trace contributes N^2 times a nondegenerate one-dimensional Gaussian. A tree has one more vertex than edge, so the full leading power is $N^{3/2}$. Its amplitude is nonzero. The exact first Fourier coefficient, the one reduced cut factor and the original scalar ledger then give (10.3) and (10.4).

The geometric criterion (10.9) by itself does not prove a quantum volume limit. It can hold when neither the summed quantum condition nor the local quantum condition does. For example, on arbitrarily long paths with powers chosen from 630 and 631, it holds at every vertex while the sum $\sum_v d_v/(d_v^2 + n_v^2)$ grows with the number of vertices. The appropriate knot parity still has to be selected. Theorem 10.1 supplies a separate analytic criterion. The earlier summed criterion in Theorem 9.1 remains useful, often with much smaller powers on a particular finite diagram.

Chapter 11

Checking the certificates and understanding the scope

The two theorems combine written analytic and geometric arguments with finite checks of different kinds. The long-twist proof reduces two local inequalities to exact rational certificates supplied with the paper. The quantitative circle-tree proof establishes its uniform bounds analytically; separate finite checks verify its edge formulas, face identifications and peripheral identities. This chapter distinguishes those roles and records the scope of each theorem.

11.1 Three different kinds of evidence

The manuscript contains analytic proofs, exact finite certificates, and optional finite audits. They have different logical roles. The matrix identities, support and strictness arguments, geometric incidence, complex estimates and filling analysis are proved in the text. The rational certificates are finite parts of the proof: the continuum inequalities reduce to their exact feasibility, coverage and supporting-plane checks. Finite matrix or numerical asymptotic audits provide independent checks of formulas and normalizations, but do not establish a theorem by extrapolation.

The proof is given in the chapters and appendices of this manuscript. The repository also supplies the exact certificate data and verification programs. Those programs check the finite inequalities described below; they do not formally verify the analytic or geometric arguments.

11.2 Replaying the exact certificates

From the repository root, run

```
python3 repro/verify.py --output repro/results
```

The serial runner uses the standard Python library. It replays the two-leaf block certificate, the attachment square certificate and the optional endpoint certificate, compares their exact results with the recorded data, and checks the elementary rational leaf and isolated-three-leaf margins. Its report records the source and data hashes. Candidate generation and numerical optimization are not run by verification.

The principal proof files are

Certificate	Mathematical coverage
`verify_conway_two_leaf_block_alias_certificate.py`	Sixteen slope intervals; both alias types; 288 corner/vertex inequalities.
`verify_conway_single_leaf_attachment_certificate.py`	Exact 55-rectangle cover; 880 corner/vertex inequalities on the full child/parent slope square.
Endpoint certificate in the runner's manifest	Optional stronger margin at the monotone-response endpoint; not needed for the main theorem.

All paths are resolved relative to the repository or script directory, rather than an author's temporary working directory. The runner's proof-input guide lists the written dependencies separately from executable checks.

11.3 Checking the quantitative tree formulas

The elementary edge equations and complex-strip estimates used by Chapter 9 have a portable reproduction command:

```
python3 repro/run_research_checks.py curvature-fillings
```

The runner checks exact symbolic edge identities, finite samples of the stated complex inequality, labeled octahedral attachments and the full signed peripheral matrix words. Its output separates those computations from the uniform analytic and geometric proofs. Appendix E gives the full quantitative finite-sum proof, including both uniform Newton estimates, the weighted contraction and the charge-tail bounds. Appendix F gives the complete ordered attachment, positive angular charts, primitive-period completion, volume calibration and preferred-cusp argument. The linked versions are optional separately available copies of those proofs. Such finite checks can detect sign or indexing errors; they do not replace the written proofs.

11.4 Building the manuscript and illustrated guide

The TeX source is in `manuscript/`; its native illustrations are in `manuscript/figures/`. A TeX Live or MacTeX installation with `latexmk` builds the paper:

```
python3 tools/build_manuscript.py
```

The output is `manuscript/build/main.pdf`. The build manifest records the hashes of all TeX and bibliography inputs and the generated PDF. It is a typesetting record, not a mathematical certificate.

The illustrated guide is built separately:

```
python3 tools/build_book.py
python3 tools/check_book.py
```

It is written to `docs/` and links the same authored PDF. Documentation builds do not execute research scripts. The guide also links the broader research companions and states their hypotheses separately from the two theorem classes developed here.

11.5 What has and has not been proved

Theorem 1.2 is uniform in the level for each fixed finite completed expression and each fixed sufficiently long signed filling tuple. It includes both parity sequences and composite levels. Its threshold can depend on the expression, its complete geometry and its retained half-twist pattern, but imposes no bound on ratios between the long slopes.

Theorem 9.1 uses a different sufficient hypothesis: the literal ordered circle tree and the explicit sum $\sum_v d_v/(d_v^2 + n_v^2) \leq 2^{-25}$. The full diagram, all degrees and all integer powers are fixed before the limit. The theorem gives actual hyperbolicity and volume equality, a uniform microscopic meridian profile with its exact scalar, and the preferred-cusp correction. It permits short coefficients under this condition; it does not prove a statement about arbitrary short or rational fillings.

Both results keep the original strand order, knot-pairing test, disk framings and unfilled main cusp. The octahedral attachment is proved by actual labeled cuts. A scalar equality in $\mathcal{H}_N$ is not used as a tangle isotopy. The continued boundary word used for the preferred-cusp calculation becomes the preferred longitude of the smooth filled knot only at the endpoint of the geometric path.

These are sufficient construction hypotheses, not a classification of all arborescent knots. The completed-box condition remains part of Theorem 1.2; the quarter-index states in circle trees are treated by the separate proof in Chapter 9. More general strict ordered Conway trees have a long-twist theorem in the linked research companions, with their own full source and marked-geometry proofs. No general short-filling theorem follows merely from the existence of complete augmented cells or from an upper bound on the absolute values of a larger sum.

Appendix A

Finite algebra and normalization identities

This appendix proves the finite identities on which the asymptotic argument rests. The first calculation compares the chosen crossing matrices and reduced trace with the conventional invariant. The next two calculations establish the Newton basis, its genuine geometric turn, and the exact color decomposition. The last calculation expands an arbitrary twist power while keeping derivatives at repeated roots. No limiting argument is used in these identities.

A.1 The crossing comparison and enhancement

This section supplies the finite normalization calculation used in Proposition 2.1. Put $\omega = Q$ and

$$p_j = \prod_{h=1}^{j}(1-\omega^h), \qquad \bar{p}_j = \prod_{h=1}^{j}(1-\omega^{-h}), \qquad p_0 = 1.$$

Kashaev's unbraided four-index matrix may be written by introducing nonnegative residues

$$u = [j-i-1], \quad v = [i-l], \quad t = [l-k], \quad h = [k-j].$$

Its support conditions are equivalent to $u+v+t+h = N-1$, and on that support

$$R^{o}_{(i,j);(k,l)} = \frac{N\omega^{1-(t+h)(u+h)}}{p_u \bar{p}_v p_t \bar{p}_h}. \tag{A.1}$$

One free j and such a composition specify one entry. Let $F_{ij} = \omega^{ij}/\sqrt{N}$. Its full Fourier transform has only the ordinary-integer-conserving entries

$$((F^{-1}\otimes F^{-1})R^{o}(F\otimes F))_{(a-k,b+k);(a,b)} = \omega^{1+(a-k)(b+1)}\frac{p_a p_{b+k}}{p_{a-k}p_b p_k}, \tag{A.2}$$

for $0 \le k \le \min(a, N-1-b)$.

Here is the finite summation. Define

$$S_m(X) = \sum_{j=0}^{m}\frac{X^j}{p_j \bar{p}_{m-j}} = \frac{\prod_{j=1}^{m}(X-\omega^j)}{p_m}.$$

The equality follows by induction from $\bar{p}_m = (-1)^m \omega^{-m(m+1)/2} p_m$ and $(1-\omega^m)S_m = (X - \omega^m)S_{m-1}$. The free Fourier index imposes conservation modulo N. The two remaining factors are $S_{N-1-z}(\omega^{c-z})S_z(\omega^{z-a})$; their zeros force $c \le z \le a$. With $m = a - c$, the residual sum is

$$\omega^{1+c(b+1)} \frac{p_a}{p_c} \sum_{j=0}^{m} \frac{(-1)^j \omega^{j(j-1)/2} \omega^{(b+1)j}}{p_j p_{m-j}} = \omega^{1+c(b+1)} \frac{p_a}{p_c} \frac{(\omega^{b+1};\omega)_m}{p_m}.$$

If $b + m \ge N$, the product contains $1 - \omega^N$ and vanishes. Otherwise it is p_{b+m}/p_b, proving (A.2) with integer, rather than merely residue, conservation.

Set $G_j = p_j/(1-\omega)^j = q^{j(j-1)/2}[j]_q!$, $D_j = (-q)^j$, and $\gamma = \omega q^{-d^2/2}$. Substitution into (2.2) gives exactly

$$(F^{-1} \otimes F^{-1}) R^o (F \otimes F) = \gamma (DG \otimes G) \mathscr{R} (G^{-1} \otimes G^{-1} D^{-1}).$$

For the enhanced braid conventions take

$$B_K = P(R^o)^T, \quad \mu_K = -qS, \quad \alpha_K = -q, \qquad B_q = P\mathscr{R}^T, \quad \mu_q = K^{-1}, \quad \alpha_q = q^{(N^2-1)/2},$$

where $Sv_j = v_{j+1 \bmod N}$. On b strands, the single similarity

$$U_b = F^{-\otimes b} \bigotimes_{j=1}^{b} (G^{-1} D^{j-1})$$

works for every adjacent generator because the quantum crossing conserves integer weight and commutes with $D \otimes D$. It also sends $\mu_q^{\otimes b}$ to $\mu_K^{\otimes b}$, and $\gamma\alpha_q = \alpha_K$. The writhe characters cancel exactly. The partial traces commute with the remaining module generators; distinct K weights and nonzero F arrows imply scalarity. Thus every internal similarity cancels, including the one open factor of a reduced cut. This proves the finite enhanced comparison used in Chapter 2. Large factorial gauges are never estimated by their norms.

The primal/dual change of basis is $Bv_j = (-1)^j q^{j(j-N)} v_{d-j}$. Direct generator substitution conjugates K^{-1} to the primal pivot and the crossing to the recorded primal convention. At completeness $g_{\mathrm{ADO}} = (-1)^d K$ and the twist has the corresponding sign. A b-strand closure of writhe w and c components has $w \equiv b - c \pmod 2$, so the remaining enhancement ratio is $(-1)^{d(c-1)}$. Its modulus is one, and for a knot it is one in this convention.

A.2 The block polynomial and the physical Newton turn

For completeness we justify the block minimal polynomial used in Lemma 2.2. Before specialization, the ordinary highest-weight module of highest weight d has weights $d - 2a$. In the pair, a highest vector of total degree j is obtained by solving the bidiagonal raising equation. If its coefficient at $|a, j-a\rangle$ is u_a, the equation at $|a, j-a-1\rangle$ determines u_{a+1}/u_a as the negative ratio of the two nonzero raising entries. Thus for generic q there is one such vector in each degree $j = 0, \dots, d$. Its highest weight is $2d - 2j$, its Casimir is $q^{2d-2j+1} + q^{-2d+2j-1}$, and its successive lowering vectors are nonzero through the usual length. The dimensions $2d - 2j + 1$ sum to N^2. Distinct generic Casimirs make these submodules independent.

In degree $T \le d$, precisely the highest vectors $j \le T$ contribute one descendant each. Their eigenvalues give the block characteristic polynomial. At $q = e^{\pi \mathrm{i}/N}$ it becomes

$$\prod_{j=0}^{T}(z - c_j), \qquad c_j = q^{2j+1} + q^{-2j-1}.$$

All physical matrix entries specialize regularly. The nonzero adjacent entries of the tridiagonal block make a boundary vector cyclic, so this characteristic polynomial is also minimal even when roots coincide. At $T = d$ it is P_N. The elementary identity

$$\prod_{j=0}^{N-1}(z - q^{2j+1} - q^{-2j-1}) = 2T_N(z/2) + 2$$

can also be checked by putting $z = w + w^{-1}$: the roots correspond to $w^N = -1$, and the monic degree and leading coefficient agree. The generating property of W_d propagates its vanishing to the full pair.

Now $B_r(\Omega)$ annihilates every degree block W_T with $T < r$, with multiplicities, and has first-color bandwidth at most r. In the turned matrix its first-color displacement is $d - (a + b)$, so $\mathsf{T}B_r$ annihilates every W_T with $T < d - r$. Its representing polynomial is therefore divisible by B_{d-r}.

For the reverse degree bound, its first middle-block column is

$$(\mathsf{T}B_r)_{a0} = x^{a-d}(B_r(\Omega))_{(d-a,0);(d-a,0)},$$

which vanishes for $a > d - r$. A polynomial of degree $k < N$ in the cyclic tridiagonal middle block has a nonzero kth coordinate on the first basis vector, from its unique outermost path. Thus $\deg \mathsf{T}B_r \le d - r$, proving $\mathsf{T}B_r = C_r B_{d-r}$.

Use the physical entry from $(d - r, 0)$ to $(0, d - r)$. On the turned side it is the unique r-step path in the high-degree block, whose factors are

$$Q^{j-r}(1 - Q^{r-j})(1 - Q^{j+1}), \qquad 0 \le j < r.$$

Including the leading coefficient $(-1)^r$ gives s_r^2. The corresponding path for B_{d-r} in the low block gives s_{d-r}^2. All turn monomials in this entry equal one. Hence

$$\mathsf{T}B_r = \frac{s_r^2}{s_{d-r}^2} B_{d-r}, \qquad \mathsf{T}\widehat{B}_r = \widehat{B}_{d-r}.$$

Finally $s_r s_{d-r} = N$ and (2.11) give $\phi(\widehat{B}_i \widehat{B}_j) = \delta_{i+j,d}$: use degree below d, leading coefficient at d, and full divisibility by P_N above it. This proves Proposition 2.3 by physical support and exact paths.

A.3 Circle coefficients and the signed alias

We derive the coefficient formulas rather than infer them from values at the distinct Casimir roots. That distinction is necessary because the quotient algebra also contains first derivatives at double roots. The generating-function calculation first gives a polynomial identity at a

generic parameter; regular specialization then retains all those derivatives automatically. The generic Chebyshev–Newton identity is

$$U_{j-1}(z/2) = \sum_{r=0}^{j-1} (-1)^r \begin{bmatrix} j+r \\ 2r+1 \end{bmatrix}_q B_r(z). \tag{A.3}$$

To prove it, successively divide $1 - zt + t^2 = (1 - c_r t + t^2) + t(c_r - z)$. The formal identity is

$$\frac{1}{1 - zt + t^2} = \sum_{r \geq 0} \frac{(-t)^r B_r(z)}{\prod_{i=0}^{r} (1 - c_i t + t^2)}.$$

After r steps the remainder is divisible by t^{r+1}. The denominator factors as a consecutive q-string, and the Gaussian Pascal recurrence gives its reciprocal coefficient. Taking the coefficient of t^{j-1} gives (A.3). For $j \leq \lfloor N/2 \rfloor$, every factorial index is below N; specializing and turning proves (2.18).

For higher colors use the polynomial identity

$$U_{N-h-1} = \tfrac{1}{2}(\mathsf{C}_h U_{N-1} - \mathsf{C}_N U_{h-1}).$$

Since $\mathsf{C}_N = -2$ in the full algebra,

$$L_{N-h} = (-1)^h \tfrac{1}{2} \mathsf{C}_h L_N + (-1)^N L_h.$$

The vth turned coefficient of the first term is $(-1)^h N \sum_{k=0}^{d} \cos(h(2k+1)\pi/N) \widehat{B}_v(c_k)$. This finite sum can be evaluated without a zero denominator. Put $z = w + w^{-1}$. The factorization of B_v is the consecutive string

$$B_v(w + w^{-1}) = (-1)^v w^{-v} \prod_{j=0}^{2v-1} \left(w - q^{-(2v-1)+2j} \right).$$

For $0 \leq v, h \leq M$, all its Fourier exponents after multiplication by $w^{\pm h}$ have absolute value below N. Summing at $w = q^{2k+1}$ extracts only the constant coefficient. The finite Gaussian product gives that coefficient as zero for $v < h$, and as $(-1)^h s_{2v}/(s_{v-h} s_{v+h})$ for $v \geq h$. Dividing by s_v^2 and multiplying the two factors of N proves (2.17), including $h = 0$ and both parities.

For the half twist, multiplication by z in Newton coordinates is the lower bidiagonal matrix Z with $Z_{rr} = c_r$ and $Z_{r+1,r} = -4 \sin^2((r+1)\pi/N)$. The one-crossing matrix is

$$(R(Z))_{ij} = \kappa (-1)^i q^{i(j+1)} \frac{s_i}{s_j s_{i-j}}, \qquad i \geq j.$$

Its first column is the finite Newton expansion of the crossing. Substituting the two neighboring ratios shows that it commutes with Z; since the first coordinate is cyclic, this determines the full matrix. Reversal on both sides gives (2.19). Coefficient conjugation gives the inverse crossing because its full Hermite values and logarithmic derivatives are conjugate. This argument retains all normal directions.

A.4 Fourier–Hermite interpolation of an arbitrary twist

Ordinary interpolation prescribes values. Hermite interpolation also prescribes derivatives at a repeated node. Here the crossing power supplies both data, and the change of variable $z = w + w^{-1}$ turns the interpolation problem into a finite Fourier calculation. The two Fourier equations below solve simultaneously for the coefficients that ordinary value interpolation would fail to distinguish. For an integer ℓ, let $f_{N,\ell}$ be the degree-below-N remainder of r_N^ℓ modulo P_N. Negative powers exist because all its nodal values are nonzero. At $w_k = q^{2k+1}$ define

$$h_k = \big((-1)^k q^{k(k+1)}\big)^\ell, \qquad d_k = -\frac{\ell}{2}(d-2k)h_k.$$

At a double node, $d_k = (w_k - w_k^{-1}) f'_{N,\ell}(c_k)$; at the odd simple node both sides of the logarithmic derivative condition vanish without prescribing an extra derivative. Put

$$H_n = \frac{1}{N}\sum_k h_k w_k^{-n}, \qquad D_n = \frac{1}{N}\sum_k d_k w_k^{-n}, \qquad 0 \le n < N.$$

Then

$$f_{N,\ell}(z) = a_0 + \sum_{n=1}^{d} a_n \mathsf{C}_n(z), \qquad a_0 = H_0, \quad a_n = \frac{D_n + (N-n)H_n}{N}. \tag{A.4}$$

Indeed the Laurent polynomial $f(w + w^{-1})$ has symmetric coefficients a_n at $w^{\pm n}$. At $w_k^N = -1$ its values and logarithmic derivatives have aliases

$$H_n = a_n - a_{N-n}, \qquad D_n = n a_n + (N-n) a_{N-n}.$$

Solving gives (A.4). At the even middle index $H_{N/2} = 0$ and the derivative equation still determines the coefficient. Paired nodes give the required compatibility, and the odd simple root imposes exactly one condition. Thus no root or normal is omitted.

Set $a_N = a_{N+1} = 0$ and

$$\beta_j^{\ell/2} = (-1)^j (a_j - a_{j+2}), \qquad 0 \le j < N.$$

The identity $\mathsf{C}_n = U_n - U_{n-2}$ gives the circle expansion. Since $|h_k| = 1$ and $|d_k| \le |\ell| N/2$, one has $|a_0| \le 1$, $|a_n| \le 1 + |\ell|/2$, and

$$\sum_j |\beta_j^{\ell/2}| \le 2N(1 + |\ell|/2).$$

For $\ell = 2P$, these are the coefficients used in (8.3). To obtain that formula directly, substitute the two finite Fourier sums into $a_j - a_{j+2}$, set $j = N - h - 1$, and pair h with $-h$ in an even corridor. The difference of the two exponentials gives $\sinh(2\pi \mathrm{i} z_k)$; the differentiated data give the linear factor $2PNz_k + h$, and differentiating the hyperbolic sine gives its cosine. Their sum is precisely $U'_N(z_k)/(2\pi\mathrm{i})$. The constant phase is $(-1)^d \kappa^{-P}$ and the two averaging factors give N^{-2}. This verifies (8.3) with both normal terms and both parity conventions.

Appendix B

Detailed estimates for the color sums and filling contours

The local and global estimates have different purposes. Section B.1 constructs an exponentially accurate holomorphic approximation on the strict neighborhood used for the saddle. Section B.2 bounds every remaining core coefficient, including parameters where the recurrence eigenvalues meet. The last section treats the joining contours for the filling integral. Keeping these estimates separate prevents a local asymptotic expansion from being applied outside its proven domain.

B.1 The exact local recurrence and its holomorphic asymptotic

This section proves the local source estimate used in the completed-root theorem. Its hypothesis is a fixed compact interval

$$I = [\tau_0, \tau_1] \subset (1/4, 1/2) \tag{B.1}$$

for every retained complete leaf coordinate. All constants may depend on this interval and the fixed source tree. The color neighborhood and every complex index neighborhood below are fixed independently of the level. We work with the actual finite low core and its specified factorial continuation. No homogeneous solution of a recurrence is substituted for that source. In particular, an inverse-power approximation on the real index grid will not be used as if it were an exponentially accurate complex interpolation.

B.1.1 Canonical factorials and the literal mixed coefficient

To move a color parameter off the integer grid, we must specify a function that agrees with the finite coefficient and preserves its phase. The Gamma product below supplies that function for the sine factorials. Its exact shift identity will then allow the finite recurrence to remain valid at complex colors; agreement only in a leading asymptotic term would not be sufficient.

Retain

$$d = N - 1, \quad M = \lfloor d/2 \rfloor, \quad q = e^{\pi \mathrm{i}/N}, \quad Q = q^2, \quad \kappa = q^{d^2/2}, \quad F_r = (Q;Q)_r.$$

In this section x denotes a scaled Newton index, rather than the physical-frame scalar q^d. The sine factorial has the exact continuation

$$G_N(z) = \left(\frac{2\pi}{N}\right)^z \Gamma(z+1) \prod_{k=1}^{\infty} \frac{\Gamma(Nk+z+1)}{\Gamma(Nk-z)(Nk)^{2z+1}}, \qquad -1 < \Re z < N. \tag{B.2}$$

All positive-base powers use their real logarithms. The logarithm of the kth factor is $O_K(k^{-2})$ on compact subsets of the strip: in the Gamma-ratio expansion its k^{-1} term cancels. Thus the product is holomorphic and zero-free there. Gamma's functional equation and the convergent sine product give

$$\frac{G_N(z+1)}{G_N(z)} = 2\sin\frac{\pi(z+1)}{N}, \qquad G_N(0) = 1, \qquad G_N(r) = s_r \quad (0 \le r < N). \tag{B.3}$$

The marked quantum factorial is

$$\widehat{F}_N(z) = e^{\pi \mathrm{i} z(z+1)/(2N) - \pi \mathrm{i} z/2} G_N(z). \tag{B.4}$$

Consequently

$$\widehat{F}_N(z+1) = (1 - Q^{z+1})\widehat{F}_N(z), \qquad \widehat{F}_N(r) = F_r, \qquad Q^z = e^{2\pi \mathrm{i} z/N}.$$

These identities fix the continuation and its phase, not just its asymptotic expansion. On a compact subset of $0 < \Re z < 1$, direct Stirling expansion of (B.2) gives

$$\begin{aligned} \log G_N(Nz) &= NJ(z) + \frac{1}{2}\operatorname{Log}(2N\sin\pi z) + \frac{\pi\cot\pi z}{12N} + O(N^{-3}), \\ J(z) &= \int_0^z \operatorname{Log}(2\sin\pi u)\,\mathrm{d}u. \end{aligned} \tag{B.5}$$

Here the logarithm continues the real positive factorial branch. One way to verify the uniform remainder is to expand the kth Gamma ratio: its error is $O_K(N^{-3}k^{-3})$. These errors are summable; the leading and half-log terms sum by the sine product. The first correction sums by

$$z^{-1} + \sum_{k\ge 1}\bigl((k+z)^{-1} - (k-z)^{-1}\bigr) = \pi\cot\pi z.$$

Cauchy's estimate supplies the corresponding derivative bounds on smaller compact sets.

Let h be a physical color deficit and put $t = h/N$. At an integer index $r > h$ the pure core coefficient is the literal factorial expression

$$A_N(r,t) = \frac{N^2 s_{2r}}{s_r^2 G_N(r-Nt)G_N(r+Nt)}. \tag{B.6}$$

For a retained half-twist sign $\varepsilon \in \{1,-1\}$, the exact triangular Newton transport gives

$$d_N^{\varepsilon}(r,t) = \kappa^{\varepsilon}\sum_{v=r}^{M}(-1)^v q^{\varepsilon v(r+1)}\frac{s_v}{s_r s_{v-r}}A_N(v,t), \qquad d_N^0(r,t) = A_N(r,t). \tag{B.7}$$

For complex t in a sufficiently small fixed disk, only the two color-shifted factorials are continued. All their arguments remain in regular strips. Equation (B.7) is then an exact finite,

even holomorphic function. At every physical integer $h < r$ it is the original transported core, including all first normal directions.

The cancellation inside this finite sum is encoded by

$$A_{h,r} = \sum_{v=r}^{M} \frac{Q^{-vr} F_{2v}}{F_v \widehat{F}_N(v-h) \widehat{F}_N(v+h) F_{v-r}}, \tag{B.8}$$

$$D_{h,r} = \frac{Q^{-r^2} F_{2r}}{F_r \widehat{F}_N(r-h) \widehat{F}_N(r+h)}, \qquad U_{h,r} = A_{h,r}/D_{h,r}. \tag{B.9}$$

The sum starts at r in the present corridor. For physical $r < h$ its lower endpoint would be h; those low indices belong to the separate global envelope, not to the interpolation asserted here. The terminal source in (B.9) is exactly $U_{h,M} = 1$.

B.1.2 Deriving the recurrence, including its endpoint

The identity traditionally called a Ward equation here is simply a second-order recurrence obtained by telescoping the actual finite sum. A recurrence has many solutions, so we include its upper endpoint condition in the derivation. That condition is what selects the coefficient occurring in the invariant.

Put $H = Q^h = e^{2\pi i t}$, $T_h = H + H^{-1}$ and $Y = Q^r$. The literal finite sum satisfies

$$\begin{aligned} QY^2(1-Q^2Y)A_{h,r+2} + Y(T_h Q^3 Y^2 + Q + 1)A_{h,r+1} \\ - Q(QY+1)(QY^2-1)A_{h,r} = 0, \qquad A_{h,M+1} = 0. \end{aligned} \tag{B.10}$$

Here and below an absent term beyond the finite upper support is zero. We give the summation certificate, since the endpoint fixes which solution of this second-order equation is the physical one.

For the summand of (B.8), write $z = Q^v$. Its ratio under $v \mapsto v+1$ is

$$\frac{Y^{-1}(1-Qz^2)(1+Qz)}{(1-Qz/H)(1-QHz)(1-Qz/Y)}.$$

Use the rational certificate

$$g_h(z) = \frac{(1-z/H)(1-Hz)(1-z/Y)}{z^2}.$$

The telescoping difference, divided by the original summand, is

$$\left(\frac{1}{YQ^2} - 1\right) z^{-2} + \left(\frac{1}{YQ} + H + H^{-1} + \frac{1}{Y}\right) z^{-1} - \frac{1}{YQ} - 1 - \frac{H + H^{-1}}{Y}. \tag{B.11}$$

The lower endpoint has $z = Y$ and hence vanishes. The upper endpoint contains $F_{2M+2} = 0$. The ratio for the shift $r \mapsto r+1$ is $z^{-1} - Y^{-1}$. Expressing the first two z^{-1} moments in terms of $A_{h,r+1}$ and $A_{h,r+2}$ in (B.11) gives (B.10) by multiplication of its displayed denominators. This argument remains exact for complex $h = Nt$: the shifted factorial equations are exact and the lower zero is still $z = Y$. It does not use interpolation from finitely many colors.

Normalize by the actual $D_{h,r}$ and set

$$\mathrm{den}_r = (1 - QY/H)(1 - QYH).$$

Then

$$U_r = p_N(r,t)U_{r+1} + q_N(r,t)U_{r+2}, \tag{B.12}$$

where the exact rational coefficients are

$$p_N = -\frac{Q^{-2}Y^{-1}(T_hQ^3Y^2 + Q + 1)}{\mathrm{den}_r}, \tag{B.13}$$

$$q_N = -\frac{Q^{-4}Y^{-2}(1-Q^2Y)(1-Q^3Y^2)(1+Q^2Y)}{\mathrm{den}_r\,\mathrm{den}_{r+1}}. \tag{B.14}$$

All these denominators are bounded away from zero on the corridor used below. At the last recurrence row one has the exact wall

$$q_N(M-1,t) = 0, \qquad U_{M-1} = p_N(M-1,t), \qquad U_M = 1. \tag{B.15}$$

For odd $N = 2m+1$, $M = m$ and the factor $1-Q^3Y^2$ vanishes. For even $N = 2m$, $M = m-1$ and $1+Q^2Y$ vanishes. The zero is independent of H. In particular no nonexistent $D_{h,M+1}$ is divided out and no arbitrary terminal ratio is imposed.

B.1.3 The selected analytic root and a source-exact continued fraction

We solve the recurrence backwards from its finite upper endpoint. On the strict high-index interval, one characteristic root dominates the other. The resulting contraction permits a holomorphic continued fraction. Crucially, an exact zero in the last recurrence row erases the artificial tail of this continued fraction, so it agrees with the original finite solution on the integer grid.

Regard (B.13)–(B.14) as holomorphic functions $p(x,t,\varepsilon_N)$ and $q_c(x,t,\varepsilon_N)$, where $\varepsilon_N = N^{-1}$, $Y = e^{2\pi\mathrm{i}x}$ and $Q = e^{2\pi\mathrm{i}\varepsilon_N}$. At $\varepsilon_N = 0$, put

$$\begin{aligned} D_t &= (1-Y/H)(1-YH), \\ p_0 &= -Y^{-1}\frac{(H+H^{-1})Y^2+2}{D_t}, \qquad q_0 = -Y^{-2}\frac{(1-Y^2)^2}{D_t^2}. \end{aligned} \tag{B.16}$$

The two roots of $\mu^2 - p_0\mu - q_0 = 0$ at $t = 0$ are

$$\mu_s(x,0) = -Y^{-1}, \qquad \mu_f(x,0) = Y^{-1}\cot^2(\pi x). \tag{B.17}$$

On a compact real interval in $(1/4, 3/4)$ containing $[\tau_0, 1/2]$, the selected branch $\mu = \mu_s$ is nonzero, $\Re\mu > 0$, and $|\mu_f/\mu| < 1$. The same inequalities, with uniform strict constants, hold on a fixed complex neighborhood and a sufficiently small complex t disk. Indeed at $t = 0$ these assertions follow directly from (B.17); nonvanishing and a separated quadratic root persist on a small complex thickening. Write

$$\left|\frac{\mu_f}{\mu}\right| \le k_0 < 1, \qquad \Re\mu \ge m_0 > 0. \tag{B.18}$$

The endpoint $x = 1/2$ is included. Direct substitution gives $\mu(1/2,t) = 1$, while q_0 has a double zero there and its first ε_N derivative also vanishes. The selected root and its normalization are regular at the physical terminal point.

Choose $\sigma_0 > 0$ so small that $[\tau_0 - \sigma_0, 1 - \tau_0 + 4\sigma_0] \subset (1/4, 3/4)$. Use a fixed complex neighborhood of this larger interval on which (B.18) holds, and put

$$\mathcal{L}_N = \lceil (1/2 - \tau_0 + 2\sigma_0)N \rceil, \qquad \mathcal{F}_x(z) = p(x, t, \varepsilon_N) + \frac{q_c(x, t, \varepsilon_N)}{z}.$$

Define the finite continued fraction

$$v_N(x, t) = \mathcal{F}_x \circ \mathcal{F}_{x+\varepsilon_N} \circ \cdots \circ \mathcal{F}_{x+(\mathcal{L}_N-1)\varepsilon_N}(\mu(x + \mathcal{L}_N\varepsilon_N, t)). \tag{B.19}$$

This is defined on a fixed complex neighborhood of $[\tau_0, 1/2]$. For a constant C independent of N, the disks of radius $C\varepsilon_N$ around the respective selected roots are invariant under backward iteration. To prove this, note first that

$$\mathcal{F}_x(\mu(x + \varepsilon_N, t)) - \mu(x, t) = O(\varepsilon_N).$$

On those disks the derivative in the continued-fraction argument has modulus at most $k_0 + O(\varepsilon_N)$. Choose $k_1 \in (k_0, 1)$ and then C so that the $O(\varepsilon_N)$ error plus $k_1 C\varepsilon_N$ is at most $C\varepsilon_N$. This proves the invariance inductively. The disks avoid zero, so every iterate is holomorphic, nonzero, and has uniformly bounded inverse. In particular

$$v_N(x, t) = \mu(x, t) + O(N^{-1}) \tag{B.20}$$

uniformly holomorphically on a fixed smaller domain.

At a physical $x = r/N$, the continued fraction reaches the row $M - 1$. By (B.15), that row forgets its entire remaining tail and takes exactly the value $p_N(M - 1, t)$. Backward induction from the literal source $U_M = 1$ therefore proves

$$v_N(r/N, t) = \frac{U_{Nt,r}}{U_{Nt,r+1}}, \qquad \tau_0 N \le r < M. \tag{B.21}$$

This also proves that the displayed physical source values are nonzero. The equality holds uniformly for complex t in the fixed disk. There is no assertion about a nonexistent physical ratio at $r = M$.

Adding one more future step to (B.19) changes the future value by $O(\varepsilon_N)$, and the preceding $\mathcal{L}_N$ contractions reduce that change by $k_1^{\mathcal{L}_N}$. Thus its Riccati defect is exponentially small. This explains why the analytic tail used to define the interpolant cannot hide a macroscopic change of the original source.

B.1.4 First transport correction and absolute amplitude normalization

Let $p_1 = \partial_\varepsilon p|_{\varepsilon=0}$ and $q_1 = \partial_\varepsilon q_c|_{\varepsilon=0}$, differentiating the explicit rational coefficients. Define

$$\nu = \frac{p_1 + q_1/\mu - q_0\mu_x/\mu^2}{1 + q_0/\mu^2}. \tag{B.22}$$

The denominator is $1 - \mu_f/\mu$, bounded away from zero by (B.18). Substitution of $\mu + \varepsilon_N\nu$ into the one-step Riccati equation gives a residual $O(\varepsilon_N^2)$. The same contracting backward iteration then gives

$$v_N = \mu + N^{-1}\nu + O(N^{-2}) \tag{B.23}$$

on a fixed smaller complex neighborhood. For example, subtract the trial iterate from the exact one at each step. The resulting error is bounded by the geometric sum of its $O(N^{-2})$ local defects and the contracted terminal discrepancy. Cauchy estimates give the corresponding uniform derivative expansions.

Use the logarithm with positive real part in (B.18), continuing from $\mu(1/2,t)=1$, and set

$$\begin{aligned} S(x,t) &= \int_x^{1/2} \operatorname{Log}\mu(u,t)\,\mathrm{d}u, \\ C(x,t) &= \sqrt{\mu(x,t)}\exp\left(\int_x^{1/2}\frac{\nu(u,t)}{\mu(u,t)}\,\mathrm{d}u\right). \end{aligned} \tag{B.24}$$

The square root is one at $1/2$. The integral branches are fixed by the actual selected source; C is holomorphic and nonzero.

Summing logarithms of the exact ratios in (B.21), using (B.23) and the first Euler–Maclaurin endpoint term, gives

$$U_{Nt,r} = C(r/N,t)e^{NS(r/N,t)}[1+O(N^{-1})]. \tag{B.25}$$

The product stops at $M-1$, because $U_M=1$. Replacing the true endpoint M/N by $1/2$ costs only $O(N^{-1})$: the displacement is $O(N^{-1})$ at either parity, and $\operatorname{Log}\mu(1/2,t)=0$. The remaining endpoint terms in the correction are likewise $O(N^{-1})$. In particular there is no free terminal matching constant.

At $t=0$, direct differentiation of (B.13)–(B.14) yields

$$\operatorname{Log}\mu = \pi\mathrm{i} - 2\pi\mathrm{i}x, \qquad \frac{\nu}{\mu} = -\frac{\pi}{2}\cot(\pi x) - \frac{5\pi\mathrm{i}}{2}.$$

Therefore

$$S(x,0) = \pi\mathrm{i}(1/2-x)^2, \qquad C(x,0) = \sqrt{\sin\pi x}\,e^{-3\pi\mathrm{i}(1/2-x)/2}. \tag{B.26}$$

This calculation fixes the absolute amplitude from the finite source, rather than from the value of an unknown closed invariant.

B.1.5 Exponential interpolation before an oscillatory saddle

Equation (B.25) is not sufficient by itself for a complex wire saddle. Its relative error is only inverse polynomial and could exceed a fixed oscillatory cancellation. The exact ratio (B.21) supplies the needed stronger estimate.

Put $f_N(x,t)=\operatorname{Log}v_N(x,t)$ and $b_N=M/N$. On a fixed larger complex tube the f_N are uniformly bounded and holomorphic. On an inner tube, Cauchy's estimate gives constants $B,a>0$, independent of N,t, such that

$$|\partial_x^k f_N(x,t)| \le Bk!a^{-k}. \tag{B.27}$$

For $K_N=\lfloor c_aN\rfloor$, with a sufficiently small fixed $c_a>0$, form the holomorphic expression

$$\begin{aligned} E_N(x,t) = N\int_x^{b_N} f_N(u,t)\,\mathrm{d}u + \frac{f_N(x,t)-f_N(b_N,t)}{2} \\ + \sum_{k=1}^{K_N}\frac{B_{2k}}{(2k)!}N^{1-2k}\left[\partial_x^{2k-1}f_N(b_N,t) - \partial_x^{2k-1}f_N(x,t)\right]. \end{aligned} \tag{B.28}$$

The Bernoulli polynomial bound $|B_{2k}(u)| \le 4(2k)!/(2\pi)^{2k}$ together with (B.27) bounds the Euler–Maclaurin remainder by

$$CN\left(\frac{K_N}{\pi aN}\right)^{2K_N} \le Ce^{-cN}$$

after decreasing c_a. This is uniform in the complex color disk and on the physical high grid. Since the product of the exact ratios is precisely U_r, it follows that

$$U_{Nt,r} = e^{E_N(r/N,t)}[1 + R_{N,r}(t)], \qquad |R_{N,r}(t)| \le Ce^{-cN}. \tag{B.29}$$

The empty terminal product equals one. The residual is retained as a source term, rather than reinterpreted as a homogeneous solution.

The same explicit expression is holomorphic in the index x. Using (B.23) in (B.28), the higher Bernoulli terms sum to $O(N^{-1})$ on an inner fixed tube, and hence

$$e^{E_N(x,t)} = C(x,t)e^{NS(x,t)}[1 + O(N^{-1})] \tag{B.30}$$

uniformly holomorphically. Equations (B.29) and (B.30) have distinct roles: the former is an exponentially accurate comparison with the original finite source, and the latter is a leading expansion of a specified complex function.

B.1.6 Keeping the quadratic phase and determining the signed action

The half-twist coefficient contains a quadratic exponential in the color. Such a phase is sometimes called a chirp. Its continuation is part of the finite formula, not an arbitrary choice of interpolation. We now combine it with the recurrence solution and the continued factorials; this calculation determines the local action and the nonzero amplitude used in the saddle.

Remove only the known external phase by defining

$$\ell_N^{\varepsilon}(r,t) = e^{-\mathrm{i}\varepsilon\pi d/4}d_N^{\varepsilon}(r,t), \qquad \eta_N = \exp\left(\frac{\mathrm{i}\pi d}{4}\sum_{\ell}\varepsilon_\ell\right).$$

The positive sign is the coefficientwise conjugate of the negative sign on real parameters. Its holomorphic continuation uses Schwarz reflection, not conjugation of a free holomorphic variable.

The original factorial phase gives the exact identity

$$q^{h^2}\frac{D_{h,r}}{D_{0,r}} = \frac{G_N(r)^2}{G_N(r-h)G_N(r+h)}. \tag{B.31}$$

It follows immediately from (B.4); the linear and quadratic exponents in the two shifted arguments must both be kept. Combining (B.7)–(B.9) with (B.31) gives the negative-sign interpolant

$$\begin{aligned} A_N^{\mathrm{hol}}(x,t) &= \frac{N^2 G_N(2Nx)}{G_N(Nx)^2 G_N(N(x-t))G_N(N(x+t))}, \\ \nu_N(z) &= e^{-\pi\mathrm{i}z+\pi\mathrm{i}z(z+1)/N}, \qquad P_N^{\mathrm{ch}}(x) = \frac{\kappa^{-1}e^{\pi\mathrm{i}d/4}}{\nu_N(Nx)}, \\ I_N^{-}(x,t) &= A_N^{\mathrm{hol}}(x,t)P_N^{\mathrm{ch}}(x)e^{E_N(x,t)}. \end{aligned} \tag{B.32}$$

The superscript on P_N^{ch} distinguishes this chirp factor from the Casimir polynomial. At every integer r, $\nu_N(r) = (-1)^r q^{r(r+1)}$. Its displayed analytic continuation is fixed before any contour deformation. In particular,

$$P_N^{\mathrm{ch}}(x) e^{NS(x,0)} = e^{\pi \mathrm{i}(3/4 - x - 1/(2N))}. \tag{B.33}$$

Changing the continuation of $(-1)^r$ would change a lattice frequency; that change is not permitted. Equation (B.29) proves that $I_N^-(r/N, t)$ agrees with the actual $\ell_N^-(r, t)$ up to a relative exponential residual. For the positive sign set $I_N^+(x,t) = \overline{I_N^-(\bar{x}, \bar{t})}$; for the pure sign use $I_N^0 = A_N^{\mathrm{hol}}$, which agrees exactly on the grid.

Define

$$\begin{aligned} g(x,t) &= J(2x) - 2J(x) - J(x-t) - J(x+t), \\ f_0(x,t) &= g(x,t), \qquad f_-(x,t) = g(x,t) + S(x,t) - S(x,0), \\ f_+(x,t) &= \overline{f_-(\bar{x}, \bar{t})}. \end{aligned} \tag{B.34}$$

The factorial expansion and (B.33) yield

$$I_N^{\varepsilon}(x,t) = N^{1/2} b_{\varepsilon}(x,t) e^{N f_{\varepsilon}(x,t)} [1 + O(N^{-1})], \tag{B.35}$$

uniformly holomorphically on fixed smaller domains, with

$$\begin{aligned} b_0(x,t) &= \frac{\sqrt{\sin(2\pi x)}}{2^{3/2} \sin(\pi x) \sqrt{\sin(\pi(x-t)) \sin(\pi(x+t))}}, \\ b_-(x,t) &= b_0(x,t) e^{\pi \mathrm{i}(3/4 - x)} C(x,t), \qquad b_+(x,t) = \overline{b_-(\bar{x}, \bar{t})}. \end{aligned} \tag{B.36}$$

All square roots continue their positive complete factorial branches. At completeness,

$$\frac{b_{\varepsilon}(x,0)}{b_0(x,0)} = \sqrt{\sin(\pi x)} e^{-\mathrm{i}\varepsilon \pi x/2}, \qquad \varepsilon \in \{-1, 1\}. \tag{B.37}$$

These factors are nonzero. The remaining $e^{-\pi \mathrm{i}/(2N)}$ in (B.33) belongs to the stated inverse-power error; it is not an omitted constant or parity phase.

B.1.7 An exact source action identity and its Hessian

The action above can also be characterized without integrating the quadratic root. For each physical sign $\varepsilon \in \{0, 1, -1\}$ solve near $s = t = 0$ the equation

$$t = s - \frac{\varepsilon}{2\pi \mathrm{i}} g_s(x,s). \tag{B.38}$$

Its s derivative at zero is $1 - \varepsilon \mathrm{i} \cot(\pi x)$, which is nonzero. Then the source-selected action satisfies

$$f_{\varepsilon}(x,t) = g(x,s) - \frac{\varepsilon}{4\pi \mathrm{i}} g_s(x,s)^2, \qquad (f_{\varepsilon})_x = g_x(x,s), \qquad (f_{\varepsilon})_t = g_s(x,s). \tag{B.39}$$

The derivative identities follow by differentiating (B.38). To identify the value with the actual selected source, consider the negative sign. Put $E = e^{2\pi \mathrm{i} x}$, $H = e^{2\pi \mathrm{i} t}$ and $T = e^{2\pi \mathrm{i} s}$. Exponentiating (B.38) gives

$$H = \frac{T(E - T)}{ET - 1}, \qquad T^2 + E(H-1)T - H = 0, \qquad T = 1 \text{ at } H = 1.$$

For $K = e^{-g_x(x,s)} = -(1 - E/T)(1 - ET)/(1 + E)^2$, elimination of T gives

$$K^2(1 + E)^2 + K[(H + H^{-1})E^2 + 2] + (1 - E)^2 = 0. \tag{B.40}$$

On the other hand (B.34) and the selected quadratic give $e^{-(f_-)_x} = E\mu D_t/(1 + E)^2$, satisfying exactly (B.40). At $H = 1$, both constructions take the branch $K = \tan^2(\pi x)$; their logarithms agree by their complete marks. Hence their x derivatives agree. The possible remaining function of t is zero by the limit $x \to 1/2$: there $S = 0$, $J(1/2 + t) = -J(1/2 - t)$, and the implicit shear has $s = t$, $g_s = 0$. Schwarz reflection proves the positive sign. This fixes the source action, not merely a Hessian with the same entries.

In particular, all local actions are even in t, equal $g(x, 0)$ at zero, and

$$(f_\varepsilon)_{tt}(x, 0) = -\frac{2\pi \cot(\pi x)}{1 - \varepsilon \mathrm{i} \cot(\pi x)}, \qquad \varepsilon \in \{0, 1, -1\}. \tag{B.41}$$

Their real parts are strictly negative on the fixed high interval.

B.1.8 The exact finite wire window and its lattice saddle

We now specify the function to which the complex saddle argument is applied. Let the fixed completed-root expression have H leaves and $K = 2H - 3$ primitive free integer wires. The last forced product is removed using its exact relation $i + j = d$, whose coefficient is one. Every remaining wire has the affine form

$$r_w = NL_w(y) + \delta_w, \qquad y = k/N, \quad k \in \mathbb{Z}^K,$$

where the integer offsets δ_w are fixed by the original compiler. Take a small fixed rectangular window B about the strict complete optimizer. All leaf coordinates lie in intervals of the form (B.1), and every retained gate inequality is strict on a fixed neighborhood of the closure of B.

The exact finite window is

$$\mathcal{C}_N(\mathbf{t}) = \sum_{k/N \in B} \prod_{\ell=1}^{H} \ell_N^{\varepsilon_\ell}(r_\ell, t_\ell) \prod_{g=1}^{H-2} \mu_N(r_{g,1}, r_{g,2}, r_{g,3}). \tag{B.42}$$

Its support is independent of the complex colors. It is a finite, separately even holomorphic function. For all sufficiently large N, every state in this window is a true core state at every physical small color deficit. The factors μ_N are the original positive Newton coefficients, not their leading asymptotic replacements.

Replace each exact local factor by its canonical holomorphic interpolant only after keeping its residual. The gate interpolation is the same seven-factorial expression with G_N, and agrees exactly on its integer grid. For the signed leaves, use (B.32). The total error in the finite sum is bounded absolutely by

$$\mathrm{poly}(N) \exp\{N(F_0 + C\|\mathbf{t}\|^2 - c)\} \tag{B.43}$$

for some fixed $c > 0$. Indeed there are polynomially many states, all actions are even analytic perturbations of the common complete real action, and the relative leaf errors in (B.29) are exponentially small. This estimate is made before any oscillatory cancellation. Shrink the fixed color polydisk so that (B.43) is exponentially below the nearby selected critical action.

The interpolated summand has phase

$$\mathcal{F}(y,\mathbf{t}) = \sum_{\ell} f_{\varepsilon_\ell}(L_\ell(y), t_\ell) + \sum_{g} m(L_{g,1}(y), L_{g,2}(y), L_{g,3}(y)). \tag{B.44}$$

At zero color it is the original complete action. Its wire Hessian at the strict optimizer is negative definite. Thus it has one holomorphic critical branch $y_*(\mathbf{t})$, with

$$\Phi(\mathbf{t}) = \mathcal{F}(y_*(\mathbf{t}), \mathbf{t}).$$

The local gate amplitude is

$$b_\mu(x,y,z) = 2^{-3/2}\sqrt{\frac{\sin\pi(x+y)\sin\pi z}{\sin\pi x\sin\pi y\sin\pi(z-x)\sin\pi(z-y)\sin\pi(x+y-z)}}.$$

Multiply all leaf and gate amplitudes by the exact leading offset factor

$$\exp\left[\sum_{\ell}\delta_\ell\partial_x f_{\varepsilon_\ell} + \sum_{g}\nabla m(L_g(y))\cdot\delta_g\right] \tag{B.45}$$

to obtain the nonzero holomorphic density $b_E(y,\mathbf{t})$. The changed f_ε is used in (B.45), including at nonzero complex color; a pure offset factor would be incorrect. The interpolated summand is

$$N^{3-H} b_E(y,\mathbf{t}) e^{N\mathcal{F}(y,\mathbf{t})}[1+O(N^{-1})].$$

For clarity we give the finite-lattice contour step. Choose the vertical faces of the rectangle at half-grid values. In each coordinate the exact summation kernel is $\pi N\cot(\pi N z)$, with residue one at $z = k/N$. On a vertical half-grid line, $|\cot(\pi N z)| \le 1$. Those faces lie away from the strict real optimizer and retain a fixed phase gap. On a horizontal line of fixed height $a > 0$, the cotangent differs from its constant value $\mp\mathrm{i}$ by $O(e^{-2\pi N a})$. At complete color, the real part of the phase grows under this small imaginary displacement by at most Ca^2; the even color perturbation costs at most $C\|\mathbf{t}\|^2$. Choose a so $Ca^2 < \pi a$, and then choose the color polydisk small enough. Every nonconstant kernel term has an exponential gap, including all side connectors. Applying this argument successively in the K primitive coordinates retains the full product contour and its orientations. The constant horizontal terms give N^K times the ordinary integral. Replacing the moving half-grid rectangle by B costs only exponentially small boundary strips.

Translate that integral to the nearby complex critical point. The endpoint connectors remain below its action, and the real tangent Gaussian is uniformly nondegenerate. Extend its local Gaussian range with exponentially small error. The odd first correction integrates to zero, giving a relative $O(N^{-1})$ expansion. Thus

$$\mathcal{C}_N(\mathbf{t}) = N^{3/2} a_E(\mathbf{t}) e^{N\Phi(\mathbf{t})}[1+O(N^{-1})], \qquad a_E(\mathbf{t}) = \frac{(2\pi)^{K/2} b_E(y_*(\mathbf{t}),\mathbf{t})}{\sqrt{\det[-\mathcal{F}_{yy}(y_*(\mathbf{t}),\mathbf{t})]}}. \tag{B.46}$$

The determinant square root is selected from the positive complete Gaussian. The power is $3 - H + K/2 = 3/2$. The exponential grid error (B.43) is smaller than this answer by the prior choice of color disk, and is added back only now. Both level parities have the same primitive density and displayed normalization.

At completeness,

$$\Phi(0) = F_0, \qquad a_E(0) = a_E^{\mathrm{pure}}(0) \prod_{\varepsilon_\ell \neq 0} \sqrt{\sin(\pi x_{\ell,*})} e^{-\mathrm{i}\varepsilon_\ell \pi x_{\ell,*}/2} \neq 0. \tag{B.47}$$

Evenness gives $\mathcal{F}_{yt_\ell}(y,0) = 0$, so the wire Schur correction to the color Hessian vanishes. With $c_\ell = \cot(\pi x_{\ell,*})$, one obtains

$$\partial_{t_\ell}\partial_{t_j}\Phi(0) = -\frac{2\pi c_\ell}{1-\varepsilon_\ell \mathrm{i} c_\ell}\delta_{\ell j} = 2\pi \mathrm{i}\tau_\ell \delta_{\ell j}, \qquad \tau_\ell = \frac{\mathrm{i}c_\ell}{1-\varepsilon_\ell \mathrm{i}c_\ell}, \qquad \Im\tau_\ell > 0. \tag{B.48}$$

On a smaller fixed complex box,

$$|\mathcal{C}_N(u+\mathrm{i}v)| \le \mathrm{poly}(N) e^{N(F_0 - a\|u\|^2 + B\|v\|^2)}, \qquad a > 0. \tag{B.49}$$

This bound is for the exact finite function (B.42), by (B.46), not just its leading formal action.

B.1.9 Returning from the local sum to the original evaluation

We finish by stating precisely how the independent alias estimates are used. The compositional proof in the main text gives a uniform positive gap for every nonempty alias sector on the entire physical color cube. It also gives a concave all-core majorant with the unique strict complete maximizer used above. The latter uniqueness is global: it agrees with the strict complete action near that point, and a second maximum of a concave function would give a constant segment through that strictly concave neighborhood.

Hence all core states outside the fixed wire window have a fixed gap, uniformly in physical colors. Within that window the actual local far-color recurrence estimate loses a fixed positive amount whenever one physical deficit exceeds a fixed small corridor. Polynomial state counting and the finite number of sectors therefore give constants $\delta, \gamma > 0$ such that the literal mixed scalar $\mathcal{A}_N$ obeys

$$\max_{0 \le h_\ell \le \delta N} |\mathcal{A}_N(\mathbf{h}) - \eta_N \mathcal{C}_N(\mathbf{h}/N)| \le \mathrm{poly}(N) e^{N(F_0 - \gamma)}, \tag{B.50}$$

$$\max_{\exists \ell:\, h_\ell > \delta N} |\mathcal{A}_N(\mathbf{h})| \le \mathrm{poly}(N) e^{N(F_0 - \gamma)}. \tag{B.51}$$

The maximum in the second line is over the entire original physical color cube, not just the low-core band. No alias is folded into the even analytic packet. The external η_N is exactly the retained unit phase defined above.

The global coherent-core input here is its uniform exponential-rate envelope, of the form $\exp\{N(\widetilde{g} + o(1))\}$ with the $o(1)$ uniform over all physical indices and colors. It is not a polynomial-prefactor assertion through every spectral collision. To obtain (B.50)–(B.51), first fix the strictly positive variational and far-color margins, then absorb the uniform $o(N)$ errors into, for example, half their minimum. Polynomial counting and the remaining finite prefactors are harmless after that reduction of γ. The polynomial-prefactor local packet proved above is a separate estimate on its fixed regular high interval.

These estimates preserve the whole-source remainder through the finite filling sum. For each fixed signed filling tuple, the exact Hermite coefficient bound is $\sum_j |\beta_{j-1}^{-p_\ell}| \le 2N(1+|p_\ell|)$. Thus (B.50) and (B.51) incur only a polynomial multiplier in that whole sum. The finite analytic

packet, rather than its pointwise leading approximation, is then used for the Fourier endpoint and charge arguments. Its exact evenness, complex estimate (B.49), nonzero amplitude and Hessian (B.48) are the needed inputs. The retained filling analysis supplies the smaller filled action and a nonzero leading coefficient; the source gap is not itself a replacement for that analysis.

The argument in this section requires (B.1). No continuation through a quarter-index dominance boundary, no additional root-leaf closure, and no theorem for arbitrary tangle syntax is used in the manuscript's completed-root result.

B.2 A global coefficient bound, including repeated recurrence roots

The local interpolation of Section B.1 is used only on a fixed high-index interval. Here we prove the different estimate needed before restricting to that interval: a uniform exponential-rate bound for every physical color and every core index. The proof retains the other recurrence branch, including its repeated-root colors. Here C_j^{ε} always denotes the core polynomial of a single physical leaf of color j and half-twist sign ε, as in (2.17); it does not denote the whole analytic packet.

B.2.1 Normalization and the finite upper endpoint

Use q, Q, d, M, F_r as in Section B.1. For a physical integer $0 \le h \le M$, interpret the finite sum $A_{h,r}$ in (B.8) with lower endpoint $\max(r, h)$. Thus all its factorials have nonnegative arguments. Set

$$V_{h,r} = \frac{q^{h^2} A_{h,r}}{D_{0,r}}, \qquad \chi = \frac{\pi h}{N}, \qquad H = Q^h, \qquad Z = Q^{r+1}.$$

The denominator is the complete diagonal, which is nonzero at every $0 \le r \le M$. The Ward identity, or its polynomial version before cancellation of zero endpoint factors, gives

$$\begin{aligned} V_{h,r} &= p_{h,N}(r) V_{h,r+1} + q_N(r) V_{h,r+2}, \\ p_{h,N}(r) &= -\frac{Z^{-1}[(H + H^{-1})Z^2 + 1 + Q^{-1}]}{(1-Z)^2}, \\ q_N(r) &= -\frac{Q^{-2} Z^{-2}(1 - QZ^2)(1 + QZ)}{(1-Z)^2(1-QZ)}. \end{aligned} \tag{B.52}$$

It holds also when $r < h$: the omitted lower terms of the polynomial Ward telescope have their zero numerator factors, rather than divided negative factorials. More explicitly, the certificate $g_h(z)$ in Section B.1 vanishes at the lower endpoint $z = Q^{\max(r,h)}$: its factor $1 - z/Y$ applies when $r \ge h$, and its factor $1 - z/H$ applies when $r < h$. At $r = M - 1$ one has $q_N(r) = 0$ at both parities. We may therefore set the unused auxiliary value $V_{h,M+1} = 0$. The actual remaining terminal value obeys the exact identity

$$|V_{h,M}| = \frac{s_M^2}{s_{M-h} s_{M+h}} = \begin{cases} 1, & N \text{ odd}, \\ \cos\chi, & N \text{ even}. \end{cases} \tag{B.53}$$

Indeed $s_a s_{N-1-a} = N$. For odd $N = 2M+1$, both products in (B.53) equal N. For even $N = 2m$, $M = m-1$, $s_M^2 = N/2$ and $s_{M-h}s_{M+h} = N/(2\cos\chi)$. Thus the source at the wall has modulus at most one uniformly over the entire physical color range.

The finite triangular transport and the marked factorial phases give the following exact modulus in the normalized Newton basis:

$$\left|[\widehat{B}_r]C^{\varepsilon}_{N-h}\right| = \frac{N^2 s_{2r}}{s_r^4}|V_{h,r}|, \qquad 0 \le r \le M, \quad \varepsilon \in \{-1,1\}. \tag{B.54}$$

One can obtain this directly by interchanging the two finite sums in (B.7); the coefficient in the unnormalized basis B_r is $N^2 s_{2r} V_{h,r}/s_r^6$ times a unit phase. Multiplication by s_r^2 gives (B.54). The positive sign is the conjugate of the negative sign on the physical grid. Phases have been used only to compute the modulus, not deleted from the finite source.

B.2.2 The spectral ellipse

The characteristic roots of the limiting two-by-two recurrence matrix control the exponential size of a long matrix product. They may coincide for some colors. We first locate both roots using a quadratic equation and an ellipse in its coefficient plane. The next subsection bounds powers of the matrix directly, so no singular eigenvector basis is required at a collision.

Let $T_N(r)$ be the companion matrix of (B.52). Thus

$$(V_{h,r}, V_{h,r+1})^t = T_N(r)(V_{h,r+1}, V_{h,r+2})^t.$$

Freeze $u = (r+1)\pi/N$ and let $Q \to 1$ while $Z = e^{2\mathrm{i}u}$ and $H = e^{2\mathrm{i}\chi}$ stay fixed. The limiting matrix $T_0(u,\chi)$ has eigenvalues $Z^{-1}v$, where

$$v^2 - bv - c = 0, \qquad c = \cot^2 u, \qquad b = (c-1)\cos^2\chi - 2\mathrm{i}\sqrt{c}\,\sin^2\chi. \tag{B.55}$$

We claim

$$\rho(T_0(u,\chi)) \le \rho_0(u) := \max(1, \cot^2 u), \qquad 0 < u \le \pi/2, \quad 0 \le \chi \le \pi/2. \tag{B.56}$$

For $0 < c < 1$, a root on the unit circle would give $b = v - c/v$. The unit-root locus in the b plane is therefore the ellipse

$$\left(\frac{\Re b}{1-c}\right)^2 + \left(\frac{\Im b}{1+c}\right)^2 = 1.$$

For the value in (B.55), its left side is

$$\cos^4\chi + \frac{4c}{(1+c)^2}\sin^4\chi \le 1,$$

strictly less than one when $\chi > 0$. At $b = 0$ the roots have modulus $\sqrt{c} < 1$. The line segment to an interior point of this ellipse cannot cross the unit-root locus, so both roots remain in the open unit disk. Continuity treats the ellipse boundary. For $c > 1$, apply the same argument to v/c, with parameter $1/c$. At $c = 1$ the roots of $v^2 + 2\mathrm{i}\sin^2\chi\, v - 1$ have modulus one. At $c = 0$ they are 0 and $-\cos^2\chi$. This proves (B.56) without selecting an eigenbasis.

In particular, if $\chi_0 > 0$ and $I \subset (\pi/4, \pi/2)$ is a fixed closed interval, then

$$\sup_{\substack{u\in I\\ \chi_0\le\chi\le\pi/2}} \rho(T_0(u,\chi)) \le \theta < 1. \tag{B.57}$$

The endpoint color $\chi = \pi/2$ is included: there the two roots coalesce, but their common modulus is $\sqrt{c} < 1$.

B.2.3 Frozen blocks and the singular lower endpoint

Over a fixed number of consecutive indices, the exact recurrence matrix changes by $O(N^{-1})$. We compare that short product with a power of one constant matrix. The block length is chosen first and N is then taken large. A separate factorial bound treats the small indices where the coefficients are singular. This order of estimates gives a uniform exponential rate without asserting a polynomial prefactor at every color.

Define

$$E(u) = \begin{cases} \dfrac{2}{\pi}\displaystyle\int_u^{\pi/4} \log\cot v\,\mathrm{d}v, & 0 \le u < \pi/4,\\ 0, & \pi/4 \le u \le \pi/2. \end{cases} \tag{B.58}$$

We prove the two uniform assertions

$$\limsup_{N\to\infty} \max_{0\le h,r\le M} \left\{ N^{-1}\log|V_{h,r}| - E(\pi r/N)\right\} \le 0, \tag{B.59}$$

$$\limsup_{N\to\infty} \max_{\substack{\delta N\le h\le M\\ 0\le r\le\beta N}} \left\{ N^{-1}\log|V_{h,r}| - E(\pi r/N)\right\} \le -\gamma_{\delta,\beta}, \tag{B.60}$$

where $0 < \delta < 1/2$, $0 < \beta < 1/2$, and $\gamma_{\delta,\beta} > 0$. We use $\log 0 = -\infty$.

First fix a lower cutoff $a > 0$. On $a \le u \le \pi/2$ the exact coefficients and their first u derivatives are uniformly bounded, independently of the physical color. The exact normalized matrices $B_N(r) = T_N(r)/\rho_0(u_r)$, $u_r = (r+1)\pi/N$, differ by $O_a(N^{-1})$ from the bounded Lipschitz family $B(u_r,\chi) = T_0(u_r,\chi)/\rho_0(u_r)$.

For any bounded 2 by 2 matrix B whose eigenvalues have modulus at most one, Cayley–Hamilton gives

$$\|B^L\| \le L\|B\| + L - 1 \le C_a L. \tag{B.61}$$

The coefficient of B is $\sum_{j=0}^{L-1}\lambda_1^{L-1-j}\lambda_2^j$, of modulus at most L, and the constant coefficient has modulus at most $L-1$. These expressions remain valid at a double root. If both roots instead have modulus at most $\theta < 1$, the same calculation gives $\|B^L\| \le C_a L\theta^{L-1}$.

Choose a fixed block length L. Telescoping a product of L adjacent exact normalized matrices against its frozen value gives

$$B_N(r)\cdots B_N(r+L-1) = B(u_r,\chi)^L + O_{a,L}(N^{-1}),$$

uniformly in χ. For a prescribed $\epsilon > 0$, choose L so that $C_a L < e^{\epsilon L}$, and then take N sufficiently large. Partition any propagation product into these blocks. The incomplete end blocks cost a fixed constant, and all ordinary blocks together cost at most $C_{a,\epsilon}e^{2\epsilon N}$.

For the strict estimate, choose $I \subset (\max(\pi\beta, \pi/4), \pi/2)$. Every propagation from the physical wall to $r \le \beta N$ crosses this interval. By (B.57), increase the same fixed L until

each full block in I has norm at most $e^{-\gamma_0 L}$ for some $\gamma_0 > 0$. There are $N|I|/(\pi L) + O(1)$ such blocks. Choose $\epsilon < \gamma_0|I|/(8\pi)$. Their strict negative budget then exceeds the cost of all ordinary blocks.

Only $u = 0$ has a singular coefficient. Directly from (B.52), for $u_r \le a < \pi/8$,

$$\|T_N(r)\| \le C\frac{N^2}{(r+1)^2}. \tag{B.62}$$

Here $|H + H^{-1}| \le 2$, $|1 - Z| \ge 4(r+1)/N$, and the ratio $|1 - QZ^2|/|1 - QZ|$ is bounded by an absolute constant in this range. If $K \le aN/\pi$, the product of the bounds in (B.62) is at most $C^K N^{2K}/(K!)^2$. Hence the entire omitted initial segment costs

$$\exp\{O(Na\log(1/a)) + O(\log N)\},$$

uniformly in its starting index. This cost has rate tending to zero as $a \downarrow 0$.

Finally $\log \rho_0(u)$ is decreasing until it reaches zero, so the right-endpoint Riemann sum satisfies

$$\sum_{j=r}^{M-1} \log \rho_0((j+1)\pi/N) \le NE(\pi r/N). \tag{B.63}$$

Combine this scalar product with (B.53), the frozen blocks, and the endpoint bound. For (B.59), take $N \to \infty$ first and then let $a, \epsilon \downarrow 0$. For (B.60), first fix I and its negative budget, then make $a\log(1/a)$ and ϵ small compared with that budget, and finally take N large. This proves both estimates with their stated uniformity. There is no adiabatic eigenvector argument and no polynomial bound claimed at a spectral collision.

B.2.4 The actual coefficient envelopes and the macroscopic color loss

The elementary sine-factorial estimate is uniform up to both endpoints:

$$\log s_j = NJ(j/N) + O(\log N), \qquad 0 \le j < N. \tag{B.64}$$

For example, subtract $\log t + \log(1-t)$ from $\log(2\sin \pi t)$ on $[0,1]$. The remainder is continuously differentiable, so its Riemann-sum error is bounded; the two removed terms are ordinary factorials and have $O(\log N)$ errors by Stirling's inequalities. This also proves the uniform assertion when one of the factorial indices is zero.

The complete prefactor in (B.54) has rate $g(x) = J(2x) - 4J(x)$. Since $g'(x) = 2\log\cot \pi x$,

$$g(x) + E(\pi x) = g(\max(x, 1/4)) = \widetilde{g}(x). \tag{B.65}$$

Consequently (B.59) supplies a sequence $e_N \downarrow 0$ after taking its decreasing upper envelope, such that

$$|[\widehat{B}_r]C^{\varepsilon}_{N-h}| \le \exp\{N[\widetilde{g}(r/N) + e_N]\}, \qquad 0 \le h, r \le M, \quad \varepsilon \in \{-1, 1\}.$$

The same estimate holds for the pure sign. Its coefficient is zero for $r < h$ and otherwise is $N^2 s_{2r}/(s_r^2 s_{r-h} s_{r+h})$. Its rate is $g(x) - \Delta(x,t)$, where

$$\Delta(x,t) = J(x-t) + J(x+t) - 2J(x), \qquad 0 \le t \le x \le 1/2.$$

For $t > 0$ and $x < 1/2$,

$$\partial_t \Delta = \log \frac{\sin \pi(x+t)}{\sin \pi(x-t)} > 0,$$

because the difference of the two sines is $2 \cos \pi x \sin \pi t > 0$. Continuity includes the integrable boundary $t = x$. Thus pure cores have a uniform strict loss when x lies in a fixed compact high interval and $t \geq \delta > 0$. For signed cores the same conclusion follows from (B.60), choosing β above that high interval. Both parities, the largest physical color, and all zero coefficients are retained.

For completeness, the alias bound in (5.1) requires no recurrence. We first prove the finite bound on the untransformed alias. Write $\theta = \pi/N$. Its coefficient from (2.18), for $1 \leq j \leq \lfloor N/2 \rfloor$ and $0 \leq r < j$, has modulus

$$b_{j,r} = \frac{s_r^2}{s_{2r+1}} \prod_{u=-r}^{r} 2 \sin((j+u)\theta).$$

As a function of real j in $r < j < N - r$, this positive product is log concave and symmetric about $N/2$, so its maximum is there. Cancelling $(2 \sin u\theta)(2 \cos u\theta) = 2 \sin 2u\theta$ gives the upper bound

$$B_r = 2 \frac{\prod_{u=1}^{r} 2 \sin(2u\theta)}{\prod_{u=0}^{r} 2 \sin((2u+1)\theta)}, \qquad \frac{B_{r+1}}{B_r} = \frac{\sin((2r+2)\theta)}{\sin((2r+3)\theta)}.$$

The ratio is at most one and then at least one, with only one possible switch, at $(4r+5)\theta = \pi$. Therefore the maximum over $0 \leq r \leq \lfloor N/2 \rfloor - 1$ is at an endpoint. The first is $B_0 = \csc \theta$. At the other endpoint, the even and odd sine products give $B_{N/2-1} = N/2$ when N is even; when $N = 2m + 1$, reflection of the sine factors gives $B_{m-1} = \csc \theta$. The even product identities follow by evaluating the factorizations of $z^m - 1$ and $z^m + 1$ at $z = 1$. Finally $\sin(\pi/N) \geq 2/N$. We have proved $b_{j,r} \leq N/2$ at every color and both parities.

A triangular half-twist entry has modulus $s_v/(s_r s_{v-r})$, so (B.64) gives the rate

$$J(w) - J(u) - J(w-u), \qquad u = r/N, \quad w = v/N \geq \max(u, 1/2).$$

For $0 < u < 1$ its derivative in w is $\log[\sin(\pi w)/\sin(\pi(w-u))]$; it changes from positive to negative at $w = (1+u)/2$. Reflection $J(1-z) = -J(z)$ and duplication give the maximum

$$J((1+u)/2) - J(u) - J((1-u)/2) = -2J(u/2) = k(u).$$

The two endpoint cases follow by continuity. All factorial errors are $O(\log N)$, and there are at most N summands. This proves the alias bound with a polynomial prefactor, on its full original index range $0 \leq r \leq d$.

To pass from these leaf estimates to the complete source, first fix the positive compositional alias margins and the compact core-state gap in Chapter 5. Every fixed tree has finitely many leaves and gates and only polynomially many lattice states. Absorb their uniform e_N and the $O(\log N)/N$ gate errors into half the fixed margin. Inside the remaining strict state window, the macroscopic-color loss just proved supplies the other half of the argument. This gives the whole-source bounds (B.50)–(B.51). The result is a uniform exponential-rate envelope at all colors; the polynomial leading packet remains the separate regular local theorem of Section B.1.

B.3 Nonlocal filling charges and local contour connectors

We give the additional contour details used after the infinite tails have been removed in Chapter 8. All constants below are chosen from one fixed source neighborhood. They do not depend on the relative sizes of different filling slopes.

The exact finite Fourier polynomial satisfies a strict real gap whenever one coordinate lies outside a fixed small interval about zero. Its fixed derivatives have the same property. This follows from the cotangent rectangle estimate (8.2); summing the other physical colors in absolute value costs powers of N under the source Gaussian bound. The estimate persists in a small complex strip in that coordinate, since an imaginary displacement of height v costs at most $e^{2\pi\delta N|v|}$ on the physical support.

Fix a sufficiently small local ratio radius t_0. For a finite charge with $|m_i/p_i| \geq t_0$, split the real interval in coordinate i into $|z_i| \leq t_0/4$ and its complement. In the first interval,

$$|m_i + 2p_i z_i| \geq |p_i| t_0/2.$$

Choose the sign of the imaginary shift to decrease the real part of $-2\pi\mathrm{i}(p_i z_i^2 + m_i z_i)$. On this interval its derivative in the shift direction is bounded above by $-\pi|p_i|t_0$. The Fourier source grows at most $e^{2\pi\delta N|v|}$, and its analytic local action has bounded derivatives on the fixed tube. For sufficiently long $|p_i|$, the chirp decrease dominates uniformly. A fixed small shift gives a strict gap on the displaced segment.

The two joining segments start at $z_i = \pm t_0/4$, where the real Fourier source already has a fixed loss. Choose their common height small enough that this source loss is preserved. On the remainder of the real interval the source Fourier gap applies directly, so it need not be displaced. This proves a strict nonlocal-charge loss. The other variables remain real at this stage; their chirps have modulus one. The finite number of charges depends on the fixed slopes but is independent of N.

For a local charge, its critical point satisfies the separately even factorization in Chapter 8. For real $\mathbf{t}, \mathbf{e}$, analyticity and the real complete coefficients give

$$|\Im z_i(\mathbf{t}, \mathbf{e})| \leq C|e_i||t_i|, \qquad |\Re z_i + t_i/2| \leq C|e_i||t_i|.$$

Translate a small real box to this critical point. On its real boundary the Fourier action has a quadratic loss relative to its critical value. The chirp's real change on a connector is bounded by a constant times $|p_i||\Im z_i|$ times the boundary displacement; the critical equation cancels the linear term at the saddle. Shrinking the fixed local ratio box first and then increasing the slope threshold leaves the quadratic boundary loss strict. The real part of the Hessian on the translated real tangent space is close to that of $\Psi''(0)$ and remains negative definite. Taylor expansion gives the Gaussian with its branch fixed by this translated contour. Odd cubic terms integrate to zero; the first relative correction is $O(N^{-1})$.

These connector estimates are applied only after the tails have been removed by the real-spectator telescoping argument. They therefore introduce no uncontrolled exponential factor from a larger slope at another cusp. Finally, the selected action tends to F_0 as all reciprocal slopes tend to zero. A single threshold absorbs the fixed source and contour losses. The proof never exchanges the limit $N \to \infty$ with a slope tending to infinity.

Appendix C

The exact certificate arithmetic

The inequalities in Chapter 5 concern continuous domains. Their certificates are finite because concavity bounds each alias action by a tangent plane, while a feasible pure point bounds the pure maximum from below. The remaining difference is affine in the domain variables and slope parameters. It is therefore enough to check finitely many vertices and corners, provided coverage and all transcendental errors are controlled. This appendix supplies those controls.

C.1 A finite proof object

The two main certificates consist of rational points and exact coverage data. The two-leaf block table has sixteen closed slope intervals, one rational feasible pure point and two strict alias tangent points per interval. The attachment table is a recursively subdivided dyadic cover of $[-2, 0]^2$ with 55 leaves, one pure point and one alias tangent point per leaf. The verifier checks feasibility and coverage before evaluating an inequality. It then checks every slope corner and every vertex of the appropriate product polytope.

The mathematical reason this finite object is sufficient is independent of the way candidate points were found. A feasible pure value is a lower bound for the pure maximum. A tangent to a concave alias action is a global upper bound, including at boundary points and on the core plateau. The difference is affine in the slope parameters and the product-domain point. Corner and vertex checks therefore prove the continuum inequality. An approximate optimizer could produce a poor candidate, but could not create a false verification result.

The exact files are

- `experiments/conway_two_leaf_block_alias_certificate.json`;
- `experiments/conway_single_leaf_attachment_certificate.json`;
- the corresponding `verify_conway_..._certificate.py` programs and recorded result files.

The reproduction runner supplies the complete, unambiguous path list and verifies fresh results against the committed records. The optional endpoint certificate strengthens the attachment margin but is not required by Theorem 1.2.

C.2 Outward rational intervals

Every interval endpoint is an integer multiple of 10^{-40}. Addition and subtraction are exact on this grid. Products and quotients use floor and ceiling integer division in the outward directions, with sign cases handled explicitly. A denominator interval must avoid zero. Intersections and monotone range reductions are checked, rather than inferred from floating-point values.

The constant π is enclosed by Machin's formula

$$\pi = 16\arctan(1/5) - 4\arctan(1/239).$$

Forty terms of each alternating arctangent series, with the next-term remainder, suffice for the recorded precision. Sine arguments are reflected into $(0, \pi/2]$. The Taylor polynomial through degree 41 has error at most the next degree-43 term there. Cosine is evaluated as the sine of the complementary angle.

For a positive logarithm argument, scale by an explicitly recorded power of two. On the resulting range let $z = (x-1)/(x+1)$, with the verifier checking $0 \le z < 1/2$. Then

$$\log x = 2\sum_{j=0}^{79} \frac{z^{2j+1}}{2j+1} + R, \qquad 0 \le R \le \frac{2z^{161}}{161(1-z^2)}.$$

The same series at $z = 1/3$ encloses $\log 2$. Negative logarithms are handled by reciprocal and sign, with outward intervals preserved.

C.3 The integrated sine product

For $0 < t < 1/2$, Euler's sine product gives

$$J(t) = t\log(2\pi t) - t - \sum_{n=1}^{\infty} \frac{\zeta(2n)t^{2n+1}}{n(2n+1)}. \tag{C.1}$$

To justify termwise integration, expand $\log(1 - t^2/k^2)$ on compact subintervals below one, where the absolute double series converges; the initial logarithm is integrable at zero. The remaining terms all have the same sign. Use $\zeta(2) = \pi^2/6$, and for $2 \le n \le 20$ the rational bounds

$$\sum_{k=1}^{64} k^{-2n} \le \zeta(2n) \le \sum_{k=1}^{64} k^{-2n} + \frac{64^{1-2n}}{2n-1}.$$

Since $\zeta(2n) < 2$, the tail after $n = 20$ in (C.1) is bounded by

$$\frac{2t^{43}}{21 \cdot 43(1-t^2)}.$$

The exact identities $J(0) = J(1/2) = J(1) = 0$ and $J(1-t) = -J(t)$ reduce every action argument to this range. Thus every action and derivative used by the certificates has a proved rational enclosure.

C.4 Gradients and coverage

The derivatives evaluated by the verifier are

$$g'(y) = 2\log\cot\pi y, \qquad k'(x) = -\log(2\sin(\pi x/2)),$$
$$m_x = \log\frac{\sin\pi(x+y)\sin\pi(z-x)}{\sin\pi x\sin\pi(x+y-z)}, \qquad m_y = \log\frac{\sin\pi(x+y)\sin\pi(z-y)}{\sin\pi y\sin\pi(x+y-z)},$$
$$m_z = \log\frac{\sin\pi z\sin\pi(x+y-z)}{\sin\pi(z-x)\sin\pi(z-y)}.$$

Every tangent point is strict, and the core tangent point in the block certificate lies above $1/4$, so these derivatives are finite and select the intended branch. The entire action remains concave even where a maximizer could lie on the boundary; differentiability is needed only at the chosen tangent point.

For the square certificate, each nonterminal rectangle is split into its four dyadic children. The verifier checks equality with these children, their endpoints and their depth. Thus the 55 leaves cover the square exactly. It does not infer coverage from a list of sampled centers or from an approximate total area. The 880 corner/vertex inequalities are then enclosed from below. Their minimum is the rational number printed in Chapter 5; outward error can only weaken the lower bound.

C.5 What execution establishes

A successful replay establishes the finite supporting inequalities, their exact domains, the recorded rational margins and consistency of the immutable data. It does not numerically establish strict geometry, the complex recurrence asymptotic, or the filling theorem. Those are analytic arguments in the preceding chapters and Appendix B. The optional finite matrix audits check independent instances of the exact identities; their logical role is supplementary, not an extrapolation from tested levels.

Appendix D

Face maps, peripheral markings, and the absolute volume

This appendix supplies the complete rational dictionary behind Chapter 7. Its scope is the completed grammar in Definition 1.1, with two completed cylindrical root children. The quadrilateral gluing faces are auxiliary cut data; no physical four-punctured-boundary representation is inferred from their moduli. The original closed cut fixes the link group and every peripheral path. At each crossing circle its two shaded triangles meet at their common ideal apex; their union is the configuration called a bowtie. We retain both triangles and their labels. All matrices below are projective: multiplying a matrix by a nonzero scalar does not change its Möbius transformation. All logarithms continue from the strict complete chamber. An unfilled main cusp remains unfilled after all auxiliary crossing-circle fillings.

D.1 Two global incidence checks for the complete anchor

The complete cells and ordered stack maps are constructed in Chapter 4. Two points in that construction deserve an explicit check before one uses the group presentation.

First, strict output dihedral angles keep the matching quadrilateral a maximal supported face. Cells on opposite supporting sides intersect only in that face inside hyperbolic space. Their ideal closures can meet only on its supporting circle. An additional ideal vertex on that circle would enlarge the supporting face beyond a quadrilateral edge, contradicting its strictly nonflat angle. Thus there are no unrecorded ideal coincidences.

Second, at every open stage the four output corners are four unpaired original terminal singletons. Product consumes F_2/G_1 and F_3/G_0 and retains F_0, F_1, G_2, G_3; turns only permute these singletons. Consumed pairs never reappear among the exposed vertices of a gluing face. At the root, all four pairs under $C_j^F \mapsto C_{3-j}^G$ join the disjoint root subtrees. Therefore every main vertex is exactly a two-terminal pair, not a chain of identifications, and a leaf's four corners stay distinct. Its two shaded triangles share only their original apex. An edge and its partner in the other shaded triangle are distinct, so the complete four-step cycle contains four distinct edge occurrences. The horospherical links are positive rectangles, and the canonical shaded-side decoration gives the actual complete cusp tori described in Chapter 4.

D.2 The selected color action and the complete face-pairing maps

Write the pure leaf action and its color derivative as

$$g(x,t) = J(2x) - 2J(x) - J(x-t) - J(x+t), \qquad v = g_t = \operatorname{Log}\frac{\sin\pi(x-t)}{\sin\pi(x+t)}.$$

For physical sign $\varepsilon \in \{0, 1, -1\}$, the source-selected shear is

$$\chi = t - \frac{\varepsilon v}{2\pi\mathrm{i}}, \qquad f_\varepsilon(x,\chi) = g(x,t) - \frac{\varepsilon v^2}{4\pi\mathrm{i}}, \qquad \mathrm{d}f_\varepsilon = g_x\,\mathrm{d}x + v\,\mathrm{d}\chi. \tag{D.1}$$

At completeness its implicit derivative is $1 - \varepsilon\mathrm{i}\cot\pi x \neq 0$. Here $J(u) = \int_0^u \operatorname{Log}(2\sin\pi s)\,\mathrm{d}s$, on the continued complete branch. The finite packet identifies the shear and its constant. An explicit algebraic check is useful: put $E = e^{2\pi\mathrm{i}x}$, $H = e^{2\pi\mathrm{i}\chi}$, and $K = e^{-\partial_x f_-}$. The selected source quadratic gives

$$K^2(1+E)^2 + K[(H+H^{-1})E^2 + 2] + (1-E)^2 = 0. \tag{D.2}$$

At $H = 1$, it factors as $(K+1)[K(1+E)^2 + (1-E)^2]$, and the source branch is $K = \tan^2(\pi x)$. For the negative shear in (D.1), setting $T = e^{2\pi\mathrm{i}t}$ gives

$$H = \frac{T(E-T)}{ET-1}, \qquad T^2 + E(H-1)T - H = 0, \qquad K = \frac{(E-T)(1-ET)}{T(1+E)^2}.$$

Eliminating T gives (D.2) with the same branch. This identifies the x derivative. The remaining χ-dependent constant is zero by the actual endpoint normalization. For fixed small χ, at $x = 1/2$ one has $v = 0$, so the hidden pure color tends to $t = \chi$, not to zero. On the continued branch $J(1) = J(1/2) = 0$ and $J(1/2+t) = -J(1/2-t)$, hence $g(1/2,t) = 0$ and the shear quadratic vanishes. The finite-source action is $g(x,\chi) + S(x,\chi) - S(x,0)$, where $S(x,\chi) = \int_x^{1/2} \operatorname{Log}\mu(u,\chi)\,\mathrm{d}u$. The selected root is regular with $\mu(1/2,\chi) = 1$; both integrals therefore tend to zero. This verifies the fixed-color constant on the same holomorphic branch. Schwarz reflection gives the positive sign. Thus the geometry below does not select a new source solution or discard a quadratic color phase.

Normalize the ordered gluing face of an elementary pyramid to $(0, \infty, 1, R)$, put $E = e^{2\pi\mathrm{i}x}$, $T = e^{2\pi\mathrm{i}t}$, and select

$$Z = \frac{1+E}{1+T}, \qquad R = \frac{T(1+E)^2}{E(1+T)^2}, \qquad Z^2 - (1+E)Z + ER = 0.$$

The two projection charts have apices Z and Z/E. Their four maps are listed below. A white map pairs pieces coming from the projection sphere. A completion map is the same algebraic expression at an attachment edge where the adjacent prism supplies the actual face. We distinguish these roles before using the maps in a group relation:

$$\begin{aligned} W_0 &= \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, & W_1 &= \begin{pmatrix} 1 & E-1 \\ 0 & E \end{pmatrix}, \\ W_2 &= \begin{pmatrix} E-R & R(1-E) \\ E-1 & 1-ER \end{pmatrix}, & W_3 &= \begin{pmatrix} R & 0 \\ 1-E & ER \end{pmatrix}. \end{aligned} \tag{D.3}$$

At an elementary pyramid, W_0, W_2 are its actual white-face maps. The expressions W_1, W_3 become actual maps when the adjacent prism is attached at the edges called cap edges in Chapter 4. All matrix identities here are projective. The pure shaded map and the two odd cross-copy maps are

$$S = \begin{pmatrix} 0 & -ER \\ 1 & -1-E \end{pmatrix}, \qquad A = \begin{pmatrix} R & -R \\ 1 & ER-1-E \end{pmatrix}, \qquad B = \begin{pmatrix} R/E-1-1/E & R \\ -1 & R \end{pmatrix}.$$

They send the original labeled triangle vertices to their prescribed partners. The apex equation verifies the fourth flag without any unit-modulus assumption.

For a prism, transport all W_j to the coordinates in (4.6). Direct rational substitution gives

$$\begin{aligned} W_{F,1} = W_{G,1} = W_{p,1} = W_{\mathrm{top}}, \qquad & W_{F,3} = W_{G,3} = W_{p,3} = W_{\mathrm{bottom}}, \\ W_{F,2} = W_{G,0}, \qquad & W_{p,0} = W_{F,0}, \quad W_{p,2} = W_{G,2}. \end{aligned} \tag{D.4}$$

Here p denotes the output gluing face, not a filling slope. The nine identifications are identities of entire maps, not just cross ratios. They can be verified without geometric continuation: use independent phase variables $\mathsf{a} = e^{2\pi i p}$, $\mathsf{b} = e^{2\pi i r}$, $\mathsf{c} = e^{2\pi i q}$, and the rational coordinates

$$\begin{aligned} &v_0 = 1, \quad v_1 = \mathsf{a}, \quad v_2 = \mathsf{ab}, \quad v_3 = \mathsf{abc}, \\ &P = 1 + \mathsf{a} - \frac{\mathsf{a}[\mathsf{ab}(1+\mathsf{c}) - 1 - \mathsf{a}]}{\mathsf{a}^2\mathsf{b}^2\mathsf{c} - \mathsf{a}}. \end{aligned}$$

The three wire phases are $\mathsf{bc}, \mathsf{ab}, \mathsf{abc}$. Normalize each ordered quadrilateral gluing face by

$$M_{a,b,c}(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}, \qquad R = M_{a,b,c}(d).$$

In world coordinates its white map is $\overline{M}^{-1} W_j M$, where the bar replaces $\mathsf{a}, \mathsf{b}, \mathsf{c}$ by their reciprocals, as prescribed by the second chart. Inserting these formulas into (D.4) gives equality entry by entry up to a common scalar. Thus no unit-circle restriction or missing translation is hidden in the identities.

The turn map $T_Q(z) = (R-1)/(z-1)$ and closing map $C_Q(z) = (z-R)/(z-1)$ satisfy

$$T_Q W_{j+1}(E,R) = W_j(E^{-1}, 1-R) T_Q, \qquad C_Q W_j(E^{-1}, R) = W_{2-j}(E,R) C_Q.$$

The identities show that the map on a white face is independent of which constituent cell is used to compute it. Thus every triangular piece of one original white face has the same map, even when its deformed image is not planar. For pure leaves, writing $\overline{S}$ for S with E replaced by E^{-1}, the edge words are

$$W_0 S = \overline{S} W_0, \quad W_2 S = \overline{S} W_2, \quad W_3 S = \overline{S} W_1.$$

For odd leaves they are

$$W_0^{-1} B W_2^{-1} A = I, \quad W_2^{-1} B W_0^{-1} A = I, \quad W_1^{-1} B W_3^{-1} A = I.$$

All hold for independent nearby E, R. They include translation and dilation, and the reverse words are inverses. The original closed cut therefore supplies a representation of the actual

half-twist partner group: use its dual two-skeleton presentation, with exactly these white-face seams and shaded four-step relations. No representation is inferred from the shape parameters of the gluing faces alone. The second charts depend holomorphically on the source parameters by the reflected-minus convention, rather than by conjugating those parameters.

A main meridian is $W_{j-1}^{-1}W_j$ at the corresponding corner. The two maps fix that corner and have equal derivatives there: E^{-1} at C_0, C_2, and E at C_1, C_3, using the finite coordinate at infinity. Thus its multiplier is one identically. It is nonidentity at completeness and stays parabolic nearby. Its commuting peripheral partner is consequently a translation as well. Every main cusp remains parabolic.

D.3 Primitive logs, the absolute regulator, and filling

We next compute the holonomy of the two actual peripheral curves. Their being primitive is a topological statement about the cusp lattice, so we check their complete translations before using their logarithms. The logarithms below are those of similarity multipliers; they are twice eigenvalue logarithms. This convention explains the factors of two in the volume differential and in the filling equation. Put

$$\rho = \frac{Z}{1+E-Z}, \qquad Q = \frac{E-Z}{Z-1}.$$

The original disk longitude is $W_2^{-1}W_0$, with multiplier Q^2 in the selected flag-at-infinity frame. The pure meridian is S^{-1}, with multiplier ρ. The two actual odd meridians are $A^{-1}W_0$ and $W_0^{-1}B$, with multipliers ρQ and ρ/Q. Returning to the original complete leaf coordinate, their translations are -2 and $2E$, while the disk longitude is $-2(E+1)$. The two-rectangle lattice fixes their primitive integral marking and orientation. The sign is fixed by the original threading convention: for the negative coupon, the typed loop $[2,1,1,2]$ followed by R^{-1} has auxiliary matrix

$$M_{\mathrm{aux}} = I - \frac{P+I}{1+E}, \qquad \operatorname{Tr} P = -2,$$

where P is the actual disk-product matrix and the preferred projective longitude is $-P$. Its translation ratio is $1+E$, identifying the first path above. Reversing the retained half twist changes the meridian by one disk longitude and gives the second. Thus the physical R^{-1} coupon uses the first path and R the second, with no sign inferred solely from a Hessian. Consequently

$$\rho = e^{-2\pi \mathrm{i} t}, \qquad Q = e^{-v}, \qquad U = -2\pi \mathrm{i} t - \frac{\varepsilon V}{2} = -2\pi \mathrm{i}\chi, \qquad V = -2v.$$

The half-longitude shear is a relation between the two actual primitive cusp markings. It does not assert that half a longitude is an integral loop or replace the actual link by a pure rational filling. The centered similarity multipliers used for U, V are squares of the corresponding eigenvalues. The original lift and pivot character are retained in the finite source; no additional square-root choice changes a peripheral word here.

Let $\Phi_\varepsilon(\chi)$ be the selected stationary action in all free wires. The envelope identity in (D.1) gives

$$U_\ell = -2\pi \mathrm{i}\chi_\ell, \qquad V_\ell = -2\partial_\ell \Phi_\varepsilon, \qquad G = 2\pi \mathrm{i}\Phi_\varepsilon, \qquad \mathrm{d}G = \frac{1}{2}\sum_\ell V_\ell \,\mathrm{d}U_\ell. \tag{D.5}$$

The source primitive includes the local term $-\varepsilon_\ell V_\ell^2/8$. It is retained before taking volume. At the complete point, $\operatorname{Im} G(0) = 2\pi F_0$ is the actual half-twist partner volume from the explicit complete cells and original pairings.

For the nearby framed representation, triangulate the same original cut balls and retain their peripheral decoration. The ideal-tetrahedron variation follows directly from the Bloch–Wigner formula

$$\mathrm{d}D(z) = \log|z|\,\mathrm{d}\arg(1-z) - \log|1-z|\,\mathrm{d}\arg z.$$

In the sum over the original oriented tetrahedra, face terms have equal flags and opposite orientations and cancel. On a peripheral torus, cut along the recorded primitive μ, λ to a parallelogram. The opposite sides cancel except for their two eigenline scale factors; the remaining alternating product is $-2\,\mathcal{M} \wedge \mathcal{L}$, where $\mathcal{M} = e^u, \mathcal{L} = e^v$ and $u = U/2, v = V/2$. Here the sign is the boundary orientation of the truncated original fundamental chain, and the factor two retains both eigenline flags. Applying the displayed differential to that alternating product gives $2(\Re v\,\mathrm{d}\Im u - \Re u\,\mathrm{d}\Im v)$. Different boundary tori have zero mutual intersection, and the parabolic main tori have zero log contribution. Thus

$$\mathrm{d}\operatorname{Vol}(\rho) = \frac{1}{2}\sum_\ell (\operatorname{Re} V_\ell\,\mathrm{d}\operatorname{Im} U_\ell - \operatorname{Re} U_\ell\,\mathrm{d}\operatorname{Im} V_\ell).$$

Using (D.5), this is the differential of $\operatorname{Im} G - \frac{1}{2}\sum \operatorname{Re} U \operatorname{Im} V$. The complete anchor fixes the constant on the same connected marked branch. Therefore the absolute identity is

$$\operatorname{Vol}(\rho_\chi) = \operatorname{Im} G - \frac{1}{2}\sum_\ell \operatorname{Re} U_\ell \operatorname{Im} V_\ell. \tag{D.6}$$

This uses the original relative fundamental chain and its actual nearby representation, not equal differentials on unrelated components.

D.4 Smooth completion on the actual primitive filled topology

We give the metric argument used here, including its uniformity in the independently chosen slopes. Let $M = Y_E^\varepsilon$ carry the oriented complete metric constructed above, with holonomy ρ_0. A nearby *framed* representation consists of the actual face-paired representation ρ and, at each cusp, its selected common fixed flag. In a frame sending that flag to infinity, the two peripheral maps are complex similarities. Their uniquely small multiplier logs are the U, V just computed. We construct one neighborhood of this framed complete point before choosing the filling slopes.

D.4.1 An actual positive metric on the core and its ends

The first task is to turn nearby holonomy data into local hyperbolic coordinate charts on the same manifold. On a compact truncated part we average small changes of the complete developing map. On each end we use an equivariant affine construction. Both maps are close to the original one on their common compact collar, allowing them to be joined without losing local invertibility. Choose a compact cusp-truncated core with a slightly larger collar, finitely many evenly covered balls, and a smooth partition of unity. Write $\eta_{j,\gamma}$ for its lifts, with one

chosen lift of each ball and $\gamma \in \pi_1(M)$. In the future unit hyperboloid model of hyperbolic space put

$$Y_\rho(x) = \sum_{j,\gamma} \eta_{j,\gamma}(x)\rho(\gamma)\rho_0(\gamma)^{-1}x, \qquad E_\rho(x) = \frac{Y_\rho(x)}{\sqrt{-\langle Y_\rho(x), Y_\rho(x)\rangle}}.$$

Every summand is future timelike; their convex average is future timelike too, so the normalization is defined. Reindexing the sum proves $E_\rho(\rho_0(\delta)x) = \rho(\delta)E_\rho(x)$. Only finitely many terms occur on a compact fundamental part, and at ρ_0 the map is the identity. Thus it is C^1-close to the identity and is an orientation-preserving local diffeomorphism in one fixed representation neighborhood. This does not assume that the deformed action is already discrete or free.

For one complete cusp write its universal coordinates as $(w, s) \in \mathbb{C}\times[s_0, \infty)$, with translation lattice Λ, and complete development (w, e^s) in upper half-space. Denote the deformed affine peripheral action by $A_\lambda(z) = a_\lambda z + b_\lambda$. Choose a nonnegative compactly supported smooth η with $\sum_{\lambda\in\Lambda} \eta(w - \lambda) = 1$, and define

$$d_\rho(w) = \sum_{\lambda\in\Lambda} \eta(w - \lambda)A_\lambda(w - \lambda).$$

Then $d_\rho(w + \lambda) = A_\lambda d_\rho(w)$. On a fixed closed fundamental parallelogram it is C^1-close to w, so its real Jacobian is positive; equivariance propagates that fact everywhere. Let ξ_ρ be the real-linear extension of $\lambda \mapsto \log|a_\lambda|$. The map

$$F_\rho(w, s) = \big(d_\rho(w), e^{s+\xi_\rho(w)}\big)$$

is exactly peripheral-equivariant. Its differential has positive determinant $e^{s+\xi_\rho(w)} \det_{\mathbb{R}} Dd_\rho(w)$ for every $s \geq s_0$. Restore the moving flag frame and extend to all cusp lifts by equivariance. Geodesically interpolate E_ρ and F_ρ on the compact collar with an invariant smooth cutoff. There both maps are C^1-close to the complete development, so the interpolation remains a positive local diffeomorphism after one further uniform shrink. Pullback of the hyperbolic metric now gives an actual oriented nonsingular metric on the original manifold with its ideal ends deleted.

D.4.2 The logarithmic link and its primitive angular period

At a filled cusp, the nearby peripheral similarities have two common fixed points. Sending one to infinity and translating the other to zero makes them multiplications. Their logarithms then act by translations on a Euclidean plane. We must show that the translations are exactly the selected small logs U, V, with no unnoticed multiple of 2πi. Only after this check may the large integer p be used in the relation $U + pV = -2\pi$i. At a cusp to be filled, the equation $U + pV = -2\pi$i ensures that at least one multiplier differs from one. Commuting affine maps then have a common finite fixed point q, in addition to infinity. For either generator with $a \neq 1$, $q = b/(1 - a)$. Its numerator stays away from zero near completeness, whereas both generator multipliers tend to one; hence $|q| \to \infty$ uniformly, even if one multiplier equals one. The developed fundamental parallelogram and fixed generator paths have uniformly bounded images and lengths. After shrinking the neighborhood they miss q, and equivariance implies that the entire development misses q.

On the simply connected link cover choose $f = \operatorname{Log}(d_\rho - q)$. Its differential is invertible as a real map and is invariant under deck transformations. Consequently $|\mathrm{d}f|^2$ descends to a nonsingular Euclidean metric on the compact torus, hence is complete. Lifting Euclidean segments proves that its developing local isometry f covers $\mathbb{C}$; because the plane is simply connected, this is a global isometry. Its periods form an injective real rank-two lattice. A fixed generator path of developed length at most L, contained in $|d_\rho| \le R$, satisfies

$$|\Delta f| \le \frac{L}{|q| - R} \longrightarrow 0.$$

Exponentiating its increment gives the corresponding similarity multiplier. The increment and its prescribed small log therefore differ by an integer multiple of $2\pi\mathrm{i}$ which is exactly zero near completeness. Thus the actual two log-link periods are U, V, before they are multiplied by any large slope. This is why the same representation neighborhood works for all slopes.

Translate q to zero. On the compact link quotient the function $e^{s_0+\xi_\rho(w)}/|d_\rho(w) - q|$ is bounded above. The explicit end therefore contains an entire equivariant epigraph $y > A|d_\rho - q|$. With

$$r = \frac{y}{|d_\rho - q|}, \qquad R = \operatorname{arsinh}(1/r), \qquad \theta = \Im f, \qquad X = \Re f + \tfrac{1}{2}\log(1 + r^2),$$

direct substitution into the upper-half-space metric gives

$$\mathrm{d}s^2 = \mathrm{d}R^2 + \cosh^2 R\,\mathrm{d}X^2 + \sinh^2 R\,\mathrm{d}\theta^2. \tag{D.7}$$

The epigraph contains a whole punctured tube $0 < R < R_0$. A log period L acts by $(X, \theta) \mapsto (X + \Re L, \theta + \Im L)$. The actual primitive curve $\mu + p\lambda$ has period $-2\pi\mathrm{i}$, exactly one angular turn. Its complementary primitive period can be taken to be V, which has nonzero real part since U, V are real-linearly independent. Adding the axis $R = 0$ in (D.7) is therefore a smooth hyperbolic solid torus, whose meridian is exactly $\mu + p\lambda$. Its affine holonomy is the identity, since it fixes both q and infinity with multiplier one. The representation descends to the actual filled group. Reversing the angular sign changes the parametrization of the meridian, not the orientation of the metric.

At an unfilled main cusp both multipliers are one. The same construction gives a translation-equivariant Euclidean link development, globally the plane by completeness of the compact link; $\xi_\rho = 0$. Its end contains a full horoball quotient, so it is complete and has finite volume. The remaining region is compact. Attaching the finitely many bounded-radius solid tori therefore gives a complete finite-volume hyperbolic metric on precisely the filled topology. Its complete universal developing map covers simply connected hyperbolic space, so the holonomy is discrete and faithful. This proof used one fixed framed neighborhood and no comparison between different slope sizes.

D.4.3 The source saddle and its absolute knot-complement volume

The full source filling transform selects a critical point of

$$S_{\varepsilon,\mathbf{p}}(\chi) = \Phi_\varepsilon(\chi) + \frac{\pi\mathrm{i}}{2}\sum_\ell \frac{(\chi_\ell - 1)^2}{p_\ell}.$$

Its equations are exactly $U_\ell + p_\ell V_\ell = -2\pi\mathrm{i}$. For existence set $r_\ell = 1/p_\ell$ and write them as $V_\ell + r_\ell U_\ell = -2\pi\mathrm{i} r_\ell$. At $r = 0$ their χ-Jacobian is $-4\pi\mathrm{i}\,\mathrm{diag}(\tau_\ell)$, with $\Im\tau_\ell > 0$, by the source packet. It is invertible. The holomorphic implicit-function theorem thus gives one nearby solution for all r in a fixed polydisk, independently of their relative sizes and signs. It lies in the framed neighborhood just constructed after that polydisk is shrunk once more. For independently sufficiently long fixed signed integral p_ℓ, the marked geometric completion gives smooth angle 2π on these primitive filled curves, with the unfilled main cusp parabolic. The physical leaf power is $R^{\varepsilon_\ell - 2p_\ell}$, once, in its original disk framing. At this saddle, (D.6) is

$$\mathrm{Vol}\big(S^3 \setminus K_{F,G,\varepsilon}(\mathbf{p})\big) = 2\pi\,\mathrm{Re}\,S_{\varepsilon,\mathbf{p}}(\chi_*).$$

All auxiliary circles have been filled and the chosen main component is the knot. The knot cusp is unfilled. The separate full-source alias, complex-packet and entire Fourier-period arguments are what identify this geometric action with the original invariant's limiting rate.

Appendix E

The quantitative finite-sum estimate

This appendix proves the analytic part of Theorem 9.1. The finite ordered tree, the crossing power at every vertex, the strand orientations and the knot pairing are fixed throughout. The integer N alone tends to infinity. The proof gives explicit estimates on the original finite invariant; Appendix F identifies the resulting critical action with the volume of the actual filled knot complement.

E.1 Polynomial coordinates and the finite sum

We first specify the finite algebra used below, including the data at repeated roots. Put $d = N - 1$, $\theta = \pi/N$, and $q = e^{i\theta}$. The main meridian parameter is $M = q^s$, with s in a fixed disk $|s| \le R < 1$. Let T_N be the normalized Chebyshev polynomial characterized by $\mathsf{T}_N(t + t^{-1}) = t^N + t^{-N}$. The polynomial relation for two strands with the same orientation is

$$P_s(z) = \mathsf{T}_N(z) + 2\cos(2\pi s).$$

For opposite orientations it is $P_0(z)$. A canonical polynomial means the representative of degree less than N obtained by division by the appropriate monic relation. These two divisions are kept distinct. The fixed basis used to express both representatives is

$$S_k = \prod_{r=1}^{k} 2\sin(r\theta), \qquad c_j = 2\cos((2j+1)\theta), \qquad e_k(z) = S_k^{-2} \prod_{j=0}^{k-1} (c_j - z), \quad 0 \le k < N. \tag{E.1}$$

An empty product is one. The low indices are $0 \le k \le \lfloor d/2 \rfloor$; the remaining indices are high. These words describe a division of the fixed basis, rather than a change in the physical strand color.

The full norm of a canonical polynomial records its value at every distinct complete node and its first derivative at every double node:

$$\|F\|_{\mathrm{J}} = \max\{|F(c_j)| : \ 0 \le j \le \lfloor d/2 \rfloor; \ |F'(c_j)| : \ c_j \text{ is a double root of } P_0\}. \tag{E.2}$$

At odd levels the endpoint -2 is simple and has no derivative coordinate. The derivative coordinates are the first normal data referred to below. Hermite interpolation, which uses values and these derivatives together, makes this norm polynomially equivalent in N to the

sup norm on $[-2,2]$ of the canonical polynomial and to its Chebyshev coefficient norm. A polynomial cost means a bound CN^A; C, A can depend on the fixed diagram and on R.

The normalized closing functional on either algebra is

$$\phi_\sigma(F) = (-1)^d N^2 [z^d] \operatorname{rem}_{P_\sigma} F, \qquad \phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}, \qquad \sigma = 0, s, -s. \tag{E.3}$$

Here $[z^d]$ extracts a monomial coefficient. To see that the pairing is unchanged when the relation moves, divide $e_i e_j = P_0 A + B$. Its degree is at most $2N-2$, so $\deg A \le N-2$. Since $P_s = P_0 + 2\cos(2\pi s) - 2$, its moving remainder is $B - (2\cos(2\pi s) - 2)A$, with exactly the same top coefficient. This calculation retains the repeated-root data.

An actual quarter-turn reverses the index $k \mapsto d-k$ and multiplies it by a specified scalar and a paired sine product. Set

$$W_k(s) = \prod_{r=1}^{k} \left(1 - \frac{\sin^2(\pi s/N)}{\sin^2(\pi r/N)}\right), \qquad H_N(s) = \frac{\sin(\pi s)}{N \sin(\pi s/N)}, \qquad W_k W_{d-k} = H_N^2. \tag{E.4}$$

The removable value is $H_N(0) = 1$. The prescribed orientation type supplies $\gamma_e(s) \in \{1, q^{-s^2/2}, q^{s^2/2}\}$. After lowering an index by the pairing in (E.3), a physical edge identifies equal indices and carries $\gamma_e(s) H_N(s)/W_k(s)$. Thus the edge factors are retained in the finite contraction, even when its tensor entries are compared with a positive reference. On every fixed disk $R < 1$, these factors and their inverses are bounded uniformly in N and k. The exact orientation and framing products give the scalar $A_N(s)$ in (9.4); they are not replaced by their limits.

The finite sum over the internal edge indices is called the packet B_N. Its angular variables τ_v are the Fourier variables of the circle traces. A Fourier charge ℓ_v labels the frequency $N\ell_v$ produced by the translated root grid; it is not a new color or a new filling coefficient. The polynomial U_j used for its coefficient satisfies $U_j(\cos t) = \sin((j+1)t)/\sin t$, with removable values at the endpoints. The fixed crossing operator $R_\sigma^{n_v}$ always includes its physical scalar, the specified strand order and the pivotal closing map. Negative powers mean its actual inverse.

For clarity, the interpolation used to move the finite index contour has a fixed normalization. The sine factorial S_r is continued by

$$G_N(z) = \left(\frac{2\pi}{N}\right)^z \Gamma(z+1) \prod_{k \ge 1} \frac{\Gamma(Nk+z+1)}{\Gamma(Nk-z)(Nk)^{2z+1}}, \qquad -1 < \Re z < N. \tag{E.5}$$

The powers use positive-base logarithms. This function is nonzero on the strip, satisfies $G_N(0) = 1$, $G_N(r) = S_r$, and $G_N(z+1)/G_N(z) = 2\sin(\pi(z+1)/N)$. For $x = k/N$ and $u = Nt/(2\pi)$, the interpolant of $e_k(-2\cos t)$ is

$$\frac{G_N(d/2+u) G_N(d/2-u)}{G_N(d/2-Nx+u) G_N(d/2-Nx-u) G_N(Nx)^2}. \tag{E.6}$$

Gamma reflection gives $G_N(s) G_N(-s) = 1/H_N(s)$ and hence the exact analytic interpolation of the paired edge insertion is

$$\frac{H_N(s)}{W_{Nx}^{\mathrm{hol}}(s)} = \frac{G_N(Nx)^2}{G_N(Nx-s) G_N(Nx+s)}. \tag{E.7}$$

All $d/2$ offsets are retained. The product normalization, its complex bulk expansion and the finite cotangent interpolation are the ones proved in the exact factorial interpolation theorem and the regular crossing-circle calculation. A periodic factor would change the interpolation and is not allowed. The following estimates apply these fixed finite identities to the explicit angular domain required by the quantitative theorem.

E.2 Finite sums under the degree and twist criterion

The separately available version of this section is finite sums under the degree and twist criterion. The proof is reproduced here in full.

The number of branches incident to a crossing circle can supply part of the estimate previously obtained by taking its twist number large. This section proves the finite-sum limit under an explicit inequality in those two integers. The volume theorem combines it with the ordered octahedral decomposition and positive filling path. The limiting action is therefore the volume of the original knot complement throughout the stated class, including the short coefficients permitted by the inequality.

E.2.1 Assertion and scope

Consider an actual reduced Conway operation tree in which every product junction carries an original crossing-circle operator. Include ordinary terminal crossing circles as degree-one vertices. The resulting circle graph $\mathcal{T}$ is a finite tree, with physical quarter-turn edges and no bare junctions. Keep the ordered branches, orientations, knot pattern, primitive cut and zero framing. For the literal construction defined in the volume theorem, the strict angle condition follows inductively from the product intervals; it is not a further restriction on these trees.

Let $d_v \geq 1$ be the graph degree of circle v, and let $n_v \in \mathbb{Z}$ be its actual fixed crossing power. The graph degree d_v is unrelated to the quantum degree $N-1$. Put

$$\mathfrak{S}(\mathcal{T}, \mathbf{n}) = \sum_v \frac{d_v}{d_v^2 + n_v^2}. \tag{E.8}$$

The following conservative sufficient condition will be proved:

$$\mathfrak{S} \leq 2^{-25}. \tag{E.9}$$

It implies a nonzero first joint Fourier charge and the actual reduced microscopic profile through both level parities, on every $|s| \leq R < 1$. The exponential rate is the real part of the explicitly continued first-charge action. Conditions (E.8)–(E.9) are quantitative finite-action conditions, not an assumption that this saddle dominates.

Short, even zero, crossing powers are permitted by this certificate. Section E.2.11 explains how the companion geometric proofs identify the action with the actual knot volume under the same bound. The graph and all powers are fixed before N tends to infinity.

The invariant in the conclusion is the reduced, zero-framed ADO polynomial $F_N(M)$, normalized to one on the unknot. Its complete value is $F_N(1) = K_N(K)$. Set $q_N = e^{\pi i/N}$ and $M = q_N^s$. The explicit elementary factors are

$$H_N(s) = \frac{\sin(\pi s)}{N \sin(\pi s/N)}, \qquad A_N(s) = q_N^{a(D)s^2/2}, \qquad a(D) = -w(D) - \sum_v n_v.$$

Here $w(D)$ is the writhe of the fixed original diagram. The scalar calculation specifies the same framing convention, including all orientation types. The critical action in the limit is defined in Sections E.2.4 and E.2.7.

E.2.2 Finite identities and the four uses of long twists

The finite identities used below have the following roles. Their separately available proofs are linked for reference:

- the exact crossing trace, Sections 2–4 and 7: exact translated-grid trace, full crossing scalar, Fourier coefficients, normal limits and fixed-power polynomial bounds;
- the finite sum over internal edge indices, Sections 2 and 4–8: primitive lattice, canonical G_N, finite cotangent currents and the saddle insertion rule;
- the joint Fourier calculation, Sections 3–5: real-spectator telescoping and the exact coefficient of charge one;
- bounds for the full polynomial traces, Sections 1–4, with its cited full spectral-jet and Chebyshev folding bounds;
- the crossing and framing scalar: the unchanged exact scalar identity for fixed integer powers.

These are mathematical inputs, not numerical diagnostics. The earlier source transfer theorem and the pretzel filling calculation use reciprocal-slope implicit functions; the geometric source-regulator theorem uses a neighborhood of the complete framed representation.

The long-twist hypothesis enters the joint Fourier argument in four analytic places: existence and charge ordering of a nearby critical branch; a nonempty finite charge cutoff with a fixed tail loss; the angular connectors; and comparison of the total critical drop with fixed discarded-source losses. The geometric completion is a separate fifth use. This proof replaces the four analytic uses in the all-circle class.

E.2.3 A graph-independent discarded-source gap

Use the fixed complete Newton basis e_k, $0 \le k < N$, and write

$$J(x) = \int_0^x \log(2\sin \pi u)\, du, \quad g(x) = J(2x) - 4J(x), \quad G_0 = g(1/4). \tag{E.10}$$

The physical turn on a covariant edge retains equal indices and the actual factor $\gamma_e(s)H_N(s)/W_k(s)$. This factor and its inverse are bounded on any fixed $|s| \le R < 1$, including all indices. Reference, pivot, framing and the single reduced-cut scalar are kept.

At a circle vertex the actual typed tensor is the prescribed version of

$$\phi_\sigma\left(R_\sigma^{n_v} \prod_{e\sim v} e_{k_e}\right), \qquad \sigma = 0, s, -s. \tag{E.11}$$

The canonical polynomial of a fixed $R_\sigma^{n_v}$ has polynomial complete spectral-jet norm. Low e_k have norm rate $g(k/N)$; high e_k have polynomial norm, including their first normals.

Finite-degree Chebyshev folding and exact top-coefficient extraction therefore give rate at most the sum of g over low incidences in (E.11). The polynomial exponent can depend on the fixed valence and power. This is the full carrier-normalized tensor, not a distinct-node trace. Opposite-color types retain their explicit outside scalar.

Each physical edge has two incidences. The whole absolute rate is at most $2\sum_{\text{low } e} g(k_e/N)$, and the positive complete action is

$$F_0 = 2|E(\mathcal{T})|G_0. \tag{E.12}$$

Retain the box $k_e/N \in I = [1/5, 3/10]$. Since

$$g'(1/4) = 0, \qquad g''(x) = -4\pi/\sin 2\pi x \le -4\pi, \qquad 2[G_0 - g(x)] \ge 4\pi(x - 1/4)^2, \tag{E.13}$$

one low edge outside I loses at least $\pi/100$. A high edge loses $2G_0 \ge \pi/4$. Closed factorial bounds and polynomial state counting consequently give a grouped scalar error

$$N^{A_{\mathcal{T},\mathbf{n},R}} e^{N(F_0-\eta_0)}, \qquad \eta_0 = \pi/200. \tag{E.14}$$

This loss is independent of the tree size and the fixed powers; prefactors and starting levels are not uniform in those data. There are no bare-tensor replacement errors. No generic conversion from a small spectral norm to a low/high coefficient cone is used: the physical vertex trace (E.11) is bounded directly.

E.2.4 Exact edge action

The retained packet factorizes over physical edges. Apart from its edge scalar, each factor is

$$K_N(t, u; s) = \sum_{k/N \in I} e_k(-2\cos t) e_k(-2\cos u) \frac{H_N(s)}{W_k(s)}. \tag{E.15}$$

The canonical interpolation has phase

$$b(x, t) = \int_0^x \log(2\cos 2\pi v + 2\cos t)\, dv - 2J(x), \qquad \psi(t, u) = \operatorname{stat}_x[b(x, t) + b(x, u)]. \tag{E.16}$$

All branches are continued from $x = 1/4, t = u = 0$. Writing $c = \cos 2\pi x$, the stationary equation gives exactly

$$c = \frac{1 - \cos t \cos u}{2 + \cos t + \cos u}, \qquad \tan^2 \pi x = \frac{1}{1 + \tan^2(t/2) + \tan^2(u/2)}. \tag{E.17}$$

Substitution $z = \tan \pi v$ in b_t yields

$$\psi_t = -\pi^{-1} \operatorname{arsinh}(\tan(t/2)\cos(u/2)). \tag{E.18}$$

For $Q = \sqrt{1 - \sin^2(t/2)\sin^2(u/2)}$,

$$\psi_{tt} = -\frac{\cos(u/2)}{2\pi\cos(t/2)Q}, \qquad \psi_{tu} = \frac{\sin(t/2)\sin(u/2)}{2\pi Q}. \tag{E.19}$$

The Hessian determinant is exactly $1/(4\pi^2)$. Thus

$$\Psi(\boldsymbol{\tau}) = \sum_{\{v,w\} \in E} \psi(\tau_v, \tau_w) = F_0 - \tfrac{1}{2}\sum_v a_v \tau_v^2 + \mathcal{E}(\boldsymbol{\tau}), \qquad a_v = d_v/(2\pi). \tag{E.20}$$

In particular the diagonal curvature grows with graph degree.

E.2.5 An available complex packet and nonlinear bounds

Take $r = 1/4$. Formula (E.17) gives $|x(t,u) - 1/4| < 0.003$ on $|t|, |u| \le r$: $|\tan(t/2)| < 0.128$, hence $|c| \le 2(0.128)^2/[2 - 2(0.128)^2] < 0.017$, and $x = 1/4 - \arcsin(c)/(2\pi)$.

Here the radius is valid for the finite packet, not only its formal action. Put $A = b(x,t) + b(x,u)$. On $\Re x \in I$, $|\Im x| \le 1/50$, $|t|, |u| \le 1/4$,

$$A_{xx} = \sum_{\xi = t,u} \left[-\frac{2\pi \sin 2\pi x}{\cos 2\pi x + \cos \xi} - 2\pi \cot \pi x \right], \qquad |A_{xx}| < 44, \quad \Re A_{xx} < -12. \tag{E.21}$$

For completeness, $|\cos 2\pi x| < 0.338$, $|\cos \xi - 1| < 0.032$, $|\sin 2\pi x| < 1.009$. Also $\Re \cot \pi x > 0.72$, while the real part of the first fraction without 2π exceeds 0.328: its conjugated numerator is at least $0.95(0.655) - 0.040(0.159)$ and its squared denominator is at most 1.37^2. These bounds imply (E.21).

Use a finite cotangent contour of height $h = 1/500$ for the original integer sum. With $X = \Re(x - x_*)$, $Y = \Im(x - x_*)$, Taylor's formula gives

$$\Re[A(x) - A(x_*)] \le -6X^2 + 44|XY| + 22Y^2. \tag{E.22}$$

On a vertical endpoint side, $|X| \ge 0.047$, $|Y| \le 0.005$; the right side is below -0.002. On a horizontal side its maximum in X is at most $(308/3)Y^2 \le 77/30000$, below the nonconstant cotangent-mode loss $2\pi h = \pi/250$. The constant-mode integral translates to x_* with the same endpoint loss, and (E.21) gives a nondegenerate Gaussian on its real tangent. All canonical G_N arguments stay in regular bulk strips; no periodic normalization of the interpolation has been changed.

Thus each exact edge sum has a nonzero analytic amplitude and the usual uniform $O_R(N^{-1})$ relative expansion on this disk, with all midpoint offsets retained. Its leading factors and $-A_{xx}(x_*)$ never vanish. Multiplication gives the whole packet expansion with phase (E.20), uniformly for the fixed tree.

Useful explicit nonlinear constants follow from (E.19). Let $\sigma = \sinh(r/2)$, $k = \cosh(r/2) - 1$, $C = (1 - \sigma^4)^{-1/2}$. The normalized diagonal error is at most $2kC/(1-k) + C - 1 < 1/50$, and the off-diagonal bound is $\sigma^2 C$; their sum is below $1/25$. For example $k < 0.008$, $\sigma < 0.126$ prove these estimates. Set $\delta = 1/50$, $\varepsilon = 1/25$. Edgewise summation gives

$$|\mathcal{E}_v/\tau_v| \le \delta a_v, \qquad \sum_w |\mathcal{E}_{vw}| \le \varepsilon a_v, \qquad |\mathcal{E}(\boldsymbol{\tau})| \le \tfrac{\varepsilon}{2} \sum_v a_v |\tau_v|^2. \tag{E.23}$$

The quotient extends at $\tau_v = 0$. For the first bound integrate the diagonal Hessian error with other coordinates fixed. The last bound uses $2|zw| \le |z|^2 + |w|^2$ on each edge, not a dimension-dependent maximum-norm estimate.

E.2.6 A uniform raw angular Gaussian

For $k/N \in I$ the exact product is

$$\frac{e_k(-2\cos t)}{e_k(-2)} = \prod_{j<k} \left(1 - \frac{\sin^2(t/2)}{\cos^2((j+1/2)\pi/N)} \right). \tag{E.24}$$

Put $u = \sin^2(t/2)$ for real t. The first $\lfloor N/5 \rfloor$ factors have absolute value at most $1 - \alpha_1 u$, $\alpha_1 = 2\sqrt{5} - 4$; the remaining at most $N/10 + 1$ factors are bounded by $1 + \alpha_2 u$, $\alpha_2 = 2\sqrt{5}/5$. These

follow by checking the two linear pieces of $|1-u/c|$ with $c \ge \cos^2(\pi/5)$ and $c \ge \cos^2(3\pi/10)$, respectively. Therefore

$$\left|\frac{e_k(-2\cos t)}{e_k(-2)}\right| \le e^2 e^{-\alpha N \sin^2(t/2)}, \qquad \alpha = \frac{9\sqrt{5}-20}{25} > \frac{1}{250}. \tag{E.25}$$

For complex $t+iy$, $|y| \le 1$, one factor's squared modulus is at most its real squared value plus $18y^2$. Indeed its denominator $c > 1/3$, its real absolute value is below two, $\cosh y - 1 \le y^2$, and $|\sinh y| \le 2|y|$. The real linear majorants above are at least $1/2$; consequently the additional factor is at most e^{36y^2}. Since $k \le 3N/10$,

$$|e_k(-2\cos(t+iy))| \le e^2 e_k(-2) e^{N[-\alpha \sin^2(t/2)+11y^2]}. \tag{E.26}$$

Use $\sin^2(t/2) \ge t^2/\pi^2$ for $|t| \le \pi$. Summing the positive edge references and retaining the bounded edge insertions yields

$$|B_N(\mathbf{t}+i\mathbf{y}, s)| \le N^A \exp N \left[F_0 - \frac{1}{400}\sum_v a_v t_v^2 + 70 \sum_v a_v y_v^2\right]. \tag{E.27}$$

We used $2\alpha/\pi > 1/400$, $22\pi < 70$. Endpoint translations $O_R(N^{-1})$ affect only prefactors.

E.2.7 Critical branches without reciprocal-slope smallness

Set

$$b_v = n_v/(2\pi), \quad D_v = a_v - ib_v, \quad t_0 = 1/16, \quad L_v = \lfloor t_0 |D_v| \rfloor. \tag{E.28}$$

Condition (E.9) implies $|D_v| > 32$ for all v, so $L_v \ge 1$. For real continuous $0 \le \ell_v \le L_v$, the critical equation is

$$D_v \tau_v = -i\ell_v + \mathcal{E}_v(\boldsymbol{\tau}). \tag{E.29}$$

The right-hand map after division by D_v is a contraction on the radius-$1/4$ polydisk, with Lipschitz constant at most ε. Its image has radius at most $t_0 + \delta/4 < 1/4$. Thus there is one analytic critical branch, also when some $n_v = 0$.

Writing $Q_v = \mathcal{E}_v/\tau_v$, the exact identity is

$$\tau_v = -i\ell_v/(D_v - Q_v). \tag{E.30}$$

For positive ℓ_v, $\Im \tau_v < 0$. With

$$\gamma_+ = \frac{1+\delta}{(1-\delta)^2} = \frac{2550}{2401}, \qquad \gamma_- = \frac{1-\delta}{(1+\delta)^2} = \frac{2450}{2601}, \tag{E.31}$$

one obtains

$$\gamma_- \frac{\ell_v a_v}{|D_v|^2} \le -\Im \tau_v \le \gamma_+ \frac{\ell_v a_v}{|D_v|^2}. \tag{E.32}$$

For $\Phi_\ell = \Psi + \frac{i}{2}\sum b_v \tau_v^2 - i\sum \ell_v \tau_v$ and its stationary value $\mathcal{C}(\ell)$, $\partial_{\ell_v} \Re \mathcal{C} = \Im \tau_v$. Integration in the independent charges proves

$$\Re \mathcal{C}(\ell) \le \Re \mathcal{C}(\mathbf{1}) - \frac{\gamma_-}{2} \sum_v \frac{a_v}{|D_v|^2}(\ell_v^2 - 1), \quad \ell_v \ge 1, \tag{E.33}$$

$$F_0 - \beta \le \Re\mathcal{C}(\mathbf{1}) \le F_0, \qquad \beta = \frac{\gamma_+}{2}\sum_v \frac{a_v}{|D_v|^2} = \pi\gamma_+\mathfrak{S}. \tag{E.34}$$

Each positive-charge coordinate is nonzero and has modulus below $t_0/(1-\delta) < \pi/2$, so the actual factor $\prod \sin\tau_v$ is nonzero. On a real tangent the Hessian has real part at most $-(1-\varepsilon)\,\mathrm{diag}(a_v)$. The Gaussian determinant is nonzero.

E.2.8 Infinite tails and connectors

Keep the exact full-normal charge coefficients

$$c_{\ell,N}(\sigma) = (-1)^{(N-1)(\ell-1)}U_{\ell-1}(\cos 2\pi\sigma), \qquad c_{1,N} = 1. \tag{E.35}$$

Initially use formal parameter circles avoiding their removable sine denominators, as in the joint Fourier transform. Let $A_0 = 1/400$ and choose

$$y_v = A_0 t_0 a_v/|D_v|. \tag{E.36}$$

With spectators real or on physical translated grids, the exponent for this coordinate is bounded by

$$F_0 - a_v t^2/400 + |b_v t y_v| - \ell y_v + 70 a_v y_v^2. \tag{E.37}$$

After maximizing t, the positive quadratic coefficient is $70a_v + 100b_v^2/a_v \le 100|D_v|^2/a_v$. Since $L_v + 1 > t_0|D_v|$, every omitted charge loses at least

$$\tfrac{1}{2}A_0 t_0^2 a_v \ge 1/(409600\pi). \tag{E.38}$$

Further charges carry $e^{-N(\ell-L_v-1)y_v}$. The coefficient bound $\ell e^{2\pi R'(\ell-1)}$ is absorbed for large N at this fixed tree.

Original endpoint loss is $\pi^2 a_v/400$. The chirp, finite-charge connector and imaginary Gaussian costs are bounded respectively by $\pi A_0 t_0 a_v$, $A_0 t_0^2 a_v$, and $70A_0^2 t_0^2 a_v$, less than half that loss in total. The zero-mode current obeys the same estimate. Exact telescoping of physical functionals and finite real-integral approximants therefore gives the finite-charge formula with (E.38). No spectator uses another coordinate's complex filling contour.

Next restrict real angles to $[-\rho,\rho]^m$, $\rho = 1/8$. The complement loses $1/(51200\pi)$ by (E.27). Translate this box to the imaginary part of its critical point. On any connector write $\tau_v = t_v - iv_v$, with $0 \le v_v \le -\Im\tau_{\ell,v}$. These points stay in the radius-1/4 polydisk. From (E.23) and Young's inequality,

$$\Re\Phi_\ell(\mathbf{t} - i\mathbf{v}) \le F_0 - \tfrac{1}{8}\sum_v a_v t_v^2 + \sum_v \left[\tfrac{1+\varepsilon}{2}a_v + \frac{2b_v^2}{(3-4\varepsilon)a_v}\right] v_v^2 - \sum_v \ell_v v_v. \tag{E.39}$$

The last two sums are at most $-\frac{1}{4}\sum_v \ell_v v_v$, because

$$\gamma_+ \max\{(1+\varepsilon)/2,\ 2/(3-4\varepsilon)\} < 3/4. \tag{E.40}$$

A connector face has some $|t_v| = \rho$, and hence loss at least $1/(1024\pi)$. This retains the negative Fourier contributions of all spectator coordinates instead of adding their imaginary growth costs independently. The translated real box has uniformly negative Hessian. Its translated horizontal integral has a relative Gaussian expansion at its own critical point. After

the connectors are restored, the estimate for a nonminimal charge is its saddle term plus $O(\mathrm{poly}(N)e^{N(F_0-\eta)})$, with η as chosen below. A connector need not be below that nonminimal charge's much smaller critical action. It is below the first-charge action, which is all that is required: (E.33) and this additive error give strict first-charge dominance. Only the first charge is asserted to have a relative asymptotic for the original contour.

E.2.9 The actual analytic conclusion

Allowing polynomial prefactors, all source, real-box and charge-tail errors above have loss at least

$$\eta = 1/(819200\pi). \tag{E.41}$$

The finite-sum contour errors have larger fixed edge losses from Section E.2.5. Condition (E.9) gives

$$\beta \le \pi(2550/2401)2^{-25} < \eta/2. \tag{E.42}$$

Thus all discarded errors lie below the same nonzero first-charge action. Only these fixed losses are compared with the total drop. The charge-dependent gaps (E.33) are compared directly with that filled action, not with the augmented reference.

The first coefficient (E.35) is exactly one for both parities. The entire difference of physical traces and their finite-charge approximation is holomorphic through the collisions. The maximum principle gives the estimates on every $|s| \le R < 1$, including actual constant and moving types, without assuming separate removability of divided endpoint currents.

The unchanged primitive count gives final power $N^{3/2}$. All original crossing, mixed-reference, pivot and zero-framing factors are retained exactly. The resulting reduced invariant has

$$\frac{F_N(q_N^s)}{F_N(1)} = A_N(s)H_N(s)(1 + O_{\mathcal{T},\mathbf{n},R}(N^{-1})), \qquad \frac{2\pi}{N}\log|F_N(q_N^s)| \longrightarrow 2\pi\Re\mathcal{C}(\mathbf{1}), \tag{E.43}$$

uniformly on the disk, and is nonzero there for sufficiently large levels. The edge insertion rule also gives its cotangent correction at this same continued wire saddle. Section E.2.11 supplies its geometric volume and cusp interpretation.

E.2.10 Explicit analytic nonvacuity

For a star with center degree B, arbitrary fixed center power n_0, and leaf powers $|n_j| \ge B$,

$$\mathfrak{S} = \frac{B}{B^2 + n_0^2} + \sum_{j=1}^{B} \frac{1}{1 + n_j^2} < 2/B. \tag{E.44}$$

Thus $B \ge 2^{26}$ suffices, even for short or zero n_0. For two adjacent centers each of degree $B+1$, with B leaves at each and every leaf power of magnitude at least B, both center powers may be arbitrary fixed short integers: $\mathfrak{S} < 4/B$, so $B \ge 2^{27}$ suffices. The large constants reflect deliberately coarse global estimates. The general quantity (E.8) allows any number and arrangement of short high-degree circles within the sum budget.

There is also an intrinsic interpretation at the original complete auxiliary cusps. The marked peripheral calculation gives $U = \epsilon v - 2\pi i t$, $V = -2v$, with $v = \log[\sin \pi(x-t)/\sin \pi(x+t)]$. At completeness, $v_t = -2\pi c$, where $c = \cot(\pi x)$. Consequently the primitive cusp shape is

$$\tau_{\mathrm{aux}} = \frac{dV}{dU} = \frac{2ic}{1 - i\epsilon c}.$$

The complete tree geometry, Section 5, gives $c_v = 1/d_v$ for the circle of degree d_v. For the original filling slope $\mu_v + p_v\lambda_v$, direct substitution now gives

$$\mathcal{L}_v^2 = \frac{|1 + p_v\tau_{\mathrm{aux},v}|^2}{\Im \tau_{\mathrm{aux},v}} = \frac{1 + n_v^2 c_v^2}{2c_v} = \frac{d_v^2 + n_v^2}{2d_v}, \qquad \mathfrak{S} = \frac{1}{2}\sum_v \mathcal{L}_v^{-2}. \tag{E.45}$$

Here $\mathcal{L}_v$ is the length of the actual primitive filling slope divided by the square root of cusp area; the retained half-twist is included through $n_v = \epsilon_v - 2p_v$. Thus (E.9) is equivalently $\sum_v \mathcal{L}_v^{-2} \le 2^{-24}$ for this class. This is a translation of the analytic hypothesis using the proved cusp shape, not an imported geometric filling criterion.

These examples retain the original knot/parity qualification. Some star presentations overlap the existing long-pretzel and fixed-outer-power belt-stack results; zero boxes may admit the already known topological reduction. The graph-wide analytic criterion does not use those reductions.

E.2.11 Identification with the knot volume

Small angular variables do not imply that the primitive meridian logarithms are small: a high-degree circle collects contributions from many edges. The geometric proof therefore follows the entire path of filling equations, with its logarithm branches retained.

The ordered decomposition cuts the original augmented exterior into two octahedra per tree edge. Its literal strand matchings retain every face pairing, half twist and primitive peripheral path. The angular construction subdivides these octahedra into eight positive tetrahedra per edge on the chart $|\tau_v| < 10^{-2}$. Section E.2.7 places the whole path

$$\Psi_{\tau_v} + \frac{in_v\tau_v}{2\pi} = ic, \qquad 0 \le c \le 1,$$

inside this chart. The completion proof computes the actual periods

$$U_v = -2\pi\Psi_{\tau_v} - i\epsilon_v\tau_v, \qquad V_v = 2i\tau_v, \qquad U_v + p_vV_v = -2\pi ic.$$

It verifies that the primitive slope bounds a tube of cone angle $2\pi c$. At $c = 1$ the tube is smooth, the auxiliary fillings are precisely the original surgeries, and the main knot cusp remains complete. The same proof fixes the additive volume constant at the complete augmented link and integrates the boundary-corrected action. It gives

$$2\pi \operatorname{Re}\mathcal{C}(\mathbf{1}) = \mathrm{Vol}(S^3 \setminus K). \tag{E.46}$$

Thus the analytic conclusion above is an actual uniform volume theorem under (E.9). The preferred-cusp continuation also identifies the first correction geometrically. This establishes the summed criterion, not arbitrary short fillings or a criterion for graphs with cycles.

E.2.12 Reproducing the edge calculations

The elementary Newton estimates give a second direct proof of the uniform discarded-state and complex-strip bounds. The exact edge equations and finite samples of that strip inequality can be checked with:

```
python3 repro/run_research_checks.py curvature-fillings
```

The script verifies four symbolic identities and samples the printed complex inequality at both even and odd levels. Its recorded output states that finite scope. The uniform inequalities, contour argument and asymptotic conclusion are proved above; the script does not establish geometric completion.

E.3 Uniform Newton bounds with explicit constants

The separately available version of this section is uniform newton bounds with explicit constants. The proof is reproduced here in full.

This section proves two uniform estimates used in the quantitative filling limit. It does not identify a complex critical value with the volume of a filled complement. The latter still requires the original marked geometry and smooth completion.

Fix a finite ordered tree. Every vertex carries its actual crossing-circle operator, including the degree-one vertices. Write d_v for a vertex degree and $a_v = d_v/(2\pi)$. All graph data and integer crossing powers are fixed before the level N tends to infinity. Constants in polynomial prefactors may depend on these fixed data. The exponential constants below do not depend on the tree.

E.3.1 A uniform loss outside the retained interval

Use the fixed complete Newton basis, the physical diagonal edge factors, the full value-and-first-derivative norm, and the fixed-power crossing bounds from the full polynomial trace bounds. A low basis element has norm at most a polynomial times $\exp(Ng(k/N))$; a high basis element has polynomial norm. Here

$$g(x) = J(2x) - 4J(x), \qquad J(x) = \int_0^x \log(2\sin \pi u)\, du,$$

$$g'(x) = 2\log\cot(\pi x), \qquad g''(x) = -\frac{4\pi}{\sin(2\pi x)} \le -4\pi.$$

Thus, with $G_0 = g(1/4)$,

$$g(x) \le G_0 - 2\pi(x - 1/4)^2.$$

Each physical edge index occurs at its two ends. Quotient multiplication, a fixed crossing power, and the trace cost only a polynomial in the full norm. Consequently the absolute scalar sum is bounded by a polynomial times the sum of the products of the squared basis bounds. There are only polynomially many edge-index assignments for a fixed tree. Its complete maximum is $F_0 = 2|E|G_0$.

Retain the interval $I = [1/5, 3/10]$ on every edge. A low edge outside I loses at least $\pi/100$ in this rate. A high edge loses $2G_0$, which is larger. The sum of every discarded assignment therefore has upper rate at most

$$F_0 - \pi/100 + o(1).$$

Equivalently, any fixed loss strictly smaller than $\pi/100$, such as $\pi/200$, holds with a polynomial prefactor for all sufficiently large levels. This statement retains both level parities and every first normal. It uses no comparison with a complete tensor at the same high index.

E.3.2 An elementary Gaussian bound on the complex angular strip

Let $N \geq 160$ and $k/N \in I$. Put

$$r_j = \cos \frac{(2j+1)\pi}{N}, \qquad Q_r(z) = \frac{r + \cos z}{1 + r}.$$

The exact normalized basis value is

$$\frac{e_k(-2\cos z)}{e_k(-2)} = \prod_{j=0}^{k-1} Q_{r_j}(z).$$

Although some individual factors have a negative constant Laurent coefficient, those factors can be paired before taking absolute values.

Reserve the first $m = \lfloor N/32 \rfloor$ factors. Their parameters satisfy $r_j \geq \cos(\pi/16) > 9/10$. There are at most $h \leq N/20 + 1$ negative parameters, all greater than $\cos(3\pi/5) > -2/5$. Pair each of these with a different parameter from the next h positions after the reserved ones. These positions are disjoint from the negative ones and have parameter at least $\cos(\pi/4) > 7/10$, because $m + h \leq N/8$. Every remaining unpaired parameter is nonnegative.

For a pair $r > -2/5$, $s > 7/10$, the reciprocal polynomial

$$(u^2 + 2ru + 1)(u^2 + 2su + 1)$$

has coefficients $1, 2(r+s), 2+4rs, 2(r+s), 1$, all nonnegative. After division by its value at $u = 1$ and multiplication by u^{-2}, these are the weights of a symmetric probability measure supported on $\{-2, -1, 0, 1, 2\}$. A remaining single nonnegative factor has the analogous interpretation on $\{-1, 0, 1\}$. For such a factor of Laurent degree b,

$$|Q(t + iy)| \leq \cosh(by) \leq \exp(b^2 y^2/2).$$

The reserved factors supply the real angular decay. If $r \geq 9/10$ and $w = \sin^2(t/2)$, then

$$|Q_r(t)| = \left|1 - \frac{2w}{1 + r}\right| \leq 1 - w/2.$$

For $|y| \leq 1/4$, the identity

$$|r + \cos(t + iy)|^2 = (r + \cos t)^2 + 2r\cos t(\cosh y - 1) + \sinh^2 y$$

and $2(\cosh y - 1) + \sinh^2 y \leq 4y^2$ give

$$|Q_r(t + iy)|^2 \leq 1 - w/2 + 4y^2, \qquad |Q_r(t + iy)| \leq \exp(-w/4 + 2y^2).$$

Since $m \ge N/40$, multiplication gives the deliberately conservative bound

$$\boxed{\left|\frac{e_k(-2\cos(t+iy))}{e_k(-2)}\right| \le \exp N\left\{-\frac{\sin^2(t/2)}{160}+4y^2\right\}.}$$

For $|t| \le \pi$, use $\sin^2(t/2) \ge t^2/\pi^2$. Multiply the bound over the incidence factors of the original finite packet, retain the bounded physical edge factors, and sum its polynomially many indices. One obtains

$$|B_N(\mathbf{t}+i\mathbf{y},s)| \le CN^A \exp N\left\{F_0 - \frac{1}{400}\sum_v a_v t_v^2 + 70\sum_v a_v y_v^2\right\}, \qquad |s| \le R < 1.$$

Indeed the stronger constants from the preceding calculation are $1/(80\pi)$ and 8π. The displayed constants leave room for the physical translated endpoints of size $O_R(N^{-1})$, which change the exponent by a bounded amount for a fixed graph and fixed strip. No assertion here concerns a growing graph as N tends to infinity.

E.3.3 The explicit action of one edge

The two-ended Newton action is the critical value in x of

$$R(x,t)+R(x,u), \qquad R(x,t)=\int_0^x \log(2\cos(2\pi v)+2\cos t)\,dv - 2J(x).$$

The logarithm is continued from $(x,t,u)=(1/4,0,0)$. Put $p=\tan(t/2)$, $q=\tan(u/2)$. The wire critical equation gives

$$\tan^2(\pi x) = \frac{1}{1+p^2+q^2}.$$

If $\psi(t,u)$ denotes the selected critical value, direct differentiation of the integral yields

$$\psi_t = -\frac{1}{\pi}\operatorname{arsinh}\frac{p}{\sqrt{1+q^2}},$$

$$\psi''(t,u) = \frac{1}{2\pi\sqrt{1+p^2+q^2}}\begin{pmatrix}-(1+p^2) & pq\\ pq & -(1+q^2)\end{pmatrix}, \qquad \det\psi'' = \frac{1}{4\pi^2}.$$

All square roots and inverse functions use their branches at the origin. The full packet action is the sum over its physical edges. Its Hessian at zero is therefore $-\operatorname{diag}(d_v/(2\pi))$. On the real angular chamber the edge Hessian is strictly negative definite. For the complex contours, this real statement must be supplemented by an explicit holomorphic bound; it does not prove global complex nondegeneracy or the geometric filling assertion.

Appendix F

The ordered octahedral geometry and the filled cusp

This appendix supplies the geometric proof for Theorem 9.1. It starts with the actual ordered tangle diagram and identifies its projection cut, rather than recovering a manifold from a scalar identity. Section F.1 constructs the two-octahedron block at every physical tree edge and tracks the primitive peripheral curves. Section F.2 gives positive tetrahedral shapes and the complete face maps. Section F.3 completes the original fillings and calibrates their volume. Section F.4 follows the preferred main longitude.

An ideal vertex lies at the sphere at infinity of hyperbolic space. An ordered ideal tetrahedron is specified by its cross ratio; a positive shape has positive imaginary part in the chosen orientation. A logarithmic peripheral period records the full change of a continued logarithm along a marked cusp curve. It is not reduced modulo $2\pi i$. This distinction matters here: each local angular parameter remains small, but the primitive meridian collects contributions from all incident edge blocks. Its total logarithm reaches one full turn at the smooth filling endpoint.

The shaded faces below come from cutting the original twice-punctured crossing disks; the white faces come from the projection regions. Their labels determine the gluing maps and the original cyclic order. The word "pants" denotes the two-triangle thrice-punctured sphere obtained by joining the projection copies of a disk-side face. It always retains its three labeled punctures. The local Möbius maps C and T realize the closing and turn operations in the displayed coordinate charts. The symbol R in those charts is a normalized ideal vertex parameter, distinguished from the crossing operator by its definition in Section F.2.1.

F.1 The ordered octahedral decomposition

The separately available version of this section is the ordered octahedral decomposition. The proof is reproduced here in full.

A crossing circle surrounds two strands of a link and bounds a disk punctured by them. Cutting along the projection regions and these disks opens the augmented link exterior into two balls. For the ordered tangle constructions considered here, each ball can be assembled from octahedra, with one octahedron for each edge of the tangle-operation tree. The proof identifies every boundary vertex and face, so it also identifies the original meridians and disk

longitudes after the pieces are joined.

The tangles have two strings and four ordered endpoints. Their operations are vertical stacking, quarter-turns in the projection plane, and the specified cylindrical closure. Every product junction in the reduced operation tree carries an original crossing circle; there are no junctions without a circle. The tree, the order of its branches, and all endpoint frames are retained throughout. Faces created by opening crossing disks are called shaded faces; the pieces of the projection regions are called white faces.

This is the topological input for the quantitative volume theorem. The angular cell calculation provides positive shapes for its octahedra, and the completion theorem uses the path supplied by the analytic filling criterion. The present proof concerns the augmented exterior before filling.

F.1.1 The actual ordered object and the statement

At each vertex v of a finite tree put an original two-strand crossing-circle tangle P_v. It retains the crossing R^{ϵ_v}, where $\epsilon_v \in \{-1, 0, 1\}$, and its auxiliary component bounds the original twice-punctured disk. Every edge is an actual quarter-turn between two ordered product junctions. There are no product junctions without their own original circle. Terminal circles are included as vertices of degree one. The tree has at least one edge. Write T for the actual quarter-turn, C for the cylindrical closure, and R for one positive braid crossing in the original ordered terminal frame.

The product order, the four endpoint frames, every rotation of a child ball, and every original disk orientation are part of the data. A representative recursive description has

$$W_v = P_v \prod_{w \text{ child of } v} T(W_w). \tag{F.1}$$

The product here is in its specified order. At the closing sphere one may use the actual collar move to expose a root circle. A double turn then leaves its actual half-turn decoration on the affected subtree; it is not replaced by the scalar identity $T^2 = 1$. The relevant moves are the literal closed-turn and cyclic collar isotopies in the original ordered tangle moves, equations (14)–(18). Their four collar sectors, and any reversal of the child order, are retained. Thus (F.1), with those transported frames, describes the original all-circle ordered operation tree. It does not introduce a commutativity move on open two-tangles.

For clarity, the literal root adapter is

$$C(h_0 T(E)) = C((TP_0)T(E)) \simeq C\Big(P_0\, T^2(E)\Big) = C(P_0\, \tau(E))\,. \tag{F.2}$$

Here τ is the actual half-turn of the ball. If $E = L_1 \cdots L_k$, then $\tau(E) = \tau(L_k) \cdots \tau(L_1)$, and $\tau(TW) = T\tau(W)$. Thus the adapter reverses the displayed root order and transports every child frame; it does not suppress this rotation. The two-leaf case of (F.2) is the base used below.

More generally the original circle coupon can occur between two ordered lists of child factors in a nonroot junction. Keep that position:

$$W_v = \left(\prod_{w \in I_v^-} T(W_w)\right) P_v \left(\prod_{w \in I_v^+} T(W_w)\right). \tag{F.3}$$

The lists record the cut at the parent edge in the original cyclic order. The left-insertion version of the calculation below handles this form without moving P_v through a child. Reassociation of a stack and transport of genuine turn-chain rotations put the literal circle-only tree in this form. At the root only the actual cyclic closed-collar move is used to choose its cutting position.

Write E for the number of tree edges, $V = E + 1$ for the number of circles, and d_v for the number of edges incident to circle v. The exterior admits the following labeled cut:

- There is one pair of octahedral balls per original tree edge.
- At a circle v, its incident edge blocks are joined in the original cyclic order. There are d_v disk-side pages, and exactly one of their cyclic seams retains the original disk identification and its ϵ_v crossing. All other page seams are the untwisted labeled identity seams supplied by the ordered insertion.
- Cutting open those distinguished disk seams and the projection faces gives two balls. Each is a tree of E octahedra attached along $E - 1$ triangular faces. Its remaining boundary has exactly the two original shaded triangles of every crossing circle and the original white face cycles.

Here an octahedron means the explicitly labeled topological cell. The angular charts give its positive shapes. The topological proof does not require the union of several cells to be convex.

F.1.2 How endpoint pairings determine the cut

Use the closed-cut convention of the projection and crossing-disk cut, sections 2–4. For the turned elementary leaf $h_v = T(P_v)$, label its four physical half-segments

$$(v_0, v_1, v_2, v_3) = (\text{lower left}, \text{lower right}, \text{upper right}, \text{upper left}). \tag{F.4}$$

A quarter-turn cyclically shifts this tuple. A lower factor F followed by an upper factor G joins F_2 to G_1 and F_3 to G_0, leaving exposed (F_0, F_1, G_2, G_3). Cylindrical closure joins positions 0 to 3 and 1 to 2. These are actual empty joining strips in the projection plane.

One ideal vertex is assigned to each paired main segment and one vertex O_v to each original crossing-circle semicircle. The two shaded triangles of a leaf are

$$(O_v, v_1, v_2), \qquad (O_v, v_3, v_0), \tag{F.5}$$

where a half-segment in this display denotes the main-segment vertex containing it. Around O_v the cyclic order is (v_0, v_1, v_2, v_3). At a main-segment vertex the two shaded corner sectors concatenate in the order of the actual planar joining strip. These sectors determine all white face cycles by the usual boundary walk. In particular the full oriented boundary ribbon graph follows from the literal matching; matching only face or component counts would not suffice.

This endpoint construction records the reversible projection/disk cut. Drilling a boundary-parallel auxiliary semicircle and then opening its recorded half disk restores a ball. The disjoint leaf boxes permit this for all circles. White faces reglue by their actual projection labels. The two shaded faces reglue by the original disk map, including the retained half twist. The proof of that cut uses no quantum source identity.

For the next local calculation it is useful to use an inverse turn:

$$P_v = T^{-1}(h_v), \qquad \text{its exposed tuple is } (v_3, v_0, v_1, v_2). \tag{F.6}$$

This is an exact physical identity by the definition of h_v. It avoids using $T^2 = 1$ on an arbitrary physical tangle.

F.1.3 One terminal insertion is exactly one octahedron

Suppose the unturned P_v occurs at the beginning of its specified local stack. In the current full diagram, let x_1 and x_2 be the half-segments matched to v_1 and v_2 outside that box. Insert a new terminal child h_w immediately after P_v. The literal product rule changes precisely the following two matches:

before	after	
$v_1 - x_1$	$v_1 - w_1,$	$w_2 - x_1$
$v_2 - x_2$	$v_2 - w_0,$	$w_3 - x_2.$

(F.7)

All other half-segment matches are unchanged. This remains true if x_1 or x_2 is in a nested factor or outside the local ball: the replacement only uses the two exposed output strips of P_v.

Name the old two vertices $C = (v_1, x_1)$, $B = (v_2, x_2)$, and name the new internal vertices $D = (v_1, w_1)$, $A = (v_2, w_0)$. The outside vertices C and B are now represented respectively by (w_2, x_1) and (w_3, x_2). Attach the octahedron with vertices

$$(O_v, O_w, A, B, C, D). \tag{F.8}$$

and equatorial cycle (A, B, C, D). Its four shaded triangles are

$$(O_v, B, C), \quad (O_v, D, A), \quad (O_w, D, C), \quad (O_w, B, A). \tag{F.9}$$

Glue the first one to the old shaded face (O_v, C, B). The two orientations are opposite, as required for joining oriented balls. The old face disappears. The second triangle in (F.9) is exactly the new v face in (F.5), and the last two are exactly the new w faces. The other shaded face of v is untouched.

The four white triangles are the complementary octahedral faces. Along the three edges of the glued triangle, their unions with the old white pieces are precisely the new white regions. One can verify this directly: at each of B, C, O_v remove the two glued shaded sectors and concatenate the remaining sectors; their order is exactly the planar strip substitution (F.7). The new vertices A, D carry the new identity strips in that same order. Thus every remaining shaded sector and every cyclic vertex order agrees with the endpoint construction. Following these sectors around each white face therefore gives the same boundary cycle.

Perform the same attachment in the other projection copy. This adds exactly one two-octahedron edge block and introduces one new original circle. It neither identifies two old circles nor inserts a circle at the attachment seam.

The calculation also proves the inverse operation: remove a terminal edge block, restore the deleted triangular face, and undo (F.7). The original diagram loses that terminal child and otherwise retains its actual strand connections and order.

There is also an explicit left-insertion version, needed for (F.3). Inserting h_w immediately before P_v changes

$v_3 - x_3$	$v_3 - w_3,$	$w_0 - x_3$
$v_0 - x_0$	$v_0 - w_2,$	$w_1 - x_0.$

(F.10)

This is the same octahedral attachment on the other shaded face (O_v, v_3, v_0). The two formulas are direct applications of the same lower-output/upper-input stack rule. They retain the physical rotation of each tangle and all its endpoint labels.

F.1.4 Why this covers every ordered tree

Start with one actual tree edge. The two-leaf matching gives exactly the octahedral boundary (F.9), in both projection copies. This is the labeled two-circle block, after the recorded endpoint-frame rotation. Its proof is a disk/projection cut, not an equality of two scalar sources.

For a general tree, choose a terminal vertex as root and start with its unique incident edge. Expose that root by the actual closed-collar moves from section F.1.1, retaining all transported frames and orders. The initial root then has no further children to insert; at the other endpoint the only existing edge is its parent edge. At each vertex add the children on the right of P_v in reverse final order, always immediately after P_v. Add the children on its left in their final order, always immediately before P_v. This is a construction of the specified word, not a permutation of an existing word: successive insertions produce exactly the two ordered lists in (F.3), including the special case (F.1). A new child initially has only its own crossing-circle tangle; its children are subsequently inserted inside its own ball. Its actual quarter-turn carries the whole local insertion, including the two projection sides and all four endpoint labels. Any previously recorded half-turn or input/output reversal similarly carries the complete insertion and its order. It does not remove the frame decoration.

Induction now gives exactly the original recursive word and its full boundary ribbon graph. Every step glues a ball along a triangular disk, so each projection copy stays a ball. There are $E - 1$ attachments after the initial octahedron. The shaded face count is

$$4E - 2(E - 1) = 2V. \tag{F.11}$$

The count is only a consistency check: the insertion proof has already identified each of these two faces with its particular original circle and each main-segment vertex with its literal pair.

The equality of oriented ribbon graphs gives a label-preserving homeomorphism of the final boundary spheres. Extend it over the balls by a ball extension. All original white and disk-face pairing maps are matched, hence the quotient is the original link exterior. This is stronger than an identification of the unmarked manifold: the main arcs, individual auxiliary circles, disk sides, and their specified collar paths are preserved.

The extra triangular interfaces provide the promised pages. In the whole exterior an upper interface triangle and its lower partner form a two-triangle, thrice-punctured sphere. The ends are the same original circle and its two physical main punctures. Its compactification is a twice-punctured spanning disk for that circle. These pages are obtained from the explicit cell subdivision and the preceding homeomorphism; we did not assume that a single original disk cut somehow creates all d_v pages. The insertion order places the pages consecutively in exactly the original cyclic order at that circle. There are $d_v - 1$ new seams there, plus the original distinguished disk seam.

F.1.5 Retained half twists and reversed disk frames

First perform the projection construction with the disk cuts open. A retained crossing changes their regluing, not the half-segment compiler of the open cut. At each v keep its original

distinguished seam throughout all insertions. Its original disk map is used once, after the untwisted page attachments. Thus no crossing is distributed among several edges, and no crossing is silently added at a new seam.

For $\epsilon_v = 0$ the original pure disk map pairs the two sides within the same projection copy. For $\epsilon_v = +1$ or -1 use its actual odd pairings: the punctures exchange and the two projection copies are permuted as in the original A,B disk maps. The signs +1 and -1 have their distinct primitive meridian words even though their endpoint permutation has the same parity. The explicit collar permutation is also recorded in the original half-twist collar, section 3a. A full twist changes the peripheral marking and must not be discarded simply because its endpoint permutation is trivial.

A physical rotation or reflection of a disk frame transports its puncture labels, its two pages, and its chosen meridian and longitude together. Apply the insertion table in that transported frame. In particular a reversal that exchanges the projection copies is applied to both halves of every incident block. There is no use of an arbitrary puncture permutation justified only by the rigidity of an abstract thrice-punctured sphere.

These conventions cover the original signed and reversed diagrams. The local geometric block may be relabeled by these same actual rotations. Its shape formula is not used to determine the topology.

F.1.6 Exact primitive peripheral concatenation

Fix one auxiliary circle v and orient its original disk-framed longitude λ_v and meridian μ_v as in the original marked leaf theorem. After the page cut, a small cusp torus of v consists of d_v annular sectors, one from each incident edge block. Every page boundary on this torus is the same primitive disk-longitude class. A small meridian crosses these sectors once in their cyclic order. The insertion (F.7) subdivides one such transverse passage into two consecutive passages; it neither winds the meridian a second time nor changes its longitude class.

This can also be read as a face-word statement. Each insertion replaces one original disk-face passage by two passages separated by the new triangular interface. In the final word all internal interface changes of coordinates occur in inverse consecutive pairs. After eliminating them, the retained word is the original disk meridian. Its ordered factors are one endpoint passage for each incident block, followed at the distinguished seam by the single original half-twist correction. This is a statement about the actual marked loop before evaluating its holonomy.

Use the local angular chart, with common oriented longitude variable τ_v at all pages of this circle. Its incoming/outgoing edge-ratio tables give exactly the inverse seam maps with the just-established puncture matching. In one edge block its small pure meridional increment is

$$u_{v,e} = -2\pi\,\partial_{\tau_v}\psi_e, \qquad V_v = 2i\tau_v. \tag{F.12}$$

This is the local primitive word, not an arbitrary logarithm of its multiplier. The longitude conjugates to the same element on all pages. Whenever it is loxodromic, its two fixed points are therefore carried to the same two marked points. In a coordinate with one at infinity and the other at q, integrate $d\log(z-q)$. The coordinate changes telescope in the concatenated path; the integral is the sum of the local increments (F.12). At the complete value it is the continuous limit of this equality.

The original disk words fix the half-twist correction unambiguously. In the notation of the marked-leaf theorem, the pure meridian has multiplier ρ, and the two odd primitive words have multipliers ρQ and ρ/Q. Write ν for the local leaf response, so $Q = e^{-\nu} = e^{i\tau_v}$ and $\nu = -i\tau_v$. The physical R^{-1} tangle uses the first word and R uses the second. Thus the single retained crossing contributes $\epsilon_v\nu = -i\epsilon_v\tau_v$, or equivalently $-\epsilon_v V_v/2$. Consequently the actual continued primitive logs are

$$U_v = -2\pi \sum_{e\sim v} \partial_{\tau_v}\psi_e - i\epsilon_v\tau_v, \qquad V_v = 2i\tau_v. \tag{F.13}$$

The original proof of these primitive words is the original marked face maps, sections 5–6, including its equations (15)–(18). The two-rectangle cusp lattice there certifies their primitivity. Our page insertion preserves those same paths; it does not choose a new rational meridian slope.

In particular the integral in (F.13) is not reset to a principal logarithm after several pages. Each summand follows its local small branch, and the actual concatenated loop retains their full sum. This is precisely where a large total logarithm can occur while every individual block stays in its small positive chart.

Along the continuous first-charge equations, with charge $c \in [0, 1]$,

$$\partial_{\tau_v}\Psi + \frac{in_v}{2\pi}\tau_v = ic, \qquad \Psi = \sum_e \psi_e, \qquad n_v = \epsilon_v - 2p_v, \tag{F.14}$$

(F.13) gives the exact original primitive filling word

$$U_v + p_v V_v = -2\pi ic. \tag{F.15}$$

At $c = 1$ this is one primitive turn. This calculation identifies the word and its continued logarithm. The completion proof verifies the positive-cell and nonsingular cusp-link conditions along the path and deduces the smooth complete filling.

F.1.7 From the cut to the geometric filling

This proof applies to the actual ordered circle-only tree, with all its inherited frames and disk collars. It does not assert such a cut for an arbitrary scalar tensor network, for arbitrary branching of six independent boundary slots, or for a graph with cycles. It uses neither commutativity of pair-algebra polynomials nor equality of source actions. No auxiliary component is filled or deleted in the topological identification.

The explicit terminal substitution (F.7), its ribbon-face realization (F.9), and the primitive path concatenation supply the ordered topological identification. The local angular charts give the positive cells and their full complex seam ratios. The analytic criterion constructs the charge path, and the geometric completion proves that its endpoint is a smooth complete metric on the stated filled topology.

F.2 Positive angular coordinates on each edge block

The separately available version of this section is positive angular coordinates on each edge block. The proof is reproduced here in full.

Each edge in an ordered tree of crossing circles contributes two octahedra to the cut link exterior, one in each projection copy. This section gives their vertices and tetrahedron shapes as explicit functions of the two endpoint longitude parameters. The shapes remain positive on a fixed complex disk, independently of the size of the tree. The two-triangle disk boundaries depend only on their own circle parameter, which is what allows incident blocks with different other endpoints to be joined.

The ordered decomposition identifies these cells and all their labels with the original link exterior. The completion proof then follows their actual logarithmic periods to the smooth filling. These are the geometric inputs to the quantitative volume theorem, whose angular path is supplied by the analytic criterion.

F.2.1 The actual two-leaf model

Start with the literal two-leaf closure $C(h_0T(h_1))$. At a pure leaf use the original coordinates below. Here T is the quarter-turn, C is the cylindrical closure, and h_j is the turned two-strand piece with its original crossing circle.

$$Q = (0, \infty, 1, R), \qquad Z = \frac{1+E}{1+H}, \qquad E = e^{2\pi i x}, \quad H = e^{2\pi i t}, \quad R = \frac{H(1+E)^2}{E(1+H)^2}. \tag{F.16}$$

The parameter R in this display is the fourth normalized ideal vertex of the quadrilateral. The second projection copy has apex Z/E, not a conjugation of the free parameters. The original turn and root closing maps are

$$T(z) = \frac{R_1 - 1}{z - 1}, \qquad C(z) = \frac{z - R}{z - 1}, \qquad R_1 = 1 - R. \tag{F.17}$$

These are the actual maps of the original marked face maps, equations (11), (14), and the original closed projection cut. In particular both maps must be used when placing the second leaf. Applying only the turn would put the two complete apices at the same point.

The complete two-leaf wire is $x = 1/4$. In each projection ball, the two ideal pyramids are joined along their ordered quadrilateral bases. Their six outer vertices form an octahedron. This is a literal local cut statement, not a deduction from its volume.

F.2.2 Solving in the two angular longitude variables

Write $\tau_0 = \tau$, $\tau_1 = \upsilon$, and prescribe the actual disk longitude logarithms

$$V_j = 2i\tau_j. \tag{F.18}$$

The pure leaf relation gives $v_j = -i\tau_j$, where $v_j = \log[\sin \pi(x - t_j)/\sin \pi(x + t_j)]$. On the branch through completeness, put

$$y = \tan(\tau/2), \qquad z = \tan(\upsilon/2), \qquad k = (1 + y^2 + z^2)^{-1/2}, \qquad k(0,0) = 1. \tag{F.19}$$

Then the exact solution of the two-leaf stationarity equations is

$$\begin{aligned} \tan(\pi x) &= k, \\ \tan(\pi t_0) &= iky, \qquad & \tan(\pi t_1) &= ikz, \\ E &= \frac{1+ik}{1-ik}, \qquad & H_0 &= \frac{1-ky}{1+ky}, \quad H_1 = \frac{1-kz}{1+kz}, \\ R &= \frac{1+z^2}{2+y^2+z^2}, \qquad & R_1 &= 1-R, \\ Z_j &= \frac{1+E}{1+H_j}. \end{aligned} \tag{F.20}$$

Indeed the logarithmic ratio equation first gives $\tan \pi t_j = ik \tan(\tau_j/2)$. Its ordered metric is

$$\kappa_j = \frac{k^2(1+\tan^2(\tau_j/2))}{1-k^2\tan^2(\tau_j/2)}. \tag{F.21}$$

The wire equation is $\kappa_0\kappa_1 = 1$. With $a = y^2+z^2$, it factors as $(k^2+1)((1+a)k^2-1) = 0$; the complete branch is (F.19). Thus $\kappa_0 = (1+y^2)/(1+z^2)$, which yields (F.20). No meridian-color approximation was used.

The two octahedra have the same equator $q = (0, \infty, 1, R)$, and apices

	A	B
$+$	Z_0	$\dfrac{RZ_1}{Z_1 - R_1}$
$-$	Z_0/E	$\dfrac{RZ_1}{Z_1 - ER_1}$.

(F.22)

For example, composing (F.17) gives $C(T(Z_1)) = RZ_1/(Z_1 - R_1)$. Replacing the second-copy apex by Z_1/E gives the last entry. At zero the two rows are

$$(A_+, B_+) = ((1+i)/2, (1-i)/2), \qquad (A_-, B_-) = ((1-i)/2, (1+i)/2). \tag{F.23}$$

The corresponding complete cells are the regular octahedra in their opposite labeled orientations.

F.2.3 Uniform positive tetrahedra on an explicit complex bidisk

Subdivide the plus octahedron into the four tetrahedra with ordered vertices (A_+, B_+, q_j, q_{j+1}), j mod 4. Reverse the last two vertices in the minus copy. Use the cross ratio

$$[a,b,c,d] = \frac{(a-c)(b-d)}{(a-d)(b-c)}. \tag{F.24}$$

All eight complete shapes are i. This triangulates the actual two-pyramid union, with the internal diagonal joining its two apices.

There is a uniform positive chart on

$$|\tau|, |\upsilon| \le 10^{-2}. \tag{F.25}$$

Here are deliberately loose explicit bounds. The elementary sine and cosine series give $|y|, |z| < 0.005001$. Hence $|y^2 + z^2| < 0.000051$, $|k - 1| < 0.000026$, $|E - i| < 0.00006$, $|H_j - 1| < 0.0102$, and $|R - 1/2| < 0.000026$. Substitution in (F.20) gives $|Z_j - (1+i)/2| < 0.004$. In each denominator in the B column of (F.22), the complete modulus is $1/2$, and its perturbation is less than 0.0041. The quotient estimate therefore bounds the motion of each finite vertex in (F.22) by 0.02; the remaining finite equatorial vertices are fixed or move by less than 0.000026.

Every finite pair of distinct complete vertices occurring in (F.24) has separation at least $1/2$. Each difference factor therefore changes by relative size less than 0.08. Comparing both the numerator and denominator products with one bounds the cross-ratio quotient change by

$$\frac{4(0.08) + 2(0.08)^2}{(1 - 0.08)^2} < 0.4. \tag{F.26}$$

For a factor involving infinity there are fewer factors and the same bound suffices. Thus every ordered shape lies in $|w - i| < 0.4$, and in particular has positive imaginary part. All denominators and ideal vertices remain distinct. The internal diagonal has total angle 2π: its product relation telescopes exactly, and its integer angle branch is fixed by (F.23).

This is positivity of the actual local geometric tetrahedra, not positivity of the original real (x, t_j) angle chamber. For complex angular parameters the equatorial quadrilateral need not be planar. The triangular octahedral boundary and the positive tetrahedral subdivision are the retained geometric objects.

F.2.4 The two disk-side pants depend only on their longitude

The local white matrices in a leaf frame are

$$\begin{aligned} W_0 &= \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, & W_1 &= \begin{pmatrix} 1 & E-1 \\ 0 & E \end{pmatrix}, \\ W_2 &= \begin{pmatrix} E-R & R(1-E) \\ E-1 & 1-ER \end{pmatrix}, & W_3 &= \begin{pmatrix} R & 0 \\ 1-E & ER \end{pmatrix}. \end{aligned} \tag{F.27}$$

The relation $Z^2 - (1 + E)Z + ER = 0$ is retained. Put

$$Q = \frac{E - Z}{Z - 1} = e^{-v} = e^{i\tau}. \tag{F.28}$$

The incoming shaded triangle has cyclic vertices $(Z, \infty, 1)$, and the outgoing one has $(Z, R, 0)$. A disk cut doubles each of them across its adjacent white faces, producing the actual two-triangle pants boundary. In the plus chart, transport the opposite triangle through the appropriate W_j^{-1}.

If an oriented edge has endpoints (a, b) and the third vertices on its two sides are c, d, measure its complex ratio by (F.24). Direct substitution gives the following cyclic tables:

incoming edge	(Z, ∞)	$(\infty, 1)$	$(1, Z)$
ratio	$-Q^{-1}$	$-Q$	$-Q^{-1}$
outgoing edge	(Z, R)	$(R, 0)$	$(0, Z)$
ratio	$-Q$	$-Q^{-1}$	$-Q$.

(F.29)

For example the first incoming ratio is $(1-Z)/(E-Z) = -Q^{-1}$. Across its base edge, the opposite apex is $W_1^{-1}(Z/E) = Z - E + 1$, giving $-Q$. For the outgoing base, its opposite apex is $W_3^{-1}(Z/E) = ER/(2E-Z)$, giving $-Q^{-1}$. The remaining entries follow in the same way from (F.27).

This table removes the other endpoint's angular parameter completely. It also retains the shear, not just the plane angle. The two main puncture products are one; the third puncture product is $Q^{\pm 2}$, with the actual disk longitude having multiplier $Q^2 = e^{2i\tau}$. The sign of the oriented pants boundary is the inverse convention already recorded by the original marked disk word.

Consequently an outgoing disk-side boundary of one local block and an incoming boundary of another with the same τ have inverse edge ratios under the original puncture matching. Normalize one triangle by its three ordered vertices. The other triangle's position is then fixed by (F.29), so the whole two-triangle transition matches. Their edge shears cancel and their interior-angle branches, continued from π, sum to 2π. This is the local geometric seam calculation required for joining edge blocks. It does not assert an arbitrary reordering of the three punctures.

F.2.5 Local logarithmic increments and the edge action

Let $\psi(\tau, \upsilon)$ be the pure two-leaf action in the angular variables, retaining its complete constant. Its actual Legendre relation gives

$$\partial_\tau \psi = it_0 = -\frac{1}{\pi}\operatorname{arsinh}(\tan(\tau/2)\cos(\upsilon/2)), \tag{F.30}$$

with the branch through zero; interchange the endpoints for the other derivative. To verify the last equality, use $\operatorname{artanh}(ky) = \operatorname{arsinh}(y/\sqrt{1+z^2})$.

The local pure meridional increment is therefore

$$-2\pi i t_0 = -2\pi \partial_\tau \psi. \tag{F.31}$$

It is a small actual log-link increment in this fixed local chart. The ordered disk-page proof in Section F.1.6, concatenates these local paths in their original order. The total pure meridian period is therefore the sum of (F.31). A single retained half twist of sign ϵ_v adds $-\epsilon_v V_v/2$, in the original primitive basis. Consequently

$$U_v = -2\pi \sum_{e \sim v} \partial_{\tau_v} \psi_e - i\epsilon_v \tau_v, \qquad V_v = 2i\tau_v. \tag{F.32}$$

The direct first-charge saddle with $n_v = \epsilon_v - 2p_v$ makes (F.32) exactly $U_v + p_v V_v = -2\pi i$. The integer is fixed by concatenating the actual local increments and continuing the periods from the complete cusp, even when the total is no longer a small logarithm. Section F.3.4 establishes that period statement without resetting a principal log.

F.2.6 Assembly in the original ordered exterior

The ordered decomposition constructs the following finite assembly. Associate the two octahedra above to each physical edge of the circle tree. At a circle vertex, list its incident edges in the actual cyclic order and join the corresponding outgoing and incoming disk-side pants

cyclically. Insert the prescribed retained half twist at its specified disk page. Table (F.29) supplies geometric seam matching for a common angular variable at that circle. The complete model has two octahedra per edge and D meridional sectors at a degree-D circle.

The topological proof cuts each original circle neighborhood into ordered disk pages and uses terminal-vertex insertion to identify the two-leaf block at each physical quarter-turn, including its four marked main segments. The two-circle endpoint pairs are interlaced by the actual quarter-turn. The proof retains that matching and every left or right insertion in the original product order. It also proves that the primitive meridian crosses the pages once in their cyclic order, with only the single recorded half-twist shear. These facts justify the actual paths used in (F.32).

At a distinguished odd seam one uses the original puncture-and-copy exchange, rather than an arbitrary permutation of the three pants punctures. The original matrices and their three full projective edge-return words are given in Section F.3.3. Table (F.29) transports that specified map between incident blocks with a common longitude parameter.

Together, (F.25), (F.29), and the actual period equations provide the positive cells and link data used by that completion. The analytic criterion keeps every angular parameter below 0.00111, within the radius 0.01 proved here. The complete conclusion and its precise class of diagrams are stated in the quantitative volume theorem.

F.3 Completing the actual fillings and calibrating volume

The separately available version of this section is completing the actual fillings and calibrating volume. The proof is reproduced here in full.

The octahedral cells of an ordered tree of crossing circles stay positive while their angular longitude parameters vary in a fixed small disk. This section proves that the parameter path selected by the analytic filling equations completes those cells to the actual filled link exterior. It follows the full logarithmic period of each primitive filling curve, so the completed meridian makes exactly one turn at the endpoint. It then identifies the absolute volume with the continued critical action.

The ordered decomposition supplies the original topology and primitive curves. The angular charts supply the positive cells and their exact face maps. The analytic criterion supplies the path under the explicit bound in (F.36). Together these establish the geometric part of the quantitative volume theorem. Only the original crossing circles are filled; the main knot cusp remains complete and unfilled.

F.3.1 Exact topological and local inputs

Fix a finite circle-only Conway tree with at least two circle vertices. Every edge is a physical quarter-turn and every vertex is an original crossing circle. Keep its actual ribbon orders, projection-copy labels, half twists and primitive disk curves.

The ordered cut theorem supplies three marked facts.

- The original projection and disk-page cuts give two octahedra per edge, with the six vertices and face maps of the literal two-leaf angular chart. Any transported corner permutations and projection-copy changes are retained.

- At each original circle, outgoing and incoming pages are joined in their actual cyclic order. The retained half twist occurs at its distinguished original seam. The face maps keep their full three-point labels.
- The disk longitude is the boundary loop with period $V_v = 2i\tau_v$. The original meridian concatenates the local meridional page paths, with the single half-longitude shear $-\epsilon_v V_v/2$. These are the original primitive curves.

Physical T^{-1}, T, and retained planar half-turns are not identified by source involutivity. Reversing a disk frame reverses both its paths and its peripheral coordinate. All formulas below use the fixed original disk basis.

The local positive eight-tetrahedron chart is valid on

$$|\tau_v|, |\tau_w| \le 10^{-2}. \tag{F.33}$$

Its incoming pants edge ratios are $(-Q^{-1}, -Q, -Q^{-1})$, and its outgoing ratios are their inverses, where $Q = e^{i\tau_v}$. Thus the marked seam cancels the full complex shears, and its two interior-angle branches sum to 2π. This includes both projection copies. At zero these are the two regular octahedra in the actual two-leaf cut.

F.3.2 The analytic path and the actual peripheral formulas

Let F_0 be the complete positive action, normalized so that $2\pi F_0$ is the volume of the original augmented exterior. There are two regular octahedra per tree edge at completeness. Let

$$\Psi(\boldsymbol{\tau}) = \sum_{\{v,w\}} \psi(\tau_v, \tau_w), \qquad \Psi(0) = F_0, \qquad \psi_{\tau_v} = -\frac{1}{\pi}\operatorname{arsinh}\big(\tan(\tau_v/2)\cos(\tau_w/2)\big). \tag{F.34}$$

The local action is separately even, with its complete branch and constant. Fix the original signs $\epsilon_v \in \{0, \pm 1\}$, integral fillings p_v, and powers $n_v = \epsilon_v - 2p_v$. The analytic criterion supplies an analytic first-charge path $\boldsymbol{\tau}(c)$, $0 \le c \le 1$, with

$$\Psi_{\tau_v} + \frac{in_v\tau_v}{2\pi} - ic = 0, \qquad \boldsymbol{\tau}(0) = 0, \qquad |\tau_v(c)| < 10^{-2}, \qquad \Im\tau_v(c) < 0 \quad (c > 0). \tag{F.35}$$

Write D_v for the circle-tree degree. The sufficient condition and pointwise bound are

$$\sum_v \frac{D_v}{D_v^2 + n_v^2} \le 2^{-25}, \qquad |\tau_v(c)| \le \frac{2\pi c}{(1 - 1/50)\sqrt{D_v^2 + n_v^2}}, \tag{F.36}$$

The common bound is less than 0.00111, inside (F.33). The geometric proof only uses the explicit path properties (F.35).

In a pure local block the small meridional increment is $-2\pi\psi_{\tau_v}$. The ordered cut theorem consequently gives the original peripheral logarithms

$$U_v = -2\pi\Psi_{\tau_v} - i\epsilon_v\tau_v, \qquad V_v = 2i\tau_v. \tag{F.37}$$

Their exponentials are the actual holonomy multipliers. Section F.3.4 proves that the chosen lifts are also the actual log-link periods. The stationary equation gives

$$U_v + p_v V_v = -2\pi i c. \tag{F.38}$$

The primitive basis is not reset when U_v approaches $-2\pi i$.

F.3.3 Positive metric on the cut-and-glued exterior

Realize every local block by its positive tetrahedra and perform the original identifications in Section F.3.1. Within a block the rational face identities retain the full edge return map. At a page seam the inverse ratios give both zero shear and angle 2π, not only equality of multiplier moduli. Their branches are continued from the complete cells.

At the distinguished odd seam use the original cross-copy maps, rather than the pure inverse-ratio identification. In the local leaf frame of the angular chart they are

$$A = \begin{pmatrix} R & -R \\ 1 & ER - 1 - E \end{pmatrix}, \qquad B = \begin{pmatrix} R/E - 1 - 1/E & R \\ -1 & R \end{pmatrix}. \tag{F.39}$$

They send respectively $(Z, \infty, 1)$ to $(Z/E, R, 0)$ and $(Z, R, 0)$ to $(Z/E, \infty, 1)$. Thus they specify all three ideal vertices and the actual exchange of projection copies. With the four original white maps W_j, direct multiplication gives the full projective edge returns

$$W_0^{-1} B W_2^{-1} A = I, \qquad W_2^{-1} B W_0^{-1} A = I, \qquad W_1^{-1} B W_3^{-1} A = I. \tag{F.40}$$

These are identities under $Z^2 - (1+E)Z + ER = 0$, not modulus comparisons. The incoming and outgoing marked pants structures depend only on their common $Q = e^{i\tau_v}$, by the local edge-ratio calculation. Transport each to the same ordered three-vertex pants chart. Equations (F.39)–(F.40) then give the same odd seam between any two incident blocks, even when their far-end parameters differ. The allowed puncture exchange is exactly the one in (F.39), not an arbitrary pants permutation. Every resulting edge has its complete angle branch 2π, by positivity and continuation from zero. The two physical signs retain their distinct primitive words and the single correction in (F.37).

This gives a nonsingular local hyperbolic metric at every finite point of the original exterior. Face pairings are isometries, edge links have angle 2π and identity return, and a finite subdivision vertex has a nonsingular spherical metric on its topological sphere link. Its complete local spherical development is the ordinary sphere. No global injectivity of the developing map is assumed.

The actual horospherical triangles form a nonsingular similarity structure on each topological cusp torus. Main puncture holonomies are the nontrivial parabolics of the local pants and remain so in the fixed local charts. The actual primitive main meridian is one of these local words. Its commutant is the translation group at its flag. Hence each main torus has a complete Euclidean translation link.

For $c > 0$, (F.35) and (F.37) give

$$\Re V_v = -2\Im \tau_v > 0. \tag{F.41}$$

The auxiliary peripheral group has a nonunit dilation and a second common fixed point q_v. Its nonsingular similarity development misses q_v. Indeed $(z - q_v)\partial_z$ is a globally invariant vector field; each possible zero has index $+1$, whereas the total index on a torus is zero.

On the simply connected link cover put

$$f_v = \log(\mathrm{dev}_v - q_v). \tag{F.42}$$

Its real derivative is invertible. The metric $|df_v|^2$ descends to a nonsingular Euclidean metric on the compact torus, and is complete. Geodesic path lifting makes f_v a covering of the Euclidean plane; since the plane is simply connected, it is a diffeomorphism. Its periods form an injective real rank-two lattice.

F.3.4 The periods throughout the charge path

Let P_{μ_v}, P_{λ_v} be the actual periods of (F.42) on the retained primitive paths. Their exponentials are the actual peripheral multipliers, so

$$P_{\mu_v} - U_v, \quad P_{\lambda_v} - V_v \in 2\pi i\mathbb{Z}. \tag{F.43}$$

These differences are continuous for $0 < c \leq 1$. The link remains nonsingular, its development avoids the second fixed point, and the paths and flags are retained. Choose a logarithm at one basepoint and transport it continuously; rebasing adds a constant and does not change a period.

The initial integers can be fixed directly from the complete charts. Choose a peripheral frame depending continuously on the selected flag and sending it to infinity. Write the longitude as $z \mapsto a(c)z + b(c)$, with $a(c) = e^{V_v(c)}$. At completeness $a(0) = 1$ and $b(0) \neq 0$, since the original primitive longitude is a nonzero translation in the complete torus. For $c > 0$, (F.41) makes $a(c) \neq 1$, and its second fixed point is $q_v(c) = b(c)/(1 - a(c))$. Thus $|q_v(c)| \to \infty$.

For this fixed finite tree, the developed fixed-generator paths remain in one bounded region and have bounded Euclidean lengths as $c \downarrow 0$. They use finitely many actual triangles and transition maps, all continuous in the complete charts. If their images lie in $|z| \leq B$ and their lengths are at most L, their actual logarithmic increments obey

$$|P_\gamma(c)| \leq \frac{L}{|q_v(c)| - B} \longrightarrow 0. \tag{F.44}$$

Meanwhile (F.37) gives $U_v, V_v \to 0$. Each integer difference in (F.43) is therefore zero for sufficiently small c, and hence on all of $(0, 1]$. This fixes the winding without importing a small-neighborhood completion theorem or resetting principal logs at the endpoint.

Equivalently, each page increment in (F.37) is small in its local two-leaf chart. Changes of a pants frame add logarithmic constants at the ends of a page path; those constants telescope around the circle. Its meridian accumulates the sum of the actual increments, even when high degree makes that sum close to $-2\pi i$. The preceding continuity argument fixes the possible integer.

The actual slope period is thus $-2\pi ic$. Its complementary primitive curve is the original longitude: the determinant of $(\mu_v + p_v\lambda_v, \lambda_v)$ in the original basis is one. Its positive core translation length is

$$\ell_v = \Re V_v = -2\Im\tau_v. \tag{F.45}$$

F.3.5 Cone completion and the smooth endpoint

The geometric end contains all sufficiently high vertical rays over its similarity triangles. Send its fixed points to zero and infinity, put $r = y/|z|$, and use

$$R = \operatorname{arsinh}(1/r), \quad \theta = \Im f_v, \quad X = \Re f_v + \tfrac{1}{2}\log(1 + r^2). \tag{F.46}$$

The metric is exactly

$$dR^2 + \cosh^2 R\, dX^2 + \sinh^2 R\, d\theta^2. \tag{F.47}$$

Compactness of the actual similarity link shows that the end contains a whole punctured tube $0 < R < R_0$. Its primitive meridian period is $-2\pi ic$, and its independent longitude has

positive real part (F.45). Adding the axis produces a cone solid torus of angle $2\pi c$, with the specified topological meridian. At $c = 1$ it is an ordinary smooth solid torus.

Main cusps remain complete horoball quotients, and the remaining middle is compact. Thus the $c = 1$ metric is complete and has finite volume on the original filled topology. The complete universal local development covers simply connected hyperbolic space, so the resulting representation is the geometric holonomy. When the original component test gives one main component, the metric is on that original knot complement. Its main cusp is unfilled.

The intermediate cone description makes no discreteness claim for $0 < c < 1$. It records the positive metric and its actual link completion; smoothness is asserted at $c = 1$.

F.3.6 Absolute action and the primitive normalization

The calibration can be derived on this explicit geometric block family. It does not require choosing an unmarked character component by comparison of its derivatives at one point.

Put $U_v^0 = -2\pi\Psi_{\tau_v}$ and

$$G^0 = 2\pi i \left(\Psi - \sum_v \tau_v \Psi_{\tau_v} \right). \tag{F.48}$$

Direct differentiation gives

$$dG^0 = \tfrac{1}{2} \sum_v V_v \, dU_v^0. \tag{F.49}$$

The actual half twist changes U_v^0 to $U_v = U_v^0 - \epsilon_v V_v/2$. Its marked primitive is therefore

$$G = G^0 - \sum_v \frac{\epsilon_v V_v^2}{8} = 2\pi i \left(\Psi - \sum_v \tau_v \Psi_{\tau_v} \right) + \sum_v \frac{\epsilon_v \tau_v^2}{2}. \tag{F.50}$$

This is exactly the source Legendre transform with its complete constant. In particular $\Im G(0) = 2\pi F_0$.

The original boundary fundamental-chain differential, applied to these actual positive blocks and their marked paths, gives

$$\mathrm{Vol}(\rho) = \Im G - \tfrac{1}{2} \sum_v \Re U_v \Im V_v. \tag{F.51}$$

The constant is fixed at the complete assembly. Every local shape stays in a simply connected positive neighborhood of i, and Section F.3.4 retains the actual period lifts. No volume period is reset. The positive tetrahedral volume sum also shows directly that the volume tends to $2\pi F_0$ as $c \downarrow 0$.

Define the stationary charge action

$$\mathcal{C}(c) = \operatorname*{stat}_{\boldsymbol{\tau}} \left[\Psi + \frac{i}{4\pi} \sum_v n_v \tau_v^2 - ic \sum_v \tau_v \right]. \tag{F.52}$$

At its critical point, set $\Phi = G/(2\pi i)$. Substitution of (F.35) gives the exact cancellation

$$\Phi - \mathcal{C}(c) = -\frac{i}{2\pi} \sum_v p_v \tau_v^2, \qquad \tfrac{1}{2} \Re U_v \Im V_v = p_v \Im(\tau_v^2). \tag{F.53}$$

Consequently

$$\mathrm{Vol}_c = 2\pi \Re \mathcal{C}(c). \tag{F.54}$$

The primitive normalization has an independent differential check:

$$\frac{d}{dc}\,2\pi\Re\mathcal{C}(c) = 2\pi\sum_v \Im\tau_v. \tag{F.55}$$

The core is represented by the complementary primitive curve λ_v, its length is (F.45), and its cone angle is $2\pi c$. The finite-cell Schläfli differential $-\frac{1}{2}\sum_v \ell_v\, d(2\pi c)$ gives (F.55). This checks the factor and sign; (F.54) already follows from the marked regulator and the complete anchor, without cone rigidity.

At $c = 1$, (F.54) identifies the actual analytic first-charge rate with the original smooth filled volume. The ordered topological theorem and the analytic criterion supply the topology and path (F.35) used in this proof. The finite-source dominance and its exact normalization are proved in the analytic theorem.

F.3.7 Scope of the completion

The metric argument permits large global meridian logs while every edge block remains uniformly close to its regular geometry. It does not require a globally small framed-representation norm.

The conclusion applies to the ordered all-circle trees of Section F.3.1 under (F.36), or more generally whenever their actual analytic path satisfies (F.35). It does not assert that every bounded crossing power gives a hyperbolic filling. The degree-weighted bound permits some small powers because the resulting local angular changes remain small while the primitive meridian is assembled from many sectors.

The preferred-cusp continuation also follows the actual main longitude through this family. Its marking is established by the original face words and surgery homology. The complete geometric and analytic conclusions are collected in the quantitative volume theorem.

F.4 The preferred cusp along the filling path

The separately available version of this section is the preferred cusp along the filling path. The proof is reproduced here in full.

A horospherical cross-section of the knot cusp is a Euclidean torus. Normalize the meridian's translation to one. The translation of the preferred longitude, which follows the knot once and has linking number zero with it, is its preferred cusp shape.

The ordered page decomposition and its positive angular realization construct a representation of the original augmented exterior. This section identifies its preferred main-cusp translation. The argument does not use a quantum asymptotic to identify a representation or a cusp. The small-neighborhood argument below is used only to establish an identity of holomorphic germs. The identity then continues along the entire positive angular chart, even when the original auxiliary meridian logarithms are no longer small.

The geometric inputs are the ordered decomposition, the positive angular charts, and their actual filling completion. The analytic criterion constructs the path used in the quantitative volume theorem. The original marked inputs are the face-map theorem the original marked face maps and the full rational translation proof the preferred-longitude translation proof.

F.4.1 Statement

Fix an original ordered all-circle tree with at least one edge. Retain the original disk frames, half twists $\epsilon_v \in \{0, 1, -1\}$, and one main knot component. Let $\mathcal{E}$ be its physical quarter-turn edges. Set

$$y_v = \tan(\tau_v/2), \qquad k_e = (1 + y_v^2 + y_w^2)^{-1/2}, \qquad \tan(\pi x_e) = k_e, \qquad x_e(\mathbf{0}) = \frac{1}{4}, \quad e = vw. \tag{F.56}$$

All roots and inverse functions use their branches at zero. The local positive chart is $|\tau_v| < 10^{-2}$. The ordered page construction gives an actual marked representation there, with main meridian parabolic and primitive auxiliary longitude logarithm $V_v = 2i\tau_v$.

Let L_{path} be exactly the framed main path in Sections 2–6 of the preferred-longitude calculation. Let λ_K be the preferred longitude in the augmented exterior. Write w_ϵ for the oriented writhe of its retained-half-twist diagram and ℓ_v for the linking number of crossing circle v with the main knot. In the coordinate where the main meridian translates by one, their translations in the angular representation are

$$\begin{aligned} \operatorname{trn}(L_{\mathrm{path}}) &= 2i \sum_{e \in \mathcal{E}} \cot(\pi x_e) - \sum_v \epsilon_v, \\ \operatorname{trn}(\lambda_K) &= 2i \sum_{e \in \mathcal{E}} \cot(\pi x_e) - \sum_v \epsilon_v - w_\epsilon. \end{aligned} \tag{F.57}$$

Here trn denotes Euclidean translation, not matrix trace. The signs are those of the original positive checked crossing and the actual oriented writhe.

For fixed integers p_v, put $n_v = \epsilon_v - 2p_v$, and follow the positive charge path

$$U_v + p_v V_v = -2\pi i c, \qquad 0 \le c \le 1. \tag{F.58}$$

Assume the path stays in the chart and has $\operatorname{Im} \tau_v < 0$ for $c > 0$, as supplied by the curvature-supported certificate. The completion theorem gives a smooth actual filled knot complement at $c = 1$. Its preferred cusp is

$$\boxed{\tau_K = -w_{\mathbf{n}} - \sum_v n_v + 2i \sum_{e \in \mathcal{E}} \cot(\pi x_e(\boldsymbol{\tau}(1)))}. \tag{F.59}$$

Here $w_{\mathbf{n}}$ is the oriented writhe of the diagram with the filled powers n_v. Equivalently, the last sum is $2i \sum_{vw \in \mathcal{E}} \sqrt{1 + y_v^2 + y_w^2}$, with the same branches. There is no smallness condition on the total U_v in this conclusion.

F.4.2 Why actual representations can be compared near zero

There are two different geometric cell descriptions, so a source Hessian by itself does not identify their holonomies. We use the marked positive-gluing rigidity proved in the marked multicusp gluing theorem, Sections 2–3 and 5. It is also used in Section 2 of the original geometric deformation. Here are the precise parts needed.

The new complete triangulation consists of eight positive ideal tetrahedra per tree edge. Its face labels and peripheral paths are those of the original exterior, by the ordered page proof. The old complete construction gives that same marked exterior. Their complete holonomies are

conjugate by an orientation-preserving marked isometry. This is the geometric uniqueness of the complete finite-volume metric. It can also be obtained from the existing maximal-volume conjugacy argument in the geometric conjugacy theorem, Section 1: the page representation is an actual representation of the original group and its positive complete tetrahedra have volume $2|\mathcal{E}|v_8 = 2\pi F_0$, the known complete volume. This use of rigidity is after the actual topological identification; an equality of cell volumes alone would not identify exteriors.

For completeness, the positive-gluing uniqueness argument applies with an auxiliary longitude rather than a meridian prescribed at each auxiliary cusp. If there are t ideal tetrahedra and q cusps, select $t - q$ independent edge equations and one primitive peripheral row at each cusp. The integral edge/peripheral identities identify the symplectic quotient of the edge row space with $H_1(\partial M; \mathbb{R})$. Thus these rows form a full-rank real Lagrangian matrix $(A\ B)$. Choose the main meridian row and all auxiliary longitude rows. They are independent and isotropic because they are one primitive nonzero class on each different torus.

In the positive local orders write

$$u_j = \log z_j, \qquad v_j = \log(1 - z_j^{-1}), \qquad D_j = \frac{dv_j}{du_j} = \frac{1}{z_j - 1}, \qquad \operatorname{Im} D_j < 0. \tag{F.60}$$

The Jacobian at fixed chosen peripheral logs is $A + B\,\mathrm{diag}(D_j)$. If it killed a nonzero vector a, then $(a, \mathrm{diag}(D_j)a)$ would lie in the complexification of a real Lagrangian subspace. Its symplectic pairing with its conjugate would vanish. The same pairing is, up to the fixed nonzero imaginary factor, $\sum_j \operatorname{Im} D_j |a_j|^2 < 0$, a contradiction. The implicit-function theorem therefore gives one decorated shape germ for the main meridian log zero and the auxiliary logs $V_v = 2i\tau_v$. The cited multicusp theorem proves the rank and decorated-representation statements for exactly these mixed peripheral coordinates. Its polarization is the all-cusp form of the proof in the polarized gluing argument, Section 1.

This is a statement about the entire nearby marked representation, not just a formal equality of its peripheral eigenvalues. To apply it to the old family, develop the vertices of the new ideal triangulation using the old representation and its recorded cusp flags. For a lifted ideal vertex gP_v, use $\rho(g)\xi_v$, where ξ_v is the selected peripheral fixed point. This is independent of the coset representative. Every complete tetrahedron has four distinct vertices, so the resulting cross ratios are holomorphic nearby and remain positive. They obey all the same edge and peripheral equations, with the same log branches by continuity. This puts the old representation into the new shape chart and justifies applying its uniqueness. No individual tetrahedron or subtree is separately conjugated.

The old auxiliary longitude coordinates are valid near zero: the marked theorem gives a diagonal, nonzero derivative of V with respect to the original auxiliary colors. Thus the old family can indeed be parametrized by these same $V_v = 2i\tau_v$. The preceding uniqueness identifies its representation, with all original paths, with the new page representation on a neighborhood of zero.

F.4.3 Identification of the retained wire coordinates

Equality of representation germs must also use the same variables in the cotangent expression. Here that identification follows from the exact classical Legendre calculation, not a comparison of volumes.

In the original physical-edge variables the local angular action is

$$b(x,\theta)=\int_0^x \log(2\cos 2\pi u+2\cos\theta)\,du-2J(x),\qquad J(x)=\int_0^x \log(2\sin\pi u)\,du. \tag{F.61}$$

For a circle vertex, the analytic elimination of its original hidden variables gives

$$\operatorname{stat}_{\theta_v}\left[\sum_{e\sim v} b(x_e,\theta_v)+\frac{i\epsilon_v\theta_v^2}{4\pi}+i\chi_v\theta_v\right]. \tag{F.62}$$

This is the original selected circle action, with its original physical-edge indices: the one-leaf identity is derived explicitly in the regular crossing-circle action, Section 7, its circle-junction extension is in Section 9, and the joint-wire Legendre identity is (38)–(39) of the joint filling transform. The nondegenerate hidden Hessian makes these analytic eliminations unique near the original complete state. The identity retains the physical-edge variables before eliminating them.

There are no bare junctions in the present tree. Set $\theta_v=-\tau_v$, so that the original relation $V_v=-2i\theta_v$ becomes the geometric convention $V_v=2i\tau_v$. Separate evenness leaves b unchanged. Every retained edge vw therefore has the stationary equation

$$\partial_x\{b(x,\tau_v)+b(x,\tau_w)\}=0. \tag{F.63}$$

Writing $c_e=\cos 2\pi x_e$, this is

$$(c_e+\cos\tau_v)(c_e+\cos\tau_w)=(1-c_e)^2. \tag{F.64}$$

The branch through $x_e=1/4$ gives

$$c_e=\frac{1-\cos\tau_v\cos\tau_w}{2+\cos\tau_v+\cos\tau_w}, \tag{F.65}$$

which is exactly (F.56). The derivative of (F.63) at zero is nonzero. Thus these are the retained old physical-turn labels as analytic germs, not merely variables with the same complete values.

The color itself also matches the primitive geometric meridian. Let $\Psi(\boldsymbol{\tau})$ be the sum of the stationary edge actions. The θ-stationarity in (F.62) gives

$$\chi_v=-i\Psi_{\tau_v}+\frac{\epsilon_v\tau_v}{2\pi},\qquad -2\pi i\chi_v=-2\pi\Psi_{\tau_v}-i\epsilon_v\tau_v=U_v. \tag{F.66}$$

The last expression is the actual page-by-page primitive word from the ordered decomposition. Near zero it has derivative $d_v-i\epsilon_v$ in τ_v, where d_v is the circle's degree. This is nonzero. Equation (F.66) is not used as a substitute for the representation comparison in Section F.4.2.

F.4.4 Holomorphic continuation of the actual main translation

The old preferred-cusp proof establishes its rational identities before imposing any auxiliary filling equation. Its product potential, endpoint rebasing rule, actual quarter-turn maps, and pure and odd leaf words give

$$\operatorname{trn}(L_{\text{path}})=2i\sum_e \cot(\pi x_e)-\sum_v \epsilon_v. \tag{F.67}$$

on the old nearby marked family. By Sections F.4.2–F.4.3, (F.67) holds for the actual new page representation near zero.

Both sides are holomorphic on the connected angular chart. For the right side this follows directly from (F.56). For the left side, keep the actual based main meridian and longitudinal words. The meridian is nonidentity parabolic throughout the chart: it is conjugate to one of the nontrivial local pants meridians. Locally send its marked fixed point to infinity and normalize its translation to one. The commuting longitudinal holonomy is then a translation as well. Its coefficient is holomorphic and independent of the remaining translation in this normalization. These local functions agree on overlaps, so they define one holomorphic translation ratio on the entire chart.

Equivalently, choose the local trace-two unipotent lifts and write $M = I + N$, $L = I + \lambda N$. Since $N \neq 0$, the coefficient λ is a holomorphic function on any open set where a chosen entry of N is nonzero. The formulas agree on overlaps. This description uses no logarithm of the main longitude and therefore has no $2\pi i$ or integer translation ambiguity.

The identity theorem now extends (F.67) throughout that chart. A continuation along the positive charge path would suffice; no uniform neighborhood in the original summed meridian coordinates is required. The original endpoint-sign convention remains part of the word: when an arc is reversed, both its traversal and its unit-meridian orientation reverse. The affine multipliers -1 at odd seams are not replaced by $+1$.

The homological computation of the fixed word is independent of the representation. It is exactly

$$[L_{\mathrm{path}}] = w_\epsilon[\mu_K] + \sum_v \ell_v[\mu_v], \qquad \lambda_K = L_{\mathrm{path}}\mu_K^{-w_\epsilon}. \tag{F.68}$$

Hence the second identity in (F.57) continues with the first.

F.4.5 The preferred longitude after the actual fillings

The original crossing circles are pairwise unlinked, and their primitive disk longitudes satisfy $[\lambda_v] = \ell_v[\mu_K]$. Filling the original slopes $\mu_v + p_v\lambda_v$ gives

$$\lambda_{\mathrm{fill}} = \lambda_K \mu_K^{\sum_v p_v \ell_v^2}. \tag{F.69}$$

This remains a fixed word on the main boundary throughout the cone path. At $c = 1$ it is precisely the preferred zero-linking longitude of the smooth filled knot exterior. At intermediate c its translation is simply the translation of that same boundary curve; no smooth filling at an intermediate nonintegral cone angle is asserted.

The disk orientations give $\ell_v^2 = 2(1 + o_v)$ with $o_v = 1$ for parallel strands and $o_v = -1$ for antiparallel strands, while $w_\epsilon = \sum_v o_v\epsilon_v$ and $w_{\mathbf{n}} = \sum_v o_v n_v$. Therefore

$$-\sum_v \epsilon_v - w_\epsilon + \sum_v p_v\ell_v^2 = -\sum_v n_v - w_{\mathbf{n}}. \tag{F.70}$$

Adding the meridian translations in (F.69) to (F.57) proves (F.59).

The new conclusion is a geometric identity on the positive angular continuation of this actual all-circle class. It does not extend the original real angle chamber, and it does not assert a formula on unrelated algebraic solutions. Its proof uses the exact marked representation, actual path words, local geometric uniqueness, and analytic continuation. The independent scalar calculation and the new source asymptotic may subsequently use (F.59), but neither was used to identify this cusp.

Appendix G

Exact subtree polynomials and local filling estimates

This appendix proves Theorem 10.1 and the separate metric criterion, Theorem 10.2. The tree, all cyclic orders, all strand frames, all signed powers and the knot pairing are fixed. Only the integer level N tends to infinity. We use

$$\kappa = \max_v \frac{d_v}{d_v^2 + n_v^2} \le 2^{-120}$$

for the quantum induction. The last section replaces this condition by $\sqrt{d_v^2 + n_v^2} \ge 630$ for geometric completion alone.

The exact finite coordinates, pairing, crossing polynomial and translated-grid trace are those of Section E.1. Their finite edge-sum expansion on a radius $1/4$ bidisk is proved in Section E.2.5; the global Newton product bounds are in Section E.3.2. In particular the input to the present proof is an asymptotic for an actual finite sum with its endpoint currents, not only a formal critical value.

We call the polynomial left on the unique boundary of an evaluated subtree its boundary message. Its value at zero will be factored out exactly. The proof first establishes a class of limiting phases, then closes an induction for the finite polynomials themselves. Each step keeps all Newton indices and all repeated-root derivative coordinates.

G.1 The full finite kernel and exact recursion

Put $d = N - 1$, $c_j = 2\cos((2j+1)\pi/N)$, and

$$S_k = \prod_{r=1}^{k} 2\sin(\pi r/N), \qquad e_k(z) = S_k^{-2} \prod_{j<k} (c_j - z). \tag{G.1}$$

The fixed pairing is $\phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}$. A physical turn, after lowering this pairing, identifies equal indices and retains $\gamma_e(s) H_N(s)/W_k(s)$. Thus define

$$K_N(t,u;s) = \sum_{k=0}^{N-1} e_k(-2\cos t) e_k(-2\cos u) \frac{H_N(s)}{W_k(s)}. \tag{G.2}$$

This is the full kernel, not the regular-window truncation. It is an even trigonometric polynomial of degree at most $N-1$ in each variable. The paired factors and their inverses are bounded uniformly in k, N on every fixed disk $R < 1$.

Choose a terminal root. For a nonroot vertex v, its message to the parent angle u is the exact polynomial

$$M_{v,N}(u) = \gamma_{e_v}(s)\, \mathcal{D}^{(t)}_{n_v,\sigma_v,N}\left[K_N(t,u;s) \prod_{w \text{ child of } v} M_{w,N}(t)\right]. \tag{G.3}$$

The functional $\mathcal{D}_{n,\sigma,N}(F) = \phi_\sigma(R^n_\sigma F)$ uses the full physical crossing polynomial, including its Cartan scalar. The polynomial representative and the trace are defined by the exact quotient in Section E.1. The actual σ_v is $s, -s$, or 0, according to the ordered pair. The reference monodromy factors remain outside in the same ledger as Theorem 9.1. Formula (G.3) follows by expanding the original covariant contraction and doing finite sums in a different order. It does not commute physical tangles.

At the root the final trace is

$$Z_N = \mathcal{D}^{(t)}_{n_0,\sigma_0,N}\left[\prod_{w \text{ child of } 0} M_{w,N}(t)\right]. \tag{G.4}$$

The original closing/framing factor, prescribed reference factors, and one reduced cut factor $H_N(s)$ multiply (G.4). There is no discarded-source remainder in (G.3)–(G.4). Every original normal channel and every $0 \le k < N$ is still present.

In particular $M_{v,N}$ is always an even trigonometric polynomial of degree at most $N-1$ in its parent variable, regardless of the size of the subtree. This is seen directly from (G.2): the trace and all child operations act only in t.

G.2 Two full-kernel bounds

Use the following constants:

$$r = 2^{-12}, \quad c_K = 2^{-16}, \quad c = 2^{-18}, \quad y_0 = 2^{-56}, \quad \rho = 2^{-14}, \quad t_0 = 2^{-16}. \tag{G.5}$$

All bounds allow polynomial factors in N, depending on R. Let $G_0 = g(1/4)$, and let $\psi(t,u)$ be the edge action, with $\psi(0,0) = 2G_0$.

First, for real $t, u \in [-\pi, \pi]$,

$$|K_N(t,u;s)| \le C_R N^{A_R} \exp N\{2G_0 - c_K[(1-\cos t) + (1-\cos u)]\}. \tag{G.6}$$

To prove this, split (G.2) into $k/N \in I = [1/5, 3/10]$ and its complement. On I, the product bound in Section E.3.2 gives a factor $\exp[-\alpha N \sin^2(t/2)]$ at each end, with $\alpha > 1/250$. The positive endpoint sum has rate $2G_0$ and a polynomial prefactor. Outside I, full Hermite interpolation gives a real sup bound with the single-edge loss $\eta_K = \pi/200$, including high indices. Since $4c_K < \eta_K$ and $c_K < \alpha/2$, both pieces satisfy (G.6).

Second, for complex $|u| \le r$, real $|t| \le \pi$, and $|y| \le y_0$,

$$|K_N(t+iy,u;s)| \le C_R N^{A_R} \exp N\{\Re\psi(0,u) - c_K(1-\cos t) + 16y^2\}. \tag{G.7}$$

This conditional normalization is essential. Replacing $\Re\psi(0,u)$ by $2G_0+O(r^2)$ would introduce an unnecessary error larger than some of the later local tail losses.

Here are the details. The complementary kernel, by the polynomial ellipse bound at each end, is at most

$$C_R N^{A_R} \exp N\{2G_0 - \eta_K + |y| + |\Im u|\}. \tag{G.8}$$

Since $|\psi(0,u) - 2G_0| \le r^2$, its fixed gap absorbs the entire claimed angular loss in (G.7), uniformly in t. Indeed $r + r^2 + y_0 + 2c_K < \eta_K/2$.

For the retained kernel and $|t| \le 1/8$, the finite edge-sum saddle of Section E.2.5 is uniform on $|t+iy|, |u| \le 1/4$. The exact formula for ψ_{tt} gives

$$|\psi_{tt} + 1/(2\pi)| \le \frac{1}{50}\frac{1}{2\pi}. \tag{G.9}$$

Separate evenness implies $\psi_t(0,u) = 0$. Integrating (G.9) on the segment from zero to $t+iy$ proves a stronger version of the exponent in (G.7). Its polynomial amplitude and its exponentially small lattice currents are retained. For $|t| \ge 1/8$, use the explicit global bound

$$|e_k(-2\cos(t+iy))| \le e^2 e_k(-2) e^{N[-\alpha \sin^2(t/2) + 11y^2]}. \tag{G.10}$$

The product at the other end satisfies $|e_k(-2\cos u)| \le e_k(-2)e^{Nr^2}$ on I: the normalized product factors have denominators $\cos^2((j+1/2)\pi/N) \ge 1/3$, and $|\sin(u/2)|^2 < r^2/3$. The additional change from $2G_0$ to $\Re\psi(0,u)$ costs at most another r^2. Now

$$2r^2 < \frac{\alpha}{2}\sin^2(1/16), \qquad c_K < \alpha/4. \tag{G.11}$$

These inequalities prove (G.7) on the remaining real angles. The first follows already from $\alpha > 1/250$, $\sin(1/16) > 1/(8\pi)$, and $\pi^2 < 10$. No quadratic complex bound for an individual high basis element has been used in this argument.

On the small bidisk $|t|, |u| \le r$, the full kernel also has the same nonzero analytic saddle expansion as its retained part. The complementary estimate (G.8) has a fixed gap there. In particular,

$$K_N(t,u;s) = N^{-3/2} a_e(t,u) e^{N\psi(t,u)}(1 + O_R(N^{-1})), \tag{G.12}$$

with the paired insertion included in the error and with the exact midpoint normalization of Section E.1. The leading a_e is holomorphic and nonzero on a slightly larger disk. The outside $\gamma_e(s)$ in (G.3) is not included in a_e.

G.3 A closed class of boundary phases

Write $a_0 = 1/(2\pi)$. A boundary phase $h(u)$ will satisfy

$$h(0) = h'(0) = 0, \qquad h(-u) = h(u), \qquad |h''(u) + a_0| \le a_0/20, \quad |u| \le r. \tag{G.13}$$

Assume the child phases have this property. At a nonroot vertex put

$$a_v = d_v/(2\pi), \quad b_v = n_v/(2\pi), \quad D_v = a_v - ib_v,$$
$$A_v(t,u) = \psi(t,u) + \sum_w h_w(t), \qquad C_{v,\ell}(u) = \operatorname{stat}_t\{A_v(t,u) + ib_v t^2/2 - i\ell t\}. \tag{G.14}$$

There are exactly d_v incidences in the t variable. Consequently

$$A_v(t,u) = \psi(0,u) - a_v t^2/2 + E_v(t,u), \qquad |E_{v,tt}| \le a_v/20, \quad |E_{v,t}/t| \le a_v/20. \tag{G.15}$$

The last quotient is removable at zero, by evenness in t. For $0 \le \ell \le L_v = \lfloor t_0|D_v| \rfloor$, the equation

$$D_v t = -i\ell + E_{v,t}(t,u) \tag{G.16}$$

is a contraction on $|t| \le r$. Its Lipschitz constant is at most $1/20$, and its image has radius $t_0 + r/20 < r$. The solution has

$$|t_\ell| \le \frac{\ell}{(1-1/20)|D_v|}, \qquad \frac{380}{441}\frac{\ell a_v}{|D_v|^2} \le -\Im t_\ell \le \frac{420}{361}\frac{\ell a_v}{|D_v|^2}. \tag{G.17}$$

These inequalities hold also for complex u. Indeed $t_\ell = -i\ell/(D_v - Q_v)$, where $|Q_v| \le a_v/20$; the real part of $D_v - Q_v$ is positive. No reality assertion about a child phase is needed.

The normalized outgoing phase is

$$h_v(u) = C_{v,1}(u) - C_{v,1}(0). \tag{G.18}$$

Stationarity gives the exact derivative

$$h_v'(u) = \psi_u(t_1(u),u), \qquad h_v''(u) = \psi_{uu} - \frac{\psi_{ut}^2}{A_{v,tt} + ib_v}. \tag{G.19}$$

The denominator has modulus at least $(19/20)|D_v|$. The explicit edge bounds on the radius $1/4$ disk give $|\psi_{uu} + a_0| \le a_0/50$ and $|\psi_{ut}| \le a_0/50$. Since $|D_v| \ge a_v \ge a_0$, the right side differs from $-a_0$ by at most

$$a_0\left(\frac{1}{50} + \frac{20}{19\cdot 2500}\right) < a_0/20. \tag{G.20}$$

Thus (G.13) closes without a depth or vertex-count constant. The same calculation starts the recursion when there are no children.

Charge ordering and the current vertex's action drop follow directly from $\partial_\ell \Re C_{v,\ell} = \Im t_\ell$:

$$\Re C_{v,\ell}(u) \le \Re C_{v,1}(u) - \frac{190}{441}\frac{a_v}{|D_v|^2}(\ell^2 - 1), \tag{G.21}$$

$$0 \le \Re\psi(0,u) - \Re C_{v,1}(u) \le \frac{210}{361}\frac{a_v}{|D_v|^2} = \frac{420\pi}{361}\frac{d_v}{d_v^2 + n_v^2} < 4\kappa. \tag{G.22}$$

There is no sum over descendants in (G.22).

G.4 The exact finite-message induction

The inductive assertions are these, for all sufficiently large N depending on the fixed subtree, uniformly on each fixed main-parameter disk $R < 1$.

1. $M_{v,N}(0) \ne 0$, and on $|u| \le r$,

$$\frac{M_{v,N}(u)}{M_{v,N}(0)} = \frac{\alpha_v(u)}{\alpha_v(0)} e^{N h_v(u)}(1 + O(N^{-1})). \tag{G.23}$$

The amplitude is holomorphic and nonzero; (G.13) holds.

2. For real $|u| \le \pi$,
$$\left|\frac{M_{v,N}(u)}{M_{v,N}(0)}\right| \le CN^A e^{-Nc(1-\cos u)}. \tag{G.24}$$

3. For real $|t| \le \pi$, $|y| \le y_0$,
$$\left|\frac{M_{v,N}(t+iy)}{M_{v,N}(0)}\right| \le CN^A e^{N[-(c/2)(1-\cos t)+y^2]}. \tag{G.25}$$

Only the phase bound and displayed exponential constants are uniform in the subtree. Amplitude bounds, polynomial powers and thresholds are allowed to depend on the fixed subtree.

In (G.23) the normalization is the exact value $M_{v,N}(0)$. It retains all N-dependent crossing, carrier and edge phases. More explicitly one may write
$$M_{v,N}(u) = \Omega_{v,N}(s,\boldsymbol{\sigma})\,\alpha_v(u)e^{N\mathcal{F}_v(u)}(1+O(N^{-1})). \tag{G.26}$$
Here $\Omega_{v,N}$ is the original nonzero scalar ledger for that subtree, including the full $q^{n_v(d^2/4-\sigma_v^2/2)}$ factors. It need not converge as N varies. The ratio in (G.23) cancels it exactly. The leading normalized amplitude $\alpha_v(u)/\alpha_v(0)$ is independent of the microscopic parameters: the first charge coefficient is one and the paired edge insertion has leading value one. The next coefficients may depend on them. No oscillatory scalar has been hidden in a claimed N-independent amplitude.

To obtain (G.25) from (G.23)–(G.24), use the exact degree bound following (G.4). For a Laurent polynomial f of degree at most $N-1$, the polynomial $z^{N-1}f(z)$, and its reciprocal, give the elementary Poisson bound
$$|f(t+iy)| \le CN^A e^{N[-c(1-\cos t)+(1+c)|y|]} \tag{G.27}$$
when its real bound is (G.24). Here are the details of the maximum-principle step. Write $f(t) = P(e^{it})$, where the Laurent polynomial P has powers between $-m$ and m, with $m \le N-1$. On the unit circle the subharmonic function $\log|z^m P(z)|$ is bounded by the harmonic function
$$\log(CN^A) - Nc + Nc\Re z.$$
The same inequality holds inside the disk, including at zeros of the polynomial. For $y \ge 0$, substitute $z = e^{-y+it}$ and divide by $|z|^m$. This gives
$$|f(t+iy)| \le CN^A \exp\{-Nc(1-e^{-y}\cos t) + my\}.$$
For $y \le 0$, apply the same argument to $z^m P(z^{-1})$. Finally $1-e^{-|y|} \le |y|$ and $m \le N$ give (G.27). This argument needs no positivity or reality of the polynomial coefficients. For $|t| \ge r/2$, our y_0 satisfies
$$(1+c)y_0 < \frac{c}{2}(1-\cos(r/2)). \tag{G.28}$$
For $|t| < r/2$, use (G.23) and (G.13):
$$\Re h_v(t+iy) \le -\frac{19a_0}{40}t^2 + \frac{21a_0}{40}y^2. \tag{G.29}$$

Since $|t+iy|<r$, these two estimates prove (G.25). The amplitude costs only a fixed polynomial factor.

In applying the next trace, factor out the exact constants $\prod_w M_{w,N}(0)$. Do not replace them by absolute complete-source bounds. On the local disk, the exact normalized child message has a holomorphic logarithm. More explicitly, (G.23) implies, on any fixed smaller disk,

$$\frac{1}{N}\log\frac{M_{w,N}(t)}{M_{w,N}(0)} = h_w(t) + \frac{1}{N}\log\frac{\alpha_w(t)}{\alpha_w(0)} + O(N^{-2}). \tag{G.30}$$

The branches vanish at zero, and all fixed derivatives have the same orders by Cauchy. The thresholds in (G.30) may depend on the subtree. For a fixed vertex the sum of these corrections tends to zero. It therefore perturbs the local nondegenerate saddle by $O(N^{-1})$, or equivalently is retained in its analytic amplitude. This is why inherited errors do not spend another exponential gap at every ancestor.

G.5 Conditional trace tails and connectors

After extracting the exact child constants, (G.7) and (G.25) give the global bound for the actual input of the next trace:

$$|K_N(t+iy,u)\prod_w M_{w,N}(t+iy)/M_{w,N}(0)| \le CN^A e^{N[\Re\psi(0,u)-2^{-20}a_vt^2+128a_vy^2]}. \tag{G.31}$$

Indeed $1-\cos t\ge 2t^2/\pi^2$, $d_v=2\pi a_v$, and $2\pi<8$. At a root with no outgoing edge, omit K_N, replace $\psi(0,u)$ by zero, and retain the same bound because there are d_v incoming messages.

Set

$$A_0=2^{-41},\qquad y_v=A_0t_0a_v/|D_v|,\qquad L_v=\lfloor t_0|D_v|\rfloor. \tag{G.32}$$

Condition (10.2) gives $L_v\ge 1$, and $y_v\le 2^{-57}<y_0$. The exact trace formula retains

$$c_{\ell,N}(\sigma)=(-1)^{(N-1)(\ell-1)}U_{\ell-1}(\cos 2\pi\sigma),\qquad c_{1,N}=1. \tag{G.33}$$

For a charge $\ell>L_v$, the chirp and (G.31), maximized in t, cost at most

$$\left(128a_v+\frac{b_v^2}{4\cdot 2^{-20}a_v}\right)y_v^2\le 2^{18}\frac{|D_v|^2}{a_v}y_v^2. \tag{G.34}$$

The negative Fourier term is at least $t_0|D_v|y_v$. Since $2^{18}A_0<1/2$, the omitted tail has loss at least

$$\frac{1}{2}A_0t_0^2a_v=2^{-74}a_v>2^{-77} \tag{G.35}$$

relative to $\Re\psi(0,u)$. Each additional charge contributes $e^{-N(\ell-L_v-1)y_v}$. For the fixed subtree, this dominates the exact coefficient growth $\ell e^{2\pi R'(\ell-1)}$ for all sufficiently large N.

At the original real endpoints the Gaussian loss is $2^{-20}\pi^2a_v$. The chirp and finite-charge connectors cost at most $\pi A_0t_0a_v+A_0t_0^2a_v+128A_0^2t_0^2a_v$. Their sum is less than half the endpoint loss. The exact zero-mode current obeys that endpoint bound. Thus the original finite functional, including its endpoints and normals, is its finite charge sum plus a conditional error with loss (G.35).

For each retained charge, restrict the real integral to $[-\rho,\rho]$. Its complement loses at least $2^{-20}a_v\rho^2 > 2^{-51}$. Translate the interval to $\Im t_\ell$; its endpoints remain in $|t| < r$. With $\delta = 1/20$, the same elementary Young inequality as in Section E.2.8 gives, along the connectors,

$$\Re\{A_v(t-iv,u)+ib_v(t-iv)^2/2-i\ell(t-iv)\} \le \Re\psi(0,u)-\frac{1}{8}a_vt^2+\left[\frac{1+\delta}{2}a_v+\frac{2b_v^2}{(3-4\delta)a_v}\right]v^2-\ell v. \tag{G.36}$$

For $0 \le v \le -\Im t_\ell$, the last two terms are at most $-\ell v/8$, since

$$\frac{420}{361}\max\left\{\frac{21}{40},\frac{2}{14/5}\right\} = \frac{300}{361} < \frac{7}{8}. \tag{G.37}$$

Thus each connector has loss at least $a_v\rho^2/8 > 2^{-34}$. The translated horizontal integral has a nondegenerate Gaussian with real quadratic part bounded above by $-(19/20)a_v$. Its leading amplitude is nonzero: the edge amplitude, the child amplitudes and the actual $\sin t_\ell$ are all nonzero.

Combining these estimates, choose

$$\eta = 2^{-80}. \tag{G.38}$$

Every nonminimal charge is its saddle term plus an absolute conditional error $CN^Ae^{N(\Re\psi(0,u)-\eta)}$. It need not have a relative expansion at its own much lower action. But (G.22) and (10.2) give $4\kappa < \eta/2$. Consequently these errors are below the first conditional charge. Equation (G.21) orders all retained charges relative to that same charge. Its coefficient in (G.33) is exactly one, for both parities. This proves (G.23), with a nonzero leading amplitude.

The argument initially takes the formal grid parameters on circles avoiding $\sin 2\pi\sigma = 0$. The entire physical trace and its first-charge approximation are holomorphic through all collisions inside $|\sigma| < 1$. Apply the maximum principle to their whole difference on a slightly larger parameter disk. The leading orientation/crossing factors are explicit and nonzero; no divided endpoint current is declared separately removable. This extends the estimate, and the eventual nonvanishing, to every actual $\sigma = 0, \pm s$ and both complete parity sequences.

Finally, prove (G.24) freshly from the exact trace rather than from its local approximation. This step can use polynomial norms directly, so colliding-grid derivatives cause no additional assumption. The real sup norm of each canonical child polynomial, divided by $|M_{w,N}(0)|$, is polynomial by (G.24). For fixed real u, (G.6) gives the real sup norm in t of the canonical kernel polynomial, retaining its factor $\exp N[2G_0 - c_K(1-\cos u)]$. Real sup and Chebyshev coefficient norms are polynomially equivalent at degree less than N. The full crossing polynomial has polynomial norm. Multiply these polynomials, apply the exact Chebyshev folding bound in the correct P_{σ_v} quotient, and then its top-coefficient functional (E.3). All these are full polynomial identities and retain every normal channel. They give

$$|M_{v,N}(u)| \le CN^A\left|\prod_w M_{w,N}(0)\right| e^{N[2G_0-c_K(1-\cos u)]}. \tag{G.39}$$

The nonzero value just proved at $u = 0$ has the exponent $2G_0 - \Delta_v$, with $0 \le \Delta_v < 4\kappa$. For $|u| \ge r/2$,

$$4\kappa < (c_K - c)(1 - \cos(r/2)); \tag{G.40}$$

for $|u| < r/2$, (G.23) and (G.29) give the stronger bound. Hence (G.24), and then (G.25), hold with the same constants for the outgoing message. This completes the induction. The proof is unchanged for a leaf, where the child product is empty.

G.6 Root assembly and the original filled action

Apply the same local trace argument to (G.4). Its normalized input has d_0 incoming messages and no outgoing edge, so the only change to (G.14)–(G.39) is to omit $\psi(t,u)$ and its constant. The current-vertex drop, charge ordering, tails and connector estimates have the same constants.

Recover the subtree constants recursively rather than summing their upper bounds. The child envelope identity (G.19) shows that the selected angles satisfy, at every original vertex,

$$\sum_{e\sim v}\psi_{e,\tau_v}+\frac{in_v\tau_v}{2\pi}=i. \tag{G.41}$$

Their modulus is bounded by $2\pi/[(19/20)\sqrt{d_v^2+n_v^2}]$, which is much less than the geometric chart radius. Substituting the recursive critical values makes all child constants telescope. The root action is exactly

$$\mathcal{C}(\mathbf{1})=\operatorname{stat}_{\boldsymbol{\tau}}\left[\sum_e\psi_e+\frac{i}{4\pi}\sum_v n_v\tau_v^2-i\sum_v\tau_v\right]. \tag{G.42}$$

It is the original filled action, not a sum of independently optimized cap actions. Its selected branch agrees with the unique small simultaneous solution, by the same weighted contraction used in (G.16).

For the geometric path replace the right side of (G.41) by ic, $0\le c\le 1$. The explicit edge Hessian estimates in Section E.2.5 give the simultaneous contraction on the radius $1/4$ polydisk, with diagonal nonlinear bound $a_v/50$ and row bound $a_v/25$. Condition (10.2) implies $\sqrt{d_v^2+n_v^2}\ge\kappa^{-1/2}$. Consequently that solution has

$$|\tau_v(c)|\le\frac{2\pi c}{(1-1/50)\sqrt{d_v^2+n_v^2}}<10^{-2},\qquad \Im\tau_v(c)<0\quad(c>0). \tag{G.43}$$

The same factorization of the nonlinear derivative by τ_v proves the sign, just as in (G.17). These are precisely the primitive-completion hypotheses of Section F.3. All original disk periods, half twists and the calibrated action are therefore retained along the actual positive path.

Each edge sum contributes $N^{-3/2}$; each actual circle trace contributes N^2 times one nondegenerate one-dimensional Gaussian, hence $N^{3/2}$. Since a tree has one more vertex than edge, the final power is $N^{3/2}$. All its leading amplitudes are nonzero. The exact first coefficient (G.33), rather than separate evenness alone, pairs the reflected Fourier modes.

The edge insertion is still H_N/W_k, with the original Gamma interpolation and physical offsets. Its leading value is one and its first correction is the same cotangent insertion as in Theorem 9.1. The exact scalar ledger is unchanged by finite summation order. Thus the complete result is (10.3)–(10.1), and the same first correction can be obtained by the normalized insertion rule if desired.

G.7 A vertexwise geometric criterion

We now prove Theorem 10.2. All the original ordered cell maps and primitive markings remain those of Appendix F. This part of the argument does not assume the local quantum condition.

G.7.1 Exact derivatives of the nonlinear error

On the branches through zero define

$$F(t,u) = 2\,\mathrm{arsinh}(\tan(t/2)\cos(u/2))\,, \qquad E(t,u) = F(t,u) - t. \tag{G.44}$$

The edge action satisfies $-2\pi\psi_t = F$. Put

$$Q(t,u) = \sqrt{1 - \sin^2(t/2)\sin^2(u/2)}, \qquad Q(0,0) = 1. \tag{G.45}$$

Differentiation gives exactly

$$F_t = \frac{\cos(u/2)}{\cos(t/2)Q(t,u)}, \qquad F_u = -\frac{\sin(t/2)\sin(u/2)}{Q(t,u)}. \tag{G.46}$$

Indeed $1 + \tan^2(t/2)\cos^2(u/2) = Q^2/\cos^2(t/2)$; the square roots agree at zero and hence throughout the small connected chart. This fixes the branches in (G.46).

For $|t|, |u| \le 10^{-2}$, let

$$a = \frac{1}{79000}, \qquad b = \frac{1}{199}, \qquad \omega = b^4, \qquad \delta = \frac{1}{39000}. \tag{G.47}$$

The elementary series bounds with $r = 1/200$ give

$$|\sin(t/2)| \le \frac{r}{1 - r^2/6} < b, \qquad |\cos(t/2) - 1| \le \frac{r^2}{2(1 - r^2/12)} < a, \tag{G.48}$$

and the same for u. Thus $|Q|^{-1} \le (1-\omega)^{-1}$ and $|Q - 1| \le \omega$. To see the latter estimate, use $(Q-1)(Q+1) = -\sin^2(t/2)\sin^2(u/2)$; the branch has $\mathrm{Re}\, Q > 0$, so $|Q + 1| \ge 1$. Equation (G.46) now yields

$$\begin{aligned} |E_t| &\le \frac{2a/(1-a) + \omega}{1 - \omega} < \delta, \\ |E_u| &\le \frac{b^2}{1 - \omega} < \delta. \end{aligned} \tag{G.49}$$

Both final inequalities are rational comparisons. In particular, $E(0,u) = 0$, and

$$E(t,u) = tH(t,u), \qquad H(t,u) = \int_0^1 E_t(\lambda t, u)\, d\lambda, \qquad |H(t,u)| \le \delta. \tag{G.50}$$

The function H is holomorphic also at $t = 0$. No division by a possibly vanishing unknown is needed to define it.

G.7.2 A contraction independent of the size of the tree

Write $L_v = \sqrt{d_v^2 + n_v^2} = |d_v - i n_v|$. Multiplying (10.10) by -2π and using (G.44) gives

$$(d_v - i n_v)\tau_v + \sum_{w \sim v} E(\tau_v, \tau_w) = -2\pi i c. \tag{G.51}$$

Define the fixed-point map

$$\mathcal{T}_{c,v}(\boldsymbol{\tau}) = -\frac{2\pi i c}{d_v - i n_v} - \frac{\sum_{w\sim v} E(\tau_v, \tau_w)}{d_v - i n_v}. \tag{G.52}$$

On the closed polydisk of radius $\rho = 10^{-2}$, (G.50) and $d_v/L_v \le 1$ give, uniformly for complex $|c| \le 1$,

$$|\mathcal{T}_{c,v}| \le \frac{2\pi}{630} + \delta\rho < \rho. \tag{G.53}$$

For example the last strict inequality already follows from $\pi < 22/7$. The total row derivative is at most

$$\frac{d_v\delta + d_v\delta}{L_v} \le 2\delta < 1. \tag{G.54}$$

Hence (G.52) is a strict contraction in the ordinary maximum norm, with a constant independent of the graph size. It has exactly one fixed point in the polydisk. Iteration from zero converges uniformly in c, so the fixed point is holomorphic for $|c| < 1$. The strict margin in (G.53) allows $|c| \le 1 + \varepsilon$ for some positive ε, proving the stated neighborhood assertion. The selected value at $c = 0$ is zero.

The sign needed for geometric completion is not inferred from smallness. Equation (G.50) rewrites (G.51) as

$$A_v(\boldsymbol{\tau})\tau_v = -2\pi i c, \qquad A_v = d_v - i n_v + \sum_{w\sim v} H(\tau_v, \tau_w). \tag{G.55}$$

For real $c > 0$,

$$\operatorname{Re} A_v \ge (1-\delta)d_v > 0, \qquad \operatorname{Im}\tau_v = -\frac{2\pi c \operatorname{Re} A_v}{|A_v|^2} < 0. \tag{G.56}$$

Moreover $|A_v| \ge L_v - \delta d_v$, which proves (10.11). Every τ_v is nonzero for $c > 0$.

This argument allows arbitrary signs of n_v, and $n_v = 0$. It makes no assumption that neighboring powers are comparable. The same sign proof works even when a large degree makes the total meridian logarithm in (10.12) close to $-2\pi i$.

G.7.3 Actual metric and absolute volume

The octahedral chart depends only on the parameters at the ends of one physical edge. Conditions (10.11) and (G.53) keep every such pair in its proved positive disk. The actual ordered disk seams match their full complex ratios, including the single prescribed odd seam. The original cut proof retains every primitive meridian and longitude. Thus Section F.3 applies to the path just constructed, rather than to an unrelated solution of exponentiated gluing equations.

In particular, the actual logarithmic periods start at zero as $c \downarrow 0$: in a fixed complete peripheral frame the second fixed point escapes to infinity, and the increments of $\log(z - q_v)$ along the fixed generator paths tend to zero. Their differences from (10.12) are continuous integers times $2\pi i$, and hence vanish along the whole path. The slope in (10.12) makes one full primitive turn at $c = 1$. This is the winding information that exponentiated filling equations alone would not supply.

Let

$$\mathcal{C}(c) = \operatorname*{stat}_{\boldsymbol{\tau}} \left[\Psi + \frac{i}{4\pi} \sum_v n_v \tau_v^2 - ic \sum_v \tau_v \right]. \tag{G.57}$$

The same actual fundamental chain and primitive period calibration as in Section F.3.6 give

$$\mathrm{Vol}_c = 2\pi \operatorname{Re} \mathcal{C}(c), \qquad \frac{d}{dc} \mathrm{Vol}_c = 2\pi \sum_v \operatorname{Im} \tau_v(c). \tag{G.58}$$

The complete anchor is the volume of two regular octahedra per tree edge. Every complementary primitive λ_v represents the core, of length $\ell_v = -2 \operatorname{Im} \tau_v$. The estimates above give the explicit bounds

$$\frac{4\pi c(1-\delta) d_v}{(L_v + \delta d_v)^2} \le \ell_v(c) \le \frac{4\pi c(1+\delta) d_v}{(L_v - \delta d_v)^2}. \tag{G.59}$$

If $S = \sum_v d_v / L_v^2$, integration yields

$$\frac{2\pi^2(1-\delta)}{(1+\delta)^2} S \le \mathrm{Vol}_0 - \mathrm{Vol}_1 \le \frac{2\pi^2(1+\delta)}{(1-\delta)^2} S. \tag{G.60}$$

Here S is a useful bound on the volume change, not a hypothesis for existence or completion.

When the main component is a knot, the geometric preferred-cusp continuation of Section F.4 also applies. With the original oriented writhe $w_{\mathbf{n}}$, it gives

$$\tau_K = -w_{\mathbf{n}} - \sum_v n_v + 2i \sum_{vw} \sqrt{1 + \tan^2(\tau_v(1)/2) + \tan^2(\tau_w(1)/2)}. \tag{G.61}$$

Its proof uses the actual marked representation and preferred longitude. Equation (G.61) is not inferred from a quantum correction.

G.7.4 A finite criterion with different radii at different vertices

One can keep the local graph data instead of using the common threshold (10.9). Choose radii $0 < r_v \le 10^{-2}$. Set

$$s_v = \sinh(r_v/2), \qquad a_v = \cosh(r_v/2) - 1, \qquad \omega_{vw} = s_v^2 s_w^2. \tag{G.62}$$

The same derivative calculation gives

$$\begin{aligned} \alpha_{vw} &= \frac{a_v + a_w}{(1-a_v)\sqrt{1-\omega_{vw}}} + \frac{1}{\sqrt{1-\omega_{vw}}} - 1, \\ \beta_{vw} &= \frac{s_v s_w}{\sqrt{1-\omega_{vw}}}, \end{aligned} \tag{G.63}$$

as bounds for $|E_t|$ and $|E_u|$ on the corresponding bidisk. Put $h_v = \sum_{w \sim v} \alpha_{vw}$. The following finite inequalities suffice:

$$\begin{aligned} &2\pi + h_v r_v < L_v r_v, \qquad h_v < d_v, \\ &\max_v \left[\frac{h_v}{L_v} + \sum_{w \sim v} \frac{\beta_{vw} r_w}{L_v r_v} \right] < 1. \end{aligned} \tag{G.64}$$

The first line makes (G.52) preserve the product of the vertex disks and gives the sign in (G.56). The second is a contraction bound in the weighted norm $\max_v |\tau_v|/r_v$. The proof of Sections G.7.2–G.7.3 applies unchanged, with

$$|\tau_v(c)| \le \frac{2\pi c}{L_v - h_v}. \tag{G.65}$$

This is a finite numerical certificate in the integer degrees and powers and the chosen positive radii. All quantities can be bounded by rational Taylor enclosures; it does not assume the existence of a saddle or a geometric representation.

More generally, the second line of (G.64) can be replaced by a positive-vector certificate for the derivative matrix with diagonal entries h_v/L_v and off-diagonal entries β_{vw}/L_v. If a positive vector q and a number $\kappa < 1$ satisfy $Mq \le \kappa q$, contraction holds in the q-weighted norm. The invariant polydisk is still supplied separately by the first line of (G.64). Thus choosing the contraction norm does not change the geometric parameter region.

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